



HISO-8583 (87)  
Host-to-Host  
Merchant  
Specification

|               |                  |
|---------------|------------------|
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## 1 INTRODUCTION

### 1.1 PURPOSE OF THIS DOCUMENT

This document describes an ISO 8583 based Host-to-Host financial card transactions message protocol which is supported by Tandem Base24 POS application system. It describes the different message types, message structures, message contents and tokens used in the ISO 8583 format. This message protocol is based on ISO 8583:1987 version which is bitmap driven, variable length, variable content message that can be configured differently, based on the type of message being sent.

### 1.2 GLOSSARY OF TERMS

| Term         | Description                                                                                                                                                                                               |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3DES         | Triple Data Encryption Standard                                                                                                                                                                           |
| ANSI         | American National Standards Institute                                                                                                                                                                     |
| ARC          | Authorisation Response Cryptogram                                                                                                                                                                         |
| ARPC         | Authorisation Response Cryptogram                                                                                                                                                                         |
| ARQC         | Authorisation Request Cryptogram                                                                                                                                                                          |
| ASCII        | American Standard Code for Information Interchange                                                                                                                                                        |
| BASE24       | Applications software (card authorisations) supplied by ACI Ltd                                                                                                                                           |
| COTS         | Consumer off-the shelf device                                                                                                                                                                             |
| Cryptography | Cryptography is the art of Secret writing. It is art and science of using mathematics to secure information and high degree of trust in the electronic realm                                              |
| CVM          | Cardholder Verification Method                                                                                                                                                                            |
| Data Key     | Is an encryption Key which is created from a number of selected fields within the message (fields are agreed between the two communicating parties) and is used to encrypt / decrypt the complete message |
| Decryption   | Decryption is the reverse of encryption. It is the transformation of encrypted data back into an intelligible form                                                                                        |
| DUKPT        | Derived Unique Key Per Transaction                                                                                                                                                                        |
| EMV          | Europay International MasterCard International and Visa International manage, maintain and enhance the EMV Integrated Circuit Card specifications for Payment Systems                                     |
| Encryption   | Encryption is the transformation of data into a form that is as close to impossible as possible to read without the appropriate knowledge                                                                 |

|                   |                                                                                                                                                                                                                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HSM               | Hardware Security Module. A secure device used to perform various cryptographic functions                                                                                                                                                                                                                                                        |
| IFSF              | International Fuel Standards Forum. A body to produce a standardised set of specifications for the Fuel Industry                                                                                                                                                                                                                                 |
| IP                | The Internet Protocol is responsible for routing an Internet datagram (data packet) between hosts                                                                                                                                                                                                                                                |
| ISO8583           | International Organisation for Standardisation                                                                                                                                                                                                                                                                                                   |
| LMK               | Local Master Key. LMKs are set of Data Encryption Standard (DES) Keys created, using a Chip Card, and stored on the HSM when the HSM is installed                                                                                                                                                                                                |
| NRID              | Network Reference Id. A numeric Identification generated by Diners/Discover and remains unchanged for the life of the transaction                                                                                                                                                                                                                |
| OCT               | Original Credit Transaction – Where a card acceptor is paying funds onto a cardholder's account. VISA refer to this an Original Credit (OC) transaction and MasterCard as a Payment transaction                                                                                                                                                  |
| PAC               | PIN Authentication Code                                                                                                                                                                                                                                                                                                                          |
| PAN               | Primary Account Number                                                                                                                                                                                                                                                                                                                           |
| PAR               | Payment Account Reference                                                                                                                                                                                                                                                                                                                        |
| PIN               | Personalised Identification Number                                                                                                                                                                                                                                                                                                               |
| PIN Block Formats | Denotes the standards to create a 16 digit PIN block                                                                                                                                                                                                                                                                                             |
| PIN Key           | Is a Data Encrypting key which is used to encrypt / decrypt the PINs for transfer between communicating parties                                                                                                                                                                                                                                  |
| Session Key       | A key for symmetric key cryptosystems which is used for the duration of one message or communication session                                                                                                                                                                                                                                     |
| SCA               | Strong Customer Authentication                                                                                                                                                                                                                                                                                                                   |
| SDES              | Single Data Encryption Standard                                                                                                                                                                                                                                                                                                                  |
| ZMK               | Zone Master Key. A Zone Master Key is a key encrypting key which is generated by entering 1 or up to 3 components on to the console. The resultant key is manually distributed between two communicating sites. The simplest model is to have a 'symmetric' system where the Zone Master Key is the same in clear for both communicating parties |
| MPG ID            | Merchant Payment Gateway - An ID provided for registering as a service provider                                                                                                                                                                                                                                                                  |

### 1.3 REFERENCES

- Base24 External Message Manual, R6.0v8
- Base24 Tokens Manual, R6.0v10

## 1.4 DATA ELEMENT FIELD DEFINITIONS

|              |                                                                                                                                                                |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a            | alphabetic characters                                                                                                                                          |
| n            | numeric characters                                                                                                                                             |
| s            | special characters                                                                                                                                             |
| an           | alphabetic and numeric characters                                                                                                                              |
| as           | alphabetic and special characters                                                                                                                              |
| ns           | numeric and special characters                                                                                                                                 |
| ans          | alphabetic, numeric and special characters                                                                                                                     |
| ..           | Variable length data element                                                                                                                                   |
| YY or YYYY   | Year                                                                                                                                                           |
| MM           | Month                                                                                                                                                          |
| DD           | Day                                                                                                                                                            |
| HH           | Hour                                                                                                                                                           |
| MM           | Minute                                                                                                                                                         |
| SS           | Second                                                                                                                                                         |
| M            | Mandatory data element                                                                                                                                         |
| O            | Optional data element                                                                                                                                          |
| C            | Conditional data element                                                                                                                                       |
| CE           | Conditional Echo data element                                                                                                                                  |
| ME           | Mandatory Echo data element                                                                                                                                    |
| <del>b</del> | Represents a 'space' value. Where this is advised as a valid value it may be used if not specified then by default 'space' is not a valid value for the field. |



## 2 TRANSACTION OVERVIEW

### 2.1 CARD TRANSACTIONS

#### 2.1.1 AUTHORISATION TRANSACTION

##### *2.1.1.1 Authorisation Request and Response Message*

0100 Authorisation request message:

An authorisation request (0100) message requests approval authorisation for the transaction to proceed further. This transaction is not intended to charge or bill the cardholder account.

An authorisation request response (0110) message is expected in return for the 0100 message, either approving or denying the request.

0110 Authorisation Response Message:

An authorisation request response message is returned in response to an Authorisation Request (0100) message to approve or deny the request.

#### 2.1.2 FINANCIAL TRANSACTION

##### *2.1.2.1 Financial Request and Response Message*

0200 Financial Request Message:

A Financial Transaction Request (0200) message requests approval for a transaction that, if approved, can be immediately applied to the account of the customer for billing or statement purposes.

A Financial Transaction Request Response (0210) message is expected in return for the 0200 message, either approving or denying the request.

0210 Financial Response Message:

The Financial Transaction Request Response (0210) message is returned in response to a Financial Transaction Request (0200) message to approve or deny the request.

Prior agreement with Barclaycard Payment Solutions must be sought prior to using these message types.

### 2.1.3 AUTHORISATION ADVICES

#### *2.1.3.1 Authorisation Advice Request and Response Message*

0120 Authorisation Advice Request Message:

The Authorisation Advice (0120) Message contains information pertaining to a previous transaction that was approved on behalf of the Card Issuer. The purpose is to inform the Card Issuer about the details of this transaction. It is not intended to permit application of the transaction to the cardholder account for the purpose of issuing a bill or statement.

0121 Authorisation Advice Repeat Request Message:

An Authorisation Advice Repeat (0121) message is identical to an Authorisation Advice (0120) message, except that it denotes to the receiver that it is a possible duplicate message. A 0121 message is used when an acknowledgment was expected to a 0120 message but never received.

0130 Authorisation Advice Response Message:

An Authorisation Advice Response (0130) acknowledges the receipt of an Authorisation Advice (0120) message or an Authorisation Advice Repeat (0121) message.

### 2.1.4 FINANCIAL ADVICES

#### *2.1.4.1 Financial Advice Request and Response Message*

0220 Financial Advice Request Message:

A Financial Transaction Advice (0220) message advises of a previously completed financial transaction.

0221 Financial Advice Repeat Request Message:

A Financial Transaction Advice Repeat (0221) message is identical to a Financial Transaction Advice (0220) message, except that it denotes to the receiver that it is a possible duplicate message. A 0221 message is used when an acknowledgment was expected to a 0220 message but never received.

0230 Financial Advice Response Message:

A Financial Transaction Advice Response (0230) message acknowledges the receipt of a Financial Transaction Advice (0220) or Financial Transaction Advice Repeat (0221) message.

## 2.1.5 REVERSAL ADVICES

### *2.1.5.1 Reversal Advice Request and Response Message*

#### 0420 Reversal Request Message:

The reversal transaction can originate from the Terminal, Merchant Host System or from the Base24 to the Card Issuer. The purpose is to restore an account when an approved response has been received but e.g. the purchase could not be finalised.

An Acquirer Reversal Advice (0420) message reverses an earlier transaction or authorisation.

#### 0421 Reversal Advice Repeat Request Message:

A Reversal Advice Repeat (0421) message is identical to a Reversal Advice (0420) message, except that it denotes to the receiver that it is a possible duplicate message. A 0421 message is used when an acknowledgment was expected to a 0420 message but never received.

#### 0430 Reversal Advice Response Message:

A Reversal Advice Response (0430) message acknowledges an Acquirer Reversal Advice (0420) message or an Acquirer Reversal Advice Repeat (0421) message.

## 2.2 NETWORK MANAGEMENT

### 2.2.1 NETWORK MANAGEMENT MESSAGE REQUEST AND RESPONSE

#### 0800 Network Management Request Message:

A Network Management Request (0800) message is used to send echo-test and logon messages.

#### 0810 Network Management Response Message:

A Network Management Request Response (0810) message is returned in response to a Network Management Request (0800) message.

### 3 H2H INTERFACE

The message format used is ASCII for all message types.

#### 3.1 MESSAGE TYPES IDENTIFIER

Message Type Identifier (MTI) identifies which transaction type is being handled. This field is 4 bytes and the following values are allowed:

| Message Type | Message Name                                 | Usage                                                                          |
|--------------|----------------------------------------------|--------------------------------------------------------------------------------|
| 0100         | Authorisation request                        | Reservation of funds, no financial impact                                      |
| 0110         | Authorisation request response               | Returned in response to a 0100 Authorisation request                           |
| 0120/0121    | Authorisation advice / repeat advice         | Fulfills a previous authorisation request or advises of a stand-in transaction |
| 0130         | Authorisation advice response                | Returned in response to a 0120/0121                                            |
| 0200         | Financial transaction request                | Sale and Refund Financial transaction                                          |
| 0210         | Financial transaction request response       | Returned in response to a 0200                                                 |
| 0220/0221    | Financial transaction advice / repeat advice | Fulfills a previous authorisation request or advises of a stand-in transaction |
| 0230         | Financial transaction advice response        | Returned in response to a 0220/0221                                            |
| 0420/0421    | Reversal advice / repeat advice              | Reversal of a previous authorisation or financial transaction request.         |
| 0430         | Reversal advice response                     | Returned in response to a 0420/0421                                            |
| 0800         | Network Management request message           | Used to perform echo tests                                                     |
| 0810         | Network management response message          | Returned in response to a 0800                                                 |

## 3.2 MESSAGE HEADER

### 3.2.1 START OF MESSAGE HEADER

The start of message Header indicator is used to mark the start of the actual message. This field must contain the literal "ISO". For incoming request messages into the Barclaycard host they are always required, for outgoing response messages from the Barclaycard host, they are always present.

### 3.2.2 MESSAGE HEADER

The BASE24 Header contains only mandatory fields and must have a fixed length so the message can be parsed correctly.

| Position | Length | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-2      | 2      | Product Indicator<br>02= BASE24 – POS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3-4      | 2      | Release Number<br>60 = BASE24-6.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5-7      | 3      | Status<br>Indicates whether there was a problem with the interpretation of the message.<br>If the message was rejected because of a security failure, this field indicates the reason. Valid values are as follows:<br><br>000 = Normal status – no error or failure<br>199 = Security device failure<br>210 = MSG key synchronization error<br>220 = MAC key synchronization error<br>230 = PIN key synchronization error<br><br>If the message was rejected because of bad data in the message, the ISO Host Interface process loads the bit map element number of the offending data element into this field and returns the message to the host |
| 8        | 1      | Originator Code<br>Indicates the network entity that introduced the transaction.<br>Valid values are as follows:<br><br>1 = Device controlled by a BASE24 process<br>5 = Host                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Position | Length | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9        | 1      | <p>Responder Code</p> <p>Indicates the network entity that created the (authorisation) response. Valid values are as follows:</p> <p>0 = Undetermined<br/>1 = Device controlled by a BASE24 process<br/>2 = Device Handler process<br/>3 = Authorisation process<br/>4 = ISO Host Interface process<br/>5 = Host<br/>6 = Interchange Interface process or remote banking standard unit interface process<br/>7 = Interchange or remote banking endpoint device<br/>8 = From Host Maintenance process<br/>9 = XPNET Billpay Server process</p> |

Refer to APPENDIX B: EXAMPLE FOR AUTH REQUEST AND RESPONSE

### 3.3 HANDLING OF ERRONEOUS MESSAGES

If the ISO Host Interface process receives an external message that it cannot process or reformat for internal use (because of bad data or a security related failure), it rejects the message as follows:

1. Changes the first position of the message type to 9 (for example, a 0200 message would be changed to a 9200 message, and a 0420 message would be changed to a 9420 message).
2. Indicates the reason the message was rejected in the status field of the BASE24 external message header. If the message was rejected because of a security failure, the status field is set to a value of 199. If the message was rejected because of bad data, this field is set to the bit number of the data element causing rejection (for example, if the Track 2 data in P-35 cannot be parsed, the status field in the header would be set to 035).
3. Returns the message to the host.

The above actions are taken on any message type that cannot be processed, not just those that require a response.

### 3.4 PRIMARY BITMAP

This field is used to identify which data components P1 to P64 that are included in the message. It is a 64 bit array (8 byte) where the bits are numbered 1 to 64, left to right. If e.g. bit 17 is '1', the data field P-17 is present in the message, if bit 17 is '0', the data field P-17 is not present in the message. Since the BASE24 external message implementation of the ISO 8583 standard doesn't allow binary data fields, the bit map is transmitted in display format such that for each 4-bit nibble of the bit map, the display character code of the corresponding hexadecimal digit is transmitted. This means that the 8 byte binary bit map will be extended to 16 byte, each containing the ASCII character code representing the display characters 0-9 or A-F.

Example: A possible primary bit map for an authorisation response (purchase)

|      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 1011 | 0010 | 0011 | 1000 | 1100 | 0010 | 1001 | 1010 | 1110 | 1100 | 0000 | 1000 | 0000 | 0111 | 0000 | 0010 |
| B    | 2    | 3    | 8    | C    | 2    | 9    | A    | E    | C    | 0    | 8    | 0    | 7    | 0    | 2    |



### 3.5 MESSAGE STRUCTURE & CONTENT

The following sections describe the message type structure and contents for the supported message types. In the sections that follow, the tables included only show the message fields that are required. The tables do not include fields that are not used or that are intended for future use.

The table below identifies the elements that are included or expected by default in ISO POS messages

| Data Element |                             | Message Class |      |        |      |                       |      |        |      |          |      |                    |       |
|--------------|-----------------------------|---------------|------|--------|------|-----------------------|------|--------|------|----------|------|--------------------|-------|
|              |                             | Authorisation |      |        |      | Financial Transaction |      |        |      | Reversal |      | Network Management |       |
|              |                             | 0100          | 0110 | 0120/1 | 0130 | 0200                  | 0210 | 0220/1 | 0230 | 0420/1   | 0430 | 0800               | 0810  |
|              | Message Header              | M             | M    | M      | M    | M                     | M    | M      | M    | M        | M    | M                  | M     |
|              | Message Type                | M             | M    | M      | M    | M                     | M    | M      | M    | M        | M    | M                  | M     |
|              | Primary Bitmap              | M             | M    | M      | M    | M                     | M    | M      | M    | M        | M    | M                  | M     |
| P-1          | Secondary Bit Map           | M             | C    | M      | C    | M                     | C    | M      | C    | M        | C    | M                  | C     |
| P-2          | Primary Account Number      | C             | CE   | C      | CE   | C                     | CE   | C      | CE   | M        | ME   | N/A                | N/A   |
| P-3          | Processing Code             | M             | ME   | M      | ME   | M                     | ME   | M      | ME   | M        | ME   | N/A                | N/A   |
| P-4          | Transaction Amount          | M             | M    | M      | ME   | M                     | M    | M      | ME   | M        | ME   | N/A                | N/A   |
| P-7          | Transmission Date and Time  | M             | M    | M      | M    | M                     | M    | M      | M    | M        | M    | M                  | M     |
| P-11         | Systems Trace Audit Number  | M             | ME   | M      | ME   | M                     | ME   | M      | ME   | M        | ME   | M                  | ME    |
| P-12         | Local Transaction Time      | M             | ME   | M      | ME   | M                     | ME   | M      | ME   | M        | ME   | O                  | N/A * |
| P-13         | Local Transaction Date      | M             | ME   | M      | ME   | M                     | ME   | M      | ME   | M        | ME   | O                  | N/A * |
| P-14         | Expiration Date             | C             | N/A  | C      | N/A  | C                     | N/A  | C      | N/A  | C        | N/A  | N/A                | N/A   |
| P-15         | Date, Settlement            | N/A           | N/A  | N/A    | N/A  | N/A                   | C    | N/A    | C    | N/A      | C    | N/A                | N/A   |
| P-18         | Merchant Type               | O             | N/A  | O      | N/A  | O                     | N/A  | O      | N/A  | O        | N/A  | N/A                | N/A   |
| P-22         | Point of Service Entry Mode | M             | N/A  | M      | N/A  | M                     | N/A  | M      | N/A  | C        | N/A  | N/A                | N/A   |

| Data Element |                                     | Message Class |      |        |      |                       |      |        |      |          |      |                    |      |
|--------------|-------------------------------------|---------------|------|--------|------|-----------------------|------|--------|------|----------|------|--------------------|------|
|              |                                     | Authorisation |      |        |      | Financial Transaction |      |        |      | Reversal |      | Network Management |      |
|              |                                     | 0100          | 0110 | 0120/1 | 0130 | 0200                  | 0210 | 0220/1 | 0230 | 0420/1   | 0430 | 0800               | 0810 |
| P-23         | Card Sequence Number                | C             | N/A  | C      | N/A  | C                     | N/A  | C      | N/A  | C        | N/A  | N/A                | N/A  |
| P-25         | Point of Service Condition Code     | M             | C    | M      | C    | M                     | C    | M      | C    | M        | C    | N/A                | N/A  |
| P-32         | Acquiring Institution Id Code       | M             | ME   | M      | ME   | M                     | ME   | M      | ME   | M        | ME   | N/A                | N/A  |
| P-35         | Track 2 Data                        | C             | N/A  | C      | N/A  | C                     | N/A  | C      | N/A  | N/A      | N/A  | N/A                | N/A  |
| P-37         | Retrieval Reference Number          | M             | ME   | M      | ME   | M                     | ME   | M      | ME   | M        | ME   | N/A                | N/A  |
| P-38         | Authorisation Code                  | N/A           | C    | C      | C    | N/A                   | C    | C      | C    | M        | C    | N/A                | N/A  |
| P-39         | Response code                       | N/A           | M    | M      | M    | N/A                   | M    | M      | M    | M        | M    | N/A                | M    |
| P-41         | Terminal ID                         | M             | ME   | M      | ME   | M                     | ME   | M      | ME   | M        | ME   | N/A                | N/A  |
| P-42         | Card Acceptor Identification Code   | O             | O    | O      | O    | O                     | O    | O      | O    | O        | O    | N/A                | N/A  |
| P-43         | Card Acceptor Name Location         | M             | N/A  | M      | N/A  | M                     | N/A  | M      | N/A  | M        | O    | N/A                | N/A  |
| P-44         | Additional Response Data            | N/A           | C    | N/A    | N/A  | N/A                   | C    | N/A    | N/A  | N/A      | N/A  | N/A                | N/A  |
| P-48         | Retailer Data                       | M             | ME   | M      | ME   | M                     | ME   | M      | ME   | M        | ME   | N/A                | N/A  |
| P-49         | Currency Code                       | M             | ME   | M      | ME   | M                     | ME   | M      | ME   | M        | ME   | N/A                | N/A  |
| P-52         | PIN Data                            | C             | N/A  | N/A    | N/A  | C                     | N/A  | N/A    | N/A  | N/A      | N/A  | N/A                | N/A  |
| P-54         | Additional Amounts                  | C             | CE   | N/A    | N/A  | C                     | CE   | N/A    | N/A  | C        | CE   | N/A                | N/A  |
| P-63         | Additional Data                     | M             | C    | M      | C    | M                     | C    | M      | C    | C        | C    | N/A                | N/A  |
| S-70         | Network Management Information Code | N/A           | N/A  | N/A    | N/A  | N/A                   | N/A  | N/A    | N/A  | N/A      | N/A  | M                  | M    |

| Data Element |                                           | Message Class |      |        |      |                       |      |        |      |          |      |                    |      |
|--------------|-------------------------------------------|---------------|------|--------|------|-----------------------|------|--------|------|----------|------|--------------------|------|
|              |                                           | Authorisation |      |        |      | Financial Transaction |      |        |      | Reversal |      | Network Management |      |
|              |                                           | 0100          | 0110 | 0120/1 | 0130 | 0200                  | 0210 | 0220/1 | 0230 | 0420/1   | 0430 | 0800               | 0810 |
| S-90         | Original Data Elements                    | N/A           | N/A  | C      | C    | N/A                   | N/A  | C      | C    | M        | ME   | N/A                | N/A  |
| S-100        | Receiving Institution Identification Code | O             | O    | O      | O    | O                     | O    | O      | O    | O        | O    | N/A                | N/A  |
| S-125        | Settlement Record 3                       | M             | N/A  | M      | N/A  | M                     | N/A  | M      | N/A  | M        | N/A  | N/A                | N/A  |

\* Except for IFSF petrol interface where this is CE

## 3.5.1 0100 AUTHORISATION REQUEST

| Bit   | Field Name                                | Presence | Format    | Description             |
|-------|-------------------------------------------|----------|-----------|-------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'           |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2  |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1    |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4    |
| P-1   | Secondary Bitmap                          | M        | an16      | Refer to section 3.6.1  |
| P-2   | Primary Account Number                    | C        | n..19     | Refer to section 3.6.2  |
| P-3   | Processing code                           | M        | n6        | Refer to section 3.6.3  |
| P-4   | Transaction amount                        | M        | n12       | Refer to section 3.6.4  |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.6.5  |
| P-11  | Systems Trace Audit Number                | M        | n6        | Refer to section 3.6.6  |
| P-12  | Local Transaction Time                    | M        | n6        | Refer to section 3.6.7  |
| P-13  | Local Transaction Date                    | M        | n4        | Refer to section 3.6.8  |
| P-14  | Expiration Date                           | C        | n4        | Refer to section 3.6.9  |
| P-18  | Merchant Type                             | C        | n4        | Refer to section 3.6.10 |
| P-22  | Point Of Service Entry Mode               | M        | n3        | Refer to section 3.6.11 |
| P-23  | Card Sequence Number                      | C        | n3        | Refer to section 3.6.12 |
| P-25  | Point of Service Condition Code           | M        | n2        | Refer to section 3.6.13 |
| P-32  | Acquiring institution ID code             | M        | n..11     | Refer to section 3.6.14 |
| P-35  | Track 2 Data                              | C        | ans ..37  | Refer to section 3.6.15 |
| P-37  | Retrieval Reference Number                | M        | an12      | Refer to section 3.6.16 |
| P-41  | Card Acceptor Terminal Identification     | M        | ans16     | Refer to section 3.6.19 |
| P-42  | Card acceptor identification code         | O        | ans15     | Refer to section 3.6.20 |
| P-43  | Card acceptor name/location               | M        | ans40     | Refer to section 3.6.21 |
| P-48  | Retailer Data                             | M        | ans30     | Refer to section 3.6.22 |
| P-49  | Currency code                             | M        | n3        | Refer to section 3.6.23 |
| P-52  | PIN data                                  | C        | an16      | Refer to section 3.6.24 |
| P-54  | Additional Amounts                        | C        | n15       | Refer to section 3.6.25 |
| P-63  | Additional Data                           | M        | ans ..600 | Refer to section 3.6.26 |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.6.29 |
| S-125 | Settlement Data                           | M        | ans15     | Refer to Section 3.6.30 |

## 3.5.2 0110 AUTHORISATION RESPONSE

| Bit   | Field Name                                | Presence | Format    | Description             |
|-------|-------------------------------------------|----------|-----------|-------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'           |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2  |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1    |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4    |
| P-1   | Secondary Bitmap                          | C        | an16      | Refer to section 3.7.1  |
| P-2   | Primary Account Number                    | CE       | n..19     | Refer to section 3.7.2  |
| P-3   | Processing code                           | ME       | n6        | Refer to section 3.7.3  |
| P-4   | Transaction amount                        | M        | n12       | Refer to section 3.7.4  |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.7.5  |
| P-11  | Systems Trace Audit Number                | ME       | n6        | Refer to section 3.7.6  |
| P-12  | Local Transaction Time                    | ME       | n6        | Refer to section 3.7.7  |
| P-13  | Local Transaction Date                    | ME       | n4        | Refer to section 3.7.8  |
| P-25  | Point of Service Condition Code           | C        | n2        | Refer to section 3.7.10 |
| P-32  | Acquiring institution ID code             | ME       | n..11     | Refer to section 3.7.11 |
| P-37  | Retrieval Reference Number                | ME       | an12      | Refer to section 3.7.12 |
| P-38  | Authorisation Identification Response     | C        | an6       | Refer to section 3.7.13 |
| P-39  | Response Code                             | M        | an2       | Refer to section 3.7.14 |
| P-41  | Card Acceptor Terminal Identification     | ME       | ans16     | Refer to section 3.7.15 |
| P-42  | Card Acceptor Identification Code         | O        | ans15     | Refer to section 3.7.16 |
| P-44  | Additional Response Data                  | C        | ans4      | Refer to section 3.7.18 |
| P-48  | Retailer Data                             | ME       | ans30     | Refer to section 3.7.19 |
| P-49  | Currency code                             | ME       | n3        | Refer to section 3.7.20 |
| P-54  | Additional Amounts                        | CE       | n15       | Refer to section 3.7.21 |
| P-63  | Additional Data                           | C        | ans ..600 | Refer to section 3.7.22 |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.7.25 |

## 3.5.3 0120 AUTHORISATION ADVICE &amp; 0121 AUTHORISATION ADVICE REPEAT REQUEST

| Bit   | Field Name                                | Presence | Format    | Description                        |
|-------|-------------------------------------------|----------|-----------|------------------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'                      |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2             |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1               |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4               |
| P-1   | Secondary Bitmap                          | M        | an16      | Refer to section 3.6.1             |
| P-2   | Primary Account Number                    | C        | n..19     | Refer to section 3.6.2             |
| P-3   | Processing code                           | M        | n6        | Refer to section 3.6.3             |
| P-4   | Transaction amount                        | M        | n12       | Refer to section 3.6.4             |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.6.5             |
| P-11  | Systems Trace Audit Number                | M        | n6        | Refer to section 3.6.6             |
| P-12  | Local Transaction Time                    | M        | n6        | Refer to section 3.6.7             |
| P-13  | Local Transaction Date                    | M        | n4        | Refer to section 3.6.8             |
| P-14  | Expiration Date                           | C        | n4        | Refer to section 3.6.9             |
| P-18  | Merchant Type                             | O        | n4        | Refer to section 3.6.10            |
| P-22  | Point Of Service Entry Mode               | M        | n3        | Refer to section 3.6.11            |
| P-23  | Card Sequence Number                      | C        | n3        | Refer to section 3.6.12            |
| P-25  | Point of Service Condition Code           | M        | n2        | Refer to section 3.6.13            |
| P-32  | Acquiring institution ID code             | M        | n..11     | Refer to section 3.6.14            |
| P-35  | Track 2 Data                              | C        | ans ..37  | Refer to section 3.6.15            |
| P-37  | Retrieval Reference Number                | M        | an12      | Refer to section 3.6.16            |
| P-38  | Authorisation Identification Response     | C        | an6       | Refer to section 3.6.17            |
| P-39  | Response Code                             | M        | an2       | Refer to section 3.6.18 and 3.7.14 |
| P-41  | Card Acceptor Terminal Identification     | M        | ans16     | Refer to section 3.6.19            |
| P-42  | Card acceptor identification code         | O        | ans15     | Refer to section 3.6.20            |
| P-43  | Card acceptor name/location               | M        | ans40     | Refer to section 3.6.21            |
| P-48  | Retailer Data                             | M        | ans30     | Refer to section 3.6.22            |
| P-49  | Currency code                             | M        | n3        | Refer to section 3.6.23            |
| P-63  | Additional Data                           | M        | ans ..600 | Refer to section 3.6.26            |
| S-90  | Original Data Elements                    | C        | n42       | Refer to section 3.6.28            |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.6.29            |
| S-125 | Settlement Data                           | M        | ans15     | Refer to Section 3.6.30            |

## 3.5.4 0130 AUTHORISATION ADVICE RESPONSE

| Bit   | Field Name                                | Presence | Format    | Description             |
|-------|-------------------------------------------|----------|-----------|-------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'           |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2  |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1    |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4    |
| P-1   | Secondary Bitmap                          | C        | an16      | Refer to section 3.7.1  |
| P-2   | Primary Account Number                    | CE       | n..19     | Refer to section 3.7.2  |
| P-3   | Processing code                           | ME       | n6        | Refer to section 3.7.3  |
| P-4   | Transaction amount                        | ME       | n12       | Refer to section 3.7.4  |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.7.5  |
| P-11  | Systems Trace Audit Number                | ME       | n6        | Refer to section 3.7.6  |
| P-12  | Local Transaction Time                    | ME       | n6        | Refer to section 3.7.7  |
| P-13  | Local Transaction Date                    | ME       | n4        | Refer to section 3.7.8  |
| P-25  | Point of Service Condition Code           | C        | n2        | Refer to section 3.7.10 |
| P-32  | Acquiring institution ID code             | ME       | n..11     | Refer to section 3.7.11 |
| P-37  | Retrieval Reference Number                | ME       | an12      | Refer to section 3.7.12 |
| P-38  | Authorisation Identification Response     | C        | an6       | Refer to section 3.7.13 |
| P-39  | Response Code                             | M        | an2       | Refer to section 3.7.14 |
| P-41  | Card Acceptor Terminal Identification     | ME       | ans16     | Refer to section 3.7.15 |
| P-42  | Card Acceptor Identification Code         | O        | ans15     | Refer to section 3.7.16 |
| P-48  | Retailer Data                             | ME       | ans30     | Refer to section 3.7.19 |
| P-49  | Currency code                             | ME       | n3        | Refer to section 3.7.20 |
| P-63  | Additional Data                           | C        | ans ..600 | Refer to section 3.7.22 |
| S-90  | Original Data Elements                    | C        | n42       | Refer to section 3.7.24 |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.7.25 |

## 3.5.5 0200 FINANCIAL REQUEST

| Bit   | Field Name                                | Presence | Format    | Description             |
|-------|-------------------------------------------|----------|-----------|-------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'           |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2  |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1    |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4    |
| P-1   | Secondary Bitmap                          | M        | an16      | Refer to section 3.6.1  |
| P-2   | Primary Account Number                    | C        | n..19     | Refer to section 3.6.2  |
| P-3   | Processing code                           | M        | n6        | Refer to section 3.6.3  |
| P-4   | Transaction amount                        | M        | n12       | Refer to section 3.6.4  |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.6.5  |
| P-11  | Systems Trace Audit Number                | M        | n6        | Refer to section 3.6.6  |
| P-12  | Local Transaction Time                    | M        | n6        | Refer to section 3.6.7  |
| P-13  | Local Transaction Date                    | M        | n4        | Refer to section 3.6.8  |
| P-14  | Expiration Date                           | C        | n4        | Refer to section 3.6.9  |
| P-18  | Merchant Type                             | C        | n4        | Refer to section 3.6.10 |
| P-22  | Point Of Service Entry Mode               | M        | n3        | Refer to section 3.6.11 |
| P-23  | Card Sequence Number                      | C        | n3        | Refer to section 3.6.12 |
| P-25  | Point of Service Condition Code           | M        | n2        | Refer to section 3.6.13 |
| P-32  | Acquiring institution ID code             | M        | n..11     | Refer to section 3.6.14 |
| P-35  | Track 2 Data                              | C        | ans ..37  | Refer to section 3.6.15 |
| P-37  | Retrieval Reference Number                | M        | an12      | Refer to section 3.6.16 |
| P-41  | Card Acceptor Terminal Identification     | M        | ans16     | Refer to section 3.6.19 |
| P-42  | Card acceptor identification code         | O        | ans15     | Refer to section 3.6.20 |
| P-43  | Card acceptor name/location               | M        | ans40     | Refer to section 3.6.21 |
| P-48  | Retailer Data                             | M        | ans30     | Refer to section 3.6.22 |
| P-49  | Currency code                             | M        | n3        | Refer to section 3.6.23 |
| P-52  | PIN data                                  | C        | an16      | Refer to section 3.6.24 |
| P-54  | Additional Amounts                        | C        | n15       | Refer to section 3.6.25 |
| P-63  | Additional Data                           | M        | ans ..600 | Refer to section 3.6.26 |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.6.29 |
| S-125 | Settlement Data                           | M        | ans15     | Refer to Section 3.6.30 |



## 3.5.6 0210 FINANCIAL RESPONSE

| Bit   | Field Name                                | Presence | Format    | Description             |
|-------|-------------------------------------------|----------|-----------|-------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'           |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2  |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1    |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4    |
| P-1   | Secondary Bitmap                          | C        | an16      | Refer to section 3.7.1  |
| P-2   | Primary Account Number                    | CE       | n..19     | Refer to section 3.7.2  |
| P-3   | Processing code                           | ME       | n6        | Refer to section 3.7.3  |
| P-4   | Transaction amount                        | M        | n12       | Refer to section 3.7.4  |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.7.5  |
| P-11  | Systems Trace Audit Number                | ME       | n6        | Refer to section 3.7.6  |
| P-12  | Local Transaction Time                    | ME       | n6        | Refer to section 3.7.7  |
| P-13  | Local Transaction Date                    | ME       | n4        | Refer to section 3.7.8  |
| P-15  | Date, Settlement                          | C        | n4        | Refer to section 3.7.9  |
| P-25  | Point of Service Condition Code           | C        | n2        | Refer to section 3.7.10 |
| P-32  | Acquiring institution ID code             | ME       | n..11     | Refer to section 3.7.11 |
| P-37  | Retrieval Reference Number                | ME       | an12      | Refer to section 3.7.12 |
| P-38  | Authorisation Identification Response     | C        | an6       | Refer to section 3.7.13 |
| P-39  | Response Code                             | M        | an2       | Refer to section 3.7.14 |
| P-41  | Card Acceptor Terminal Identification     | ME       | ans16     | Refer to section 3.7.15 |
| P-42  | Card Acceptor Identification Code         | O        | ans15     | Refer to section 3.7.16 |
| P-44  | Additional Response Data                  | C        | ans4      | Refer to section 3.7.18 |
| P-48  | Retailer Data                             | ME       | ans30     | Refer to section 3.7.19 |
| P-49  | Currency code                             | ME       | n3        | Refer to section 3.7.20 |
| P-54  | Additional Amounts                        | CE       | n15       | Refer to section 3.7.21 |
| P-63  | Additional Data                           | C        | ans ..600 | Refer to section 3.7.22 |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.7.25 |

## 3.5.7 0220 FINANCIAL ADVICE &amp; 0221 FINANCIAL ADVICE REPEAT REQUEST

| Bit   | Field Name                                | Presence | Format    | Description                        |
|-------|-------------------------------------------|----------|-----------|------------------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'                      |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2             |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1               |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4               |
| P-1   | Secondary Bitmap                          | M        | an16      | Refer to section 3.6.1             |
| P-2   | Primary Account Number                    | C        | n..19     | Refer to section 3.6.2             |
| P-3   | Processing code                           | M        | n6        | Refer to section 3.6.3             |
| P-4   | Transaction amount                        | M        | n12       | Refer to section 3.6.4             |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.6.5             |
| P-11  | Systems Trace Audit Number                | M        | n6        | Refer to section 3.6.6             |
| P-12  | Local Transaction Time                    | M        | n6        | Refer to section 3.6.7             |
| P-13  | Local Transaction Date                    | M        | n4        | Refer to section 3.6.8             |
| P-14  | Expiration Date                           | C        | n4        | Refer to section 3.6.9             |
| P-18  | Merchant Type                             | O        | n4        | Refer to section 3.6.10            |
| P-22  | Point Of Service Entry Mode               | M        | n3        | Refer to section 3.6.11            |
| P-23  | Card Sequence Number                      | C        | n3        | Refer to section 3.6.12            |
| P-25  | Point of Service Condition Code           | M        | n2        | Refer to section 3.6.13            |
| P-32  | Acquiring institution ID code             | M        | n..11     | Refer to section 3.6.14            |
| P-35  | Track 2 Data                              | C        | ans ..37  | Refer to section 3.6.15            |
| P-37  | Retrieval Reference Number                | M        | an12      | Refer to section 3.6.16            |
| P-38  | Authorisation Identification Response     | C        | an6       | Refer to section 3.6.17            |
| P-39  | Response Code                             | M        | an2       | Refer to section 3.6.18 and 3.7.14 |
| P-41  | Card Acceptor Terminal Identification     | M        | ans16     | Refer to section 3.6.19            |
| P-42  | Card acceptor identification code         | O        | ans15     | Refer to section 3.6.20            |
| P-43  | Card acceptor name/location               | M        | ans40     | Refer to section 3.6.21            |
| P-48  | Retailer Data                             | M        | ans30     | Refer to section 3.6.22            |
| P-49  | Currency code                             | M        | n3        | Refer to section 3.6.23            |
| P-63  | Additional Data                           | M        | ans ..600 | Refer to section 3.6.26            |
| S-90  | Original Data Elements                    | C        | n42       | Refer to section 3.6.28            |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.6.29            |
| S-125 | Settlement Data                           | M        | ans15     | Refer to Section 3.6.30            |

## 3.5.8 0230 FINANCIAL ADVICE RESPONSE

| Bit   | Field Name                                | Presence | Format    | Description             |
|-------|-------------------------------------------|----------|-----------|-------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'           |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2  |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1    |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4    |
| P-1   | Secondary Bitmap                          | C        | an16      | Refer to section 3.7.1  |
| P-2   | Primary Account Number                    | CE       | n..19     | Refer to section 3.7.2  |
| P-3   | Processing code                           | ME       | n6        | Refer to section 3.7.3  |
| P-4   | Transaction amount                        | ME       | n12       | Refer to section 3.7.4  |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.7.5  |
| P-11  | Systems Trace Audit Number                | ME       | n6        | Refer to section 3.7.6  |
| P-12  | Local Transaction Time                    | ME       | n6        | Refer to section 3.7.7  |
| P-13  | Local Transaction Date                    | ME       | n4        | Refer to section 3.7.8  |
| P-15  | Date, Settlement                          | C        | n4        | Refer to section 3.7.9  |
| P-25  | Point of Service Condition Code           | C        | n2        | Refer to section 3.7.10 |
| P-32  | Acquiring institution ID code             | ME       | n..11     | Refer to section 3.7.11 |
| P-37  | Retrieval Reference Number                | ME       | an12      | Refer to section 3.7.12 |
| P-38  | Authorisation Identification Response     | C        | an6       | Refer to section 3.7.13 |
| P-39  | Response Code                             | M        | an2       | Refer to section 3.7.14 |
| P-41  | Card Acceptor Terminal Identification     | ME       | ans16     | Refer to section 3.7.15 |
| P-42  | Card Acceptor Identification Code         | O        | ans15     | Refer to section 3.7.16 |
| P-48  | Retailer Data                             | ME       | ans30     | Refer to section 3.7.19 |
| P-49  | Currency code                             | ME       | n3        | Refer to section 3.7.20 |
| P-63  | Additional Data                           | C        | ans ..600 | Refer to section 3.7.22 |
| S-90  | Original Data Elements                    | C        | n42       | Refer to section 3.7.24 |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.7.25 |

## 3.5.9 0420 REVERSAL ADVICE &amp; 0421 REVERSAL ADVICE REPEAT REQUEST

| Bit   | Field Name                                | Presence | Format    | Description             |
|-------|-------------------------------------------|----------|-----------|-------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'           |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2  |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1    |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4    |
| P-1   | Secondary Bitmap                          | M        | an16      | Refer to section 3.6.1  |
| P-2   | Primary Account Number                    | M        | n..19     | Refer to section 3.6.2  |
| P-3   | Processing code                           | M        | n6        | Refer to section 3.6.3  |
| P-4   | Transaction amount                        | M        | n12       | Refer to section 3.6.4  |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.6.5  |
| P-11  | Systems Trace Audit Number                | M        | n6        | Refer to section 3.6.6  |
| P-12  | Local Transaction Time                    | M        | n6        | Refer to section 3.6.7  |
| P-13  | Local Transaction Date                    | M        | n4        | Refer to section 3.6.8  |
| P-14  | Expiration Date                           | C        | n4        | Refer to section 3.6.9  |
| P-18  | Merchant Type                             | O        | n4        | Refer to section 3.6.10 |
| P-22  | Point Of Service Entry Mode               | C        | n3        | Refer to section 3.6.11 |
| P-23  | Card Sequence Number                      | C        | n3        | Refer to section 3.6.12 |
| P-25  | Point of Service Condition Code           | M        | n2        | Refer to section 3.6.13 |
| P-32  | Acquiring institution ID code             | M        | n..11     | Refer to section 3.6.14 |
| P-37  | Retrieval Reference Number                | M        | an12      | Refer to section 3.6.16 |
| P-38  | Authorisation Identification Response     | M        | an6       | Refer to section 3.6.17 |
| P-39  | Response Code                             | M        | an2       | Refer to section 3.6.18 |
| P-41  | Card Acceptor Terminal Identification     | M        | ans16     | Refer to section 3.6.19 |
| P-42  | Card acceptor identification code         | O        | ans15     | Refer to section 3.6.20 |
| P-43  | Card acceptor name/location               | M        | ans40     | Refer to section 3.6.21 |
| P-48  | Retailer Data                             | M        | ans30     | Refer to section 3.6.22 |
| P-49  | Currency code                             | M        | n3        | Refer to section 3.6.23 |
| P-54  | Additional Amounts                        | C        | n15       | Refer to section 3.6.25 |
| P-63  | Additional Data                           | C        | ans ..600 | Refer to section 3.6.26 |
| S-90  | Original Data Elements                    | M        | n42       | Refer to section 3.6.28 |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.6.29 |
| S-125 | Settlement Data                           | M        | ans15     | Refer to Section 3.6.30 |

## 3.5.10 0430 REVERSAL ADVICE RESPONSE

| Bit   | Field Name                                | Presence | Format    | Description             |
|-------|-------------------------------------------|----------|-----------|-------------------------|
|       | Start of Message Header                   | M        | a3        | Value = 'ISO'           |
|       | Message Header                            | M        | n9        | Refer to section 3.2.2  |
|       | Message Type Identifier                   | M        | n4        | Refer to section 3.1    |
|       | Primary Bitmap                            | M        | an16      | Refer to section 3.4    |
| P-1   | Secondary Bitmap                          | C        | an16      | Refer to section 3.7.1  |
| P-2   | Primary Account Number                    | ME       | n ..19    | Refer to section 3.7.2  |
| P-3   | Processing code                           | ME       | n6        | Refer to section 3.7.3  |
| P-4   | Transaction amount                        | ME       | n12       | Refer to section 3.7.4  |
| P-7   | Transmission date and time                | M        | n10       | Refer to section 3.7.5  |
| P-11  | Systems Trace Audit Number                | ME       | n6        | Refer to section 3.7.6  |
| P-12  | Local Transaction Time                    | ME       | n6        | Refer to section 3.7.7  |
| P-13  | Local Transaction Date                    | ME       | n4        | Refer to section 3.7.8  |
| P-15  | Date, Settlement                          | C        | n4        | Refer to section 3.7.9  |
| P-25  | Point of Service Condition Code           | C        | n2        | Refer to section 3.7.10 |
| P-32  | Acquiring institution ID code             | ME       | n..11     | Refer to section 3.7.11 |
| P-37  | Retrieval Reference Number                | ME       | an12      | Refer to section 3.7.12 |
| P-38  | Authorisation Identification Response     | C        | an6       | Refer to section 3.7.13 |
| P-39  | Response Code                             | M        | an2       | Refer to section 3.7.14 |
| P-41  | Card Acceptor Terminal Identification     | ME       | ans16     | Refer to section 3.7.15 |
| P-42  | Card Acceptor Identification Code         | O        | ans15     | Refer to section 3.7.16 |
| P-43  | Card acceptor name/location               | O        | ans40     | Refer to section 3.7.17 |
| P-48  | Retailer Data                             | ME       | ans30     | Refer to section 3.7.19 |
| P-49  | Currency code                             | ME       | n3        | Refer to section 3.7.20 |
| P-54  | Additional Amounts                        | CE       | n15       | Refer to section 3.7.21 |
| P-63  | Additional Data                           | C        | ans ..600 | Refer to section 3.7.22 |
| S-90  | Original Data Elements                    | ME       | n42       | Refer to section 3.7.24 |
| S-100 | Receiving Institution Identification Code | O        | n ..11    | Refer to section 3.7.25 |

## 3.5.11 0800 NETWORK MANAGEMENT REQUEST

| Bit  | Field Name                          | Presence | Format | Description             |
|------|-------------------------------------|----------|--------|-------------------------|
|      | Start of Message Header             | M        | a3     | Value = 'ISO'           |
|      | Message Header                      | M        | n9     | Refer to section 3.2.2  |
|      | Message Type Identifier             | M        | n4     | Refer to section 3.1    |
|      | Primary Bitmap                      | M        | an16   | Refer to section 3.4    |
| P-1  | Secondary Bitmap                    | M        | an16   | Refer to section 3.6.1  |
| P-7  | Transmission date and time          | M        | n10    | Refer to section 3.6.5  |
| P-11 | Systems Trace Audit Number          | M        | n6     | Refer to section 3.6.6  |
| P-12 | Local Transaction Time              | O        | n6     | Refer to section 3.6.7  |
| P-13 | Local Transaction Date              | O        | n4     | Refer to section 3.6.8  |
| S-70 | Network Management Information Code | M        | n3     | Refer to Section 3.6.27 |

## 3.5.12 0810 NETWORK MANAGEMENT RESPONSE

| Bit  | Field Name                          | Presence | Format | Description             |
|------|-------------------------------------|----------|--------|-------------------------|
|      | Start of Message Header             | M        | a3     | Value = 'ISO'           |
|      | Message Header                      | M        | n9     | Refer to section 3.2.2  |
|      | Message Type Identifier             | M        | n4     | Refer to section 3.1    |
|      | Primary Bitmap                      | M        | an16   | Refer to section 3.4    |
| P-1  | Secondary Bitmap                    | C        | an16   | Refer to section 3.7.1  |
| P-7  | Transmission date and time          | M        | n10    | Refer to section 3.7.5  |
| P-11 | Systems Trace Audit Number          | ME       | n6     | Refer to section 3.7.6  |
| P-12 | Local Transaction Time              | N/A *    | n6     | Refer to section 3.7.7  |
| P-13 | Local Transaction Date              | N/A *    | n4     | Refer to section 3.7.8  |
| P-39 | Response Code                       | M        | an2    | Refer to section 3.7.14 |
| S-70 | Network Management Information Code | M        | n3     | Refer to section 3.7.23 |

\* Except for IFSF petrol interface where this is CE

### 3.6 DATA FIELD DESCRIPTION IN REQUEST MESSAGE

#### 3.6.1 SECONDARY BITMAP (P-1)

The secondary bit map identifies the presence or absence of data elements 65 through 128 in the BASE24 external message. It functions the same as the primary bit map, except that the primary bit map identifies the presence or absence of data elements 1 through 64 and the secondary bit map identifies the presence or absence of data elements 65 through 128.

The secondary bit map is required if any of data elements 65 through 128 are included in the message. Otherwise, it is not used.

The presence or absence of the secondary bit map is identified by bit position 1 in the primary bit map. Data elements 65 through 128 cannot be included in the message if the secondary bit map is not present.

#### 3.6.2 PRIMARY ACCOUNT NUMBER (P-2)

The Primary Account Number data element contains the primary account number (PAN) of the cardholder account involved in the transaction

This data element is required in case of manual or keyed transaction and mandatory for reversal request (0420) messages.

#### 3.6.3 PROCESSING CODE (P-3)

The Processing Code data element contains a series of digits used to describe the effect of a transaction on the customer account and the accounts affected.

Valid Values:

| Position 1-2 | Position 3-4 | Position 5-6 | Description                           |
|--------------|--------------|--------------|---------------------------------------|
| 00           | nn           | nn           | Normal Purchase                       |
| 09           | nn           | nn           | Cashback                              |
| 10           | nn           | nn           | Account Funding                       |
| 20           | nn           | nn           | Refund (returns, payment of winnings) |
| 80           | nn           | nn           | MOTO Purchase                         |
| 81           | nn           | nn           | Account Verification                  |

A value of '00' in position 3-4 indicates a default From Account Type and a value of '00' in position 5-6 indicates a default To Account Type.

#### 3.6.4 TRANSACTION AMOUNT (P-4)

The Transaction Amount data element contains the amount of funds requested (either for debit or credit) in the currency of the source location of the transaction.

This data field may sometimes carry zero amount (000000000000) in 0120/0121, 0220/0221 messages.

For Diners/Discover/Visa Account Verification and MasterCard/Maestro Account Status Inquiry Service transactions this field must carry a zero amount.

This data field may contain an amount which differs (typically lower) to the original authorised amount for an 0420 reversal request.

#### 3.6.5 TRANSMISSION DATE AND TIME (P-7)

The Transmission Date and Time data element contains the time the message is initiated by the terminal. Data and time format is MMDDhhmmss.

#### 3.6.6 SYSTEM TRACE AUDIT NUMBER (P-11)

The System Trace Audit Number is mandatory for both in the request and response messages. It carries a unique number in each transaction which is incremented by one for every transaction from an acquiring system.

A reversal and advice message for a corresponding authorisation has a different STAN.

Reversal Repeats and Advice Repeats (0421/0121/0221) has the same STAN as that of an original reversal and advice messages.

#### 3.6.7 LOCAL TRANSACTION TIME (P-12)

The Local Transaction Time data element contains the local time at which the transaction began at the card acceptor location. Format for Local Transaction Time is 'HHMMSS'. This data element is mandatory in all incoming and outgoing messages apart from Network Management messages.



### 3.6.8 LOCAL TRANSACTION DATE (P-13)

The Local Transaction Date data element contains the local month and day when the transaction began at the card acceptor location. Format for Local Transaction Date is 'MMDD'. This data element is mandatory in all incoming and outgoing messages apart from Network Management messages.

### 3.6.9 EXPIRATION DATE (P-14)

The Expiration Date of the Card in the format YYMM. It is a conditional data element, not mandatory in all messages. A manual or keyed transaction requires expiration date data element for others it can be obtained from Track 2 data. A value of 22/22 is acceptable for recurring transactions.

### 3.6.10 MERCHANT TYPE (P-18)

The Merchant Type data element contains the Standard Industrial Classification (SIC) code of the retailer involved in the transaction. Do not populate without prior permission from Barclaycard Payments. This data element is mandatory in 0100 and 0200 messages for Payment Facilitator transactions, and remains optional for 0120/1, 0220/1 and 0420/1 messages.

### 3.6.11 POINT OF SERVICE ENTRY MODE (P-22)

The Point of Service Entry Mode data element is a single field that contains two codes. The first code is two digits in length and indicates the method by which Track data or the primary account number (PAN) was entered into the system. The second code is one digit in length and indicates the entry capabilities available at the point of service. This data element is mandatory in 0100, 0200, 0120/1 and 0220/1 messages. This data element is conditional in 0420/1.

| Pos 1-2 Entry Mode            | Pos. 3 Entry capability            |
|-------------------------------|------------------------------------|
| Valid values:                 | Valid values:                      |
| 00 Undetermined               | 0 Undetermined                     |
| 01 Manual input               | 1 The terminal can handle PIN      |
| 02 Magstripe was read         | 2 The terminal cannot handle PIN   |
| 05 Input from chip card (ICC) | 3 The terminal can handle software |
| 07 Contactless chip           | PIN (mPOS only)                    |
| 90 All of track 2 was read    | 8 Pin Pad Broken                   |
| 91 Contactless Magstripe      |                                    |

### 3.6.12 CARD SEQUENCE NUMBER (P-23)

The Card Sequence Number contains a number which distinguishes between different cards that have the same PAN. This number must be right justified and zero filled, or must contain three zeros. If the IC card does not have a Pan Sequence Number, then this field must not be populated

### 3.6.13 POINT OF SERVICE CONDITION CODE (P-25)

The Point of Service Condition Code data element contains a code that identifies the condition under which the transaction is taking place at the point of service.

Incoming POS messages must contain this data element, BASE24 places the value into the PT-SRV-COND-CDE field in the PSTM.

Supported values:

- 00 Normal conditions
- 01 Customer not present (Ecommerce)
- 02 Unattended terminal able to retain card (unattended terminals)
- 03 Merchant suspicious
- 06 Preauthorisation transaction requests
- 08 Mail order/telephone order
- 27 Unattended term unable to retain card (unattended terminals)

### 3.6.14 ACQUIRING INSTITUTION ID CODE (P-32)

The Acquiring Institution Identification Code data element contains a code that identifies the acquiring institution for the transaction, or its agent. The acquiring institution may be different from the card acceptor.

This data element is mandatory for all authorisation, financial and reversal messages, but optional in Network Management messages. Use default value of 492900

### 3.6.15 TRACK 2 DATA (P-35)

The Track 2 Data element is the information encoded on Track 2 of the magnetic stripe on the back of the card originating the transaction, excluding start and end sentinel and longitudinal redundancy check (LRC) characters.

This data element includes the following;

Primary account number (PAN), left justified (up to 19 digits)

Field separator (=)

Expiration date (YYMM)

Service code (if present; 3 digits)

Discretionary data (balance of available space)

### 3.6.16 RETRIEVAL REFERENCE NUMBER (P-37)

A number assigned by the message initiator to uniquely identify a transaction. The number remains unchanged for all messages related to this transaction, i.e. a reversal advice should have the same number as the associated authorisation transaction.

Optional format for P-37 - DDHHMM + STAN (6 digits)

In case of reversal, advice and completion advice messages these will use the Time and STAN from the original data elements field. The values in original data elements have the same values as that of an original authorisation transaction.

### 3.6.17 AUTHORISATION IDENTIFICATION RESPONSE (P-38)

The Authorisation Identification Response data element contains a response identification number assigned by the authorising institution. This field is required for 0120, 0220 and 0420 messages and must include the authorisation identification response data from the original authorisation message if required by scheme.

### 3.6.18 RESPONSE CODE (P-39)

The Response Code data element contains a code that indicates the disposition of a message.

This data element is mandatory for 0120, 0220, 0420 messages.

For an 0120 or 0220 please refer to section 3.7.14 in this guide.

For an 0420 valid values are:

| Reversal Codes | Description                        |
|----------------|------------------------------------|
| 00             | Approved or Completed successfully |
| 10             | Partial Approval                   |
| 17             | Customer Cancellation              |
| 22             | Suspected Malfunction              |
| 34             | Fraud alerts for issuers           |
| 40             | Requested function not supported   |
| 68             | Response received too late         |
| 96             | System Malfunction                 |

### 3.6.19 CARD ACCEPTOR TERMINAL IDENTIFICATION (P-41)

The Card Acceptor Terminal Identification data element contains a unique code identifying the terminal at the card acceptor location.

This data element is mandatory in all messages excluding Network Management messages. This field must be left justified and  filled.

### 3.6.20 CARD ACCEPTOR IDENTIFICATION CODE (P-42)

The Card Acceptor Identification Code contains the Merchant Id of a card acceptor. This value is defined by the acquirer.

This data element is optional. This field must be left justified and  filled.

### 3.6.21 CARD ACCEPTOR NAME/LOCATION (P-43)

The Card Acceptor Name/Location data element contains the name and location of the card acceptor that defines the point of service in both local and interchange environments.

The Card Acceptor Name/Location data element is mandatory in all 0100, 0120, 0121, 0200, 0220, 0221, 0420, and 0421 messages.

The structure of this field is defined as below;

| Position | Length | Description                                                                                                                                                    |
|----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-22     | 22     | Terminal Owner<br>The name of the institution owning the terminal                                                                                              |
| 23-35    | 13     | Terminal City<br>The city in which the transaction-originating terminal is located                                                                             |
| 36-38    | 3      | Terminal State<br>A code indicating the state or province in which the transaction-originating terminal is located.<br><i>b</i> is a valid value in this field |
| 39-40    | 2      | Terminal Country<br>A code indicating the country in which the transaction originating terminal is located (i.e GB, DE, IT..etc...)                            |

### 3.6.22 RETAILER DATA (P-48)

The Retailer Data element carries the information required to identify the retailer involved in the transaction. It is mandatory for all authorisation, financial transaction, and reversal messages, with the exception of 0130, 0230, and 0430 messages.

The structure of this field is defined as below;

| Position | Length | Description                                                                                                                                             |
|----------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-3      | 3      | Field Length Indicator<br><br>This field must be set to a value of 027                                                                                  |
| 4-22     | 19     | Retailer ID<br><br>The retailer ID of the retailer initiating the transaction. <i>b</i> is a valid value in this field following the Retailer ID value. |
| 23-26    | 4      | Retailer Group<br><br>The retailer group to which the retailer initiating the transaction belongs. Must be 'A' preceeded by 3 <i>b</i>                  |
| 27-30    | 4      | Retailer Region<br><br>The retailer region to which the retailer initiating the transaction belongs. Must be 'A' preceeded by 3 <i>b</i>                |

### 3.6.23 CURRENCY CODE (P-49)

The code in the Transaction Currency Code data element identifies the currency that applies to the Transaction Amount (P-4) data element. It is mandatory for all authorisation, financial transaction, and reversal messages. Currency code used must be according to the ISO standard.

### 3.6.24 PIN DATA (P-52)

This data element contains the encrypted PIN block. This data element is only used for Online PIN verification transaction. Also this data element is a conditional field in 0100 and 0200 messages. Must not be present for OCTs.

PIN data field is not used in the response messages.

Standard Triple DES is supported. For more details refer to APPENDIX A

### 3.6.25 ADDITIONAL AMOUNTS (P-54)

This data element contains the Additional Amounts which carries the cash back amount for purchase with cashback or cashback without purchase transactions where cash is being returned to the customer. This data element is only used for cashback transactions.

If the transaction is a purchase with cashback or cashback without purchase, the Additional Amounts data element must be included field in 0100/0200 and 0420 messages. Otherwise the field should not be present for non cashback transaction.

The structure of this field is defined as below;

| Position | Length | Description                                                                                |
|----------|--------|--------------------------------------------------------------------------------------------|
| 1-3      | 3      | Field Length Indicator<br><br>This field must be set to a value of 012                     |
| 4-15     | 12     | Additional amounts<br><br>Fixed-length data element carrying a single 12-character amount. |

### 3.6.26 ADDITIONAL DATA (P-63)

The Additional Data element contains Base24 message tokens. This data element is mandatory for 0100 and 0200 messages, and conditional for 0110, 0120/1, 0130, 0210, 0220/1, 0230, 0420/1, 0430. This data element is not required in the Network Management messages.

The structure of this data element is defined as below;

| Position | Length | Description                                                                                                                                                 |
|----------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-3      | 3      | Field Length Indicator<br><br>The field length indicator value is the sum of the lengths of the Header token, all token headers, and token data being used. |
| 4-15     | 12     | Header Token                                                                                                                                                |
| 16-24    | 10     | Token Header                                                                                                                                                |
| a-b      | an..   | Token data                                                                                                                                                  |
| ....     | ...    | ....                                                                                                                                                        |
| w-x      | 10     | Token Header                                                                                                                                                |
| y-z      | an..   | Token Data                                                                                                                                                  |

The ICC/EMV data is carried in this data field in the form of tokens. Refer to tokens section for more details.

For more information on Header Token, token header, and specific tokens see section 4

### 3.6.27 NETWORK MANAGEMENT INFORMATION CODE (S-70)

The Network Management Information Code data element contains a code that is used to manage the online processing status between host systems. This code identifies the purpose of a network management request message.

The following codes are supported:

001 = Logon

301 = Echo-test

This data element is mandatory for 0800 and 0810 messages.

### 3.6.28 ORIGINAL DATA ELEMENTS (S-90)

The Original Data Elements data element contains a group of five sub-elements included in a reversal or advice message. The information in these subelements identifies the original transaction being reversed.

The Original Data Elements data element is mandatory for 0420, 0421, and 0430 messages.

The structure of this data element is defined as below;

| Position | Length | Description                                                                                                            |
|----------|--------|------------------------------------------------------------------------------------------------------------------------|
| 1-4      | 4      | Original Message Type<br><br>The transaction type identifying the original transaction.                                |
| 5-16     | 12     | Retrieval Reference Number / Original Sequence Number<br><br>The sequence number identifying the original transaction. |
| 17-20    | 4      | Transaction Date<br><br>The date of the original transaction. Contents of P13                                          |
| 21-28    | 8      | Transaction Time<br><br>The time of the original transaction. Contents of P12 + '00'                                   |
| 29-32    | 4      | Original BASE24 Capture Date<br><br>The date the original transaction was posted by BASE24. Contents of P13            |
| 33-42    | 10     | Filler with '0's                                                                                                       |



### 3.6.29 RECEIVING INSTITUTION IDENTIFICATION CODE (S-100)

The Receiving Institution Identification Code data element contains a code that identifies the institution receiving a request message.

The Receiving Institution Identification Code data element is optional for 0100, 0110, 0120, 0121, 0200, 0210, 0220, 0221, 0420, and 0421 messages.

### 3.6.30 SETTLEMENT DATA (S-125)

The Settlement Data format of this data element is mandatory for authorisation, financial transaction, and reversal messages, with the exception of 0130, 0230, and 0430 messages.

The structure of this data element is defined as below;

| Position | Length | Description                                                                                                                                                                                                                                  |
|----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-3      | 3      | Field Length Indicator<br><br>This field must be set to a value of 012.                                                                                                                                                                      |
| 4-5      | 2      | Services<br><br>Set value = 00                                                                                                                                                                                                               |
| 6-9      | 4      | Originator<br><br>Set value = HOST                                                                                                                                                                                                           |
| 10-13    | 4      | Destination<br><br>Set value = B24                                                                                                                                                                                                           |
| 14       | 1      | Draft Capture Flag<br><br>Valid values are:<br><br>0 = Default value, draft not captured<br>1 = Draft captured. Valid message types are 0200/0220/0420.<br><br>For internal use only. Unless advised otherwise by BPS set this field to '0'. |
| 15       | 1      | Settlement Flag<br><br>Set value = 0                                                                                                                                                                                                         |

### 3.7 DATA FIELD DESCRIPTION IN RESPONSE MESSAGE

#### 3.7.1 SECONDARY BITMAP (P-1)

If any of the secondary bitmap data element (64-128) present in the response message, then this field is required.

#### 3.7.2 PRIMARY ACCOUNT NUMBER (P-2)

The Primary Account Number data element contains the primary account number (PAN) of the cardholder account involved in the transaction. It is optional in the response messages.

This data element is mandatory in reversal response (0430) messages.

#### 3.7.3 PROCESSING CODE (P-3)

The Processing Code data element is a mandatory field in the response message; the value in this field is echoed from the request message.

#### 3.7.4 TRANSACTION AMOUNT (P-4)

The Transaction Amount data element is a mandatory field in the response message

This data field may sometimes carry zero amount in 0130, 0230 messages.

This data field may contain an amount which differs (typically lower) to the original authorised amount for an 0110/0210 for partially approved transactions.

#### 3.7.5 TRANSMISSION DATE AND TIME (P-7)

The Transmission Date and Time data element contains the system time the message is delivered by the host. Data and time format is MMDDhhmmss.

#### 3.7.6 SYSTEM TRACE AUDIT NUMBER (P-11)

The System Trace Audit Number is mandatory for both in the request and response messages. The value in this field is echoed from the request message.

### 3.7.7 LOCAL TRANSACTION TIME (P-12)

The Local Transaction Time data element contains the local time at which the transaction began at the card acceptor location. Format for Local Transaction Time is 'HHMMSS'. This data element is mandatory for all response messages apart from Network Management messages. The value in this field is echoed from the request message.

### 3.7.8 LOCAL TRANSACTION DATE (P-13)

The Local Transaction Date data element contains the local month and day when the transaction began at the card acceptor location. Format for Local Transaction Date is 'MMDD'. This data element is mandatory for all response messages apart from Network Management messages. The value in this field is echoed from the request message.

### 3.7.9 DATE, SETTLEMENT (P-15)

Date, Settlement (DE 15) is the logical settlement date based on the Acquirer date. This data element is present in all Financial Transaction responses(0210/0230) and Reversal response(0430) messages that convey a settlement value. It contains the calendar date on which funds for a transaction will be settled. Format for Date, Settlement is 'DDMM'. Will only be returned if the draft capture flag is set to '1' in field S-125 position 14.

### 3.7.10 POINT OF SERVICE CONDITION CODE (P-25)

The Point of Service Condition Code is conditional in the response messages. The value in this field is echoed from the request message.

### 3.7.11 ACQUIRING INSTITUTION ID CODE (P-32)

This data element is mandatory for all response messages (0110, 0130, 0210, 0230 and 0430). The value in this field is echoed from the request message.

### 3.7.12 RETRIEVAL REFERENCE NUMBER (P-37)

If present, echo the value from the corresponding request message.

### 3.7.13 AUTHORISATION IDENTIFICATION RESPONSE (P-38)

The Authorisation Identification Response data element contains a response identification number assigned by the authorising institution.

The Authorisation Identification Response data element is mandatory for 0420/1 messages.

## 3.7.14 RESPONSE CODE (P-39)

The Response Code data element contains a code that indicates the disposition of a message. This data element is mandatory for all response messages.

Valid response codes listed below;

| Response Code | Definition           | Category                       |
|---------------|----------------------|--------------------------------|
| 00            | APPROVAL             | AUTHORISED                     |
| 10            | PARTIAL APPROVAL     | AUTHORISED                     |
| 11            | APPROVAL             | AUTHORISED                     |
| 77            | HONOUR WITH ID       | AUTHORISED                     |
| 78            | APPROVAL             | AUTHORISED                     |
| 04            | PICK UP CARD         | PICK UP CARD <sup>Note 3</sup> |
| 41            | LOST CARD, PICK-UP   | PICK UP CARD <sup>Note 3</sup> |
| 43            | STOLEN CARD, PICK-UP | PICK UP CARD <sup>Note 3</sup> |
| 01            | REFERRAL *           | REFERRAL *                     |
| 02            | REFERRAL *           | REFERRAL *                     |
| 03            | INVALID MERCHANT     | DECLINE                        |
| 05            | NOT AUTHORISED       | DECLINE                        |
| 06            | INVALID TRANS        | DECLINE <sup>Note 3</sup>      |
| 12            | INVALID TRANS        | DECLINE <sup>Note 3</sup>      |
| 13            | INVALID AMOUNT       | DECLINE                        |
| 14            | INVALID CARD NO.     | DECLINE <sup>Note 3</sup>      |
| 15            | INVALID ISSUER       | DECLINE <sup>Note 3</sup>      |
| 25            | LIFECYCLE-DECLINED   | DECLINE <sup>Note 4</sup>      |
| 26            | POLICY-DECLINED      | DECLINE <sup>Note 4</sup>      |
| 27            | FRAUD-DECLINED       | DECLINE <sup>Note 4</sup>      |
| 30            | BAD FORMAT           | DECLINE                        |
| 33            | EXPIRED CARD         | DECLINE                        |
| 36            | RESTRICTED CARD      | DECLINE <sup>Note 3</sup>      |
| 38            | PIN TRIES EXCEEDED   | DECLINE                        |
| 46            | CLOSED ACCOUNT       | DECLINE <sup>Note 3</sup>      |
| 51            | LOW FUNDS            | DECLINE                        |
| 54            | EXPIRED CARD         | DECLINE                        |
| 55            | INCORRECT PIN        | DECLINE                        |
| 57            | INVALID TRANS        | DECLINE <sup>Note 3</sup>      |
| 61            | EXCEEDS LIMIT        | DECLINE                        |
| 62            | RESTRICTED CARD      | DECLINE                        |
| 65            | SCA REQUIRED         | DECLINE <sup>Note 1</sup>      |
| 66            | PIN REQUIRED - 1TAP  | DECLINE <sup>Note 2</sup>      |
| 68            | TIME OUT             | DECLINE                        |
| 75            | PIN TRIES EXCEEDED   | DECLINE                        |

|    |                             |                           |
|----|-----------------------------|---------------------------|
| 82 | NOT AUTHENTICATED           | DECLINE <sup>Note 5</sup> |
| 83 | CARD IS TEMPORARILY BLOCKED | DECLINE                   |
| 86 | INVALID TRANS               | DECLINE <sup>Note 3</sup> |
| 87 | BAD TRACK2                  | DECLINE                   |
| 89 | INVALID ISSUER              | DECLINE <sup>Note 3</sup> |
| 91 | ISSUER UNAVAIL              | DECLINE                   |
| 94 | DUPLICATE TRAN              | DECLINE                   |
| 96 | SYSTEM MALFUNCTN            | DECLINE                   |
| 6P | VERIFICATION FAIURE         | DECLINE                   |
| L1 | SERVICE NOT AVAILABLE       | DECLINE                   |
| L2 | DO NOT TRY AGAIN            | DECLINE <sup>Note 3</sup> |
| N0 | UNABLE TO AUTH              | DECLINE                   |
| N1 | INVALID PAN LENGTH          | DECLINE                   |
| N8 | OVER FLOOR LIMIT            | DECLINE                   |
| N9 | OVER ALLOWABLE LIMIT        | DECLINE                   |
| O5 | PIN REQUIRED                | DECLINE                   |
| O6 | MOD 10 CHECK FAILED         | DECLINE                   |
| P1 | OVER ALLOWABLE LIMIT        | DECLINE                   |
| P9 | ENTER LESSER AMOUNT         | DECLINE                   |
| Q0 | INVALID DATE                | DECLINE                   |
| Q1 | INVALID EXP DATE            | DECLINE                   |
| Q2 | INVALID TRANS               | DECLINE <sup>Note 3</sup> |
| T1 | INVALID AMOUNT              | DECLINE                   |
| T2 | BAD FORMAT                  | DECLINE                   |
| T3 | INVALID CARD NO.            | DECLINE <sup>Note 3</sup> |
| T4 | INVALID AMOUNT              | DECLINE                   |
| T7 | OVER ALLOWABLE LIMIT        | DECLINE                   |
| T8 | UNABLE TO AUTH              | DECLINE                   |
| V1 | INVALID GAMING TRANS        | DECLINE                   |
| V2 | SCA BAD FORMAT              | DECLINE                   |

All other codes are RFU and should be considered DECLINE

Note 1: For processing information for Response Code 65 "SCA REQUIRED" please refer to the PSD2 (HISO) internal guide.

Note 2: Response Code 66 "PIN REQUIRED - 1TAP " will only be returned if Single Tap and PIN is supported (QI Token position 22 = '1'). It is an indication to request an online PIN and include it in a replay of the declined contactless transactions. For more processing information for Response Code 66 "PIN REQUIRED - 1TAP " please refer to the PSD2 (HISO) internal guide.

Note 3: If a card acceptor receives one of these Decline response codes they must not resubmit the transaction for authorisation in an attempt to receive an approval response.

Note 4: If received for Mastercard CNP transactions then the card acceptor should refer to the MAC provided in token 04 position 1.

Note 5: If received can relate to Negative online CAM, dCVV, iCVV, CVV, CAVV, dCVV2, TAVV, or DTVV results Or Offline PIN authentication interrupted.

\*: The use of response codes 01/02 is no longer supported and all referrals will be mapped to a decline.

### 3.7.15 CARD ACCEPTOR TERMINAL IDENTIFICATION (P-41)

This data element is mandatory in all messages excluding Network Management messages. The value in this field is echoed from the request message.

### 3.7.16 CARD ACCEPTOR IDENTIFICATION CODE (P-42)

This data element is optional. If present, the value echoed from the request message.

### 3.7.17 CARD ACCEPTOR NAME/LOCATION (P-43)

This data element is optional for 0430 reversal responses messages. The value in this field is echoed from the request message.

### 3.7.18 ADDITIONAL RESPONSE DATA (P-44)

Additional Response Data element carries a code indicating the result of address verification. It indicates the response to the verification result of the cardholder billing address as requested in the authorisation message.

The structure of this field is defined as below;

| Position | Length | Description                                                                                                                                                                                                                 |
|----------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-2      | 2      | Field Length Indicator<br><br>This field must be set to a value of 02.                                                                                                                                                      |
| 3        | 1      | Response Data<br><br>This field is not used by Base24-pos                                                                                                                                                                   |
| 4        | 1      | Address Verification Status<br><br>A code identifying the result of comparing address verification information received in the transaction and address verification information contained in the database for the processor |

Valid response codes listed below;

## VISA

| Code | Definition                                                                                                                                                |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| A    | Address- Addresses matched but ZIP codes did not match                                                                                                    |
| N    | No- Addresses did not match and ZIP codes did not match                                                                                                   |
| R    | Retry- Primary and secondary authorisers were unavailable or declined the transaction and address verification was not performed by the BASE24-pos system |
| U    | Unavailable- Address information was not available to the processor performing address verification                                                       |
| Y    | Yes - Address and Postal/Zip code matched                                                                                                                 |
| Z    | ZIP- Five-digit ZIP codes matched, but addresses did not match                                                                                            |
| b    | Address verification information was not included in this transaction ( b denotes a blank character)                                                      |
| 0    | Default/Unknown response                                                                                                                                  |

## MASTERCARD

| Code | Definition                                                                                                                                                |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| A    | Address- Addresses matched but ZIP codes did not match                                                                                                    |
| N    | No- Addresses did not match and ZIP codes did not match                                                                                                   |
| R    | Retry- Primary and secondary authorisers were unavailable or declined the transaction and address verification was not performed by the BASE24-pos system |
| S    | Service not supported- the BASE24-pos product authorised the transaction, but did not have the add-on Address Verification module                         |
| U    | Unavailable- Address information was not available to the processor performing address verification                                                       |
| W    | ZIP- Nine-digit ZIP codes matched, but addresses did not match                                                                                            |
| X    | Yes- Addresses and Nine-digit ZIP codes matched                                                                                                           |
| Y    | Yes- Addresses and five-digit ZIP codes matched                                                                                                           |
| Z    | ZIP- Five-digit ZIP codes matched, but addresses did not match                                                                                            |
| 0    | Default/Unknown response                                                                                                                                  |
| b    | Address verification information was not included in this transaction ( b denotes a blank character)                                                      |

### 3.7.19 RETAILER DATA (P-48)

The Retailer Data element is mandatory for all response messages (0110, 0210, 0130, 0230 and 0430).

The value in this field is echoed from the request message.

### 3.7.20 CURRENCY CODE (P-49)

The currency code data element is mandatory for all response messages (0110, 0210, 0130, 0230 and 0430). The value in this field is echoed from the request message.

### 3.7.21 ADDITIONAL AMOUNTS (P-54)

This data element is conditional for 0110/0210/0430 messages. The value in this field is echoed from the request message.

### 3.7.22 ADDITIONAL DATA (P-63)

The Additional Data element contains Base24 message tokens. This data element is conditional for response messages 0110, 0130, 0210, 0230 and 0430. This data element is not required in the Network Management messages.

The structure of this data element is defined as below;

| Position | Length | Description                                                                                                                                            |
|----------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-3      | 3      | Field Length Indicator<br>The field length indicator value is the sum of the lengths of the Header token, all token headers, and token data being used |
| 4-15     | 12     | Header Token                                                                                                                                           |
| 16-25    | 10     | Token Header                                                                                                                                           |
| a-b      | n      | Token data                                                                                                                                             |
| ...      | ...    | ...                                                                                                                                                    |
| w-x      | 10     | Token Header                                                                                                                                           |
| y-z      | n      | Token Data                                                                                                                                             |

The ICC/EMV data is carried in this data field in the form of tokens. Refer to tokens section for more details.

For more information on Header Token, token header, and specific tokens see section 4 TOKENS



### 3.7.23 NETWORK MANAGEMENT INFORMATION CODE (S-70)

This data element is mandatory for 0810 messages. The value in this field is echoed from the request message.

### 3.7.24 ORIGINAL DATA ELEMENTS (S-90)

The Original Data Elements data element is mandatory for 0430 messages. The value in this field is echoed from the request message.

### 3.7.25 RECEIVING INSTITUTION IDENTIFICATION CODE (S-100)

Variable value returned.

## 4 TOKENS

Tokens are used to send additional information in the transaction. The tokens are submitted in data element P-63. Each request token is a fixed length.

### 4.1 LIST OF TOKENS SUPPORTED

| Token Id | Token Name                                              | Length | Request | Response |
|----------|---------------------------------------------------------|--------|---------|----------|
| N/A      | Header Token                                            | 12     | Y       | Y        |
| N/A      | Token Header                                            | 10     | Y       | Y        |
| B2       | EMV Request Data Token                                  | 158    | Y       | N        |
| B3       | EMV Discretionary Data Token                            | 80     | Y       | N        |
| B4       | EMV Status Token                                        | 20     | Y       | N        |
| B5       | EMV Response Data Token                                 | 38     | N       | Y        |
| B6       | EMV Script Data Token                                   | 260    | N       | Y        |
| C4       | Point of Service Data Token                             | 12     | Y       | N        |
| C0       | BASE24-pos Release 5.1 Token                            | 26     | Y       | N        |
| CH       | POS Data1 Token                                         | 40     | Y       | Y        |
| CE       | Authentication Data Token                               | 202    | Y       | N        |
| C6       | Trans Stain XID Token                                   | 80     | Y       | N        |
| CT       | Trans Specific Data Token                               | 4      | Y       | N        |
| FC       | Interchange Program Token                               | 56     | Y       | N        |
| F1       | Additional Data Token                                   | 30     | Y       | N        |
| F3       | Transit Data Token                                      | 4      | Y       | N        |
| R3       | Payment Data Token                                      | 11     | Y       | N        |
| R4       | Instalment Token                                        | 35     | Y       | N        |
| SE       | Additional Amounts Token                                | 22     | Y       | N        |
| SO       | Payment Account Reference Token                         | 40     | Y       | Y        |
| S8       | PAN Mapping Token                                       | 30     | N       | Y        |
| 01       | Address Verification Token                              | 30     | Y       | N        |
| 04       | BASE24-POS RELEASE 5.0 TOKEN                            | 20     | N       | Y        |
| 20       | Interchange Compliance Token                            | 22     | Y       | Y        |
| QE       | Payment Token                                           | 142    | Y       | N        |
| QG       | Visa and Mastercard Additional Authorisation Data Token | 30     | Y       | N        |
| QI       | CSM Data Token                                          | 32     | Y       | N        |
| QJ       | POS Balance Token                                       | 28     | N       | Y        |
| QL       | Dynamic Currency Conversion Token                       | 16     | Y       | N        |
| QV       | Payment Facilitator Token                               | 26     | Y       | N        |
| QW       | Recipient Data Token                                    | 500    | Y       | N        |
| QX       | Miscellaneous Response Data Token                       | 40     | N       | Y        |
| QY       | Sender Data Token                                       | 500    | Y       | N        |

## 4.2 TOKEN STRUCTURE & CONTENT

### 4.2.1 HEADER TOKEN

The fields in the ASCII format Header Token are described below

| Position | Data Type | Field Name and Description                                                                                                   |
|----------|-----------|------------------------------------------------------------------------------------------------------------------------------|
| 1-12     |           | HEADER-TKN                                                                                                                   |
| 1        | an 1      | Eye-Catcher<br>Indicates the start of token data. The only valid value is an ampersand (&)                                   |
| 2        | an 1      | USER-FLD1<br>Use <i>b</i> in this sub-field                                                                                  |
| 3-7      | n 5       | CNT<br>The count of the number of tokens, including the Header token, that are present in the token data buffer.             |
| 8-12     | n 5       | LGTH<br>The length of all token data, including the Header token and token header structures, present in a token data buffer |

### 4.2.2 TOKEN HEADER

The fields in the ASCII format Token Header are described below

| Position | Data Type | Field Name and Description                                                                                  |
|----------|-----------|-------------------------------------------------------------------------------------------------------------|
| 1-10     |           | TKN-HEADER                                                                                                  |
| 1        | an 1      | Eye-Catcher<br>Indicates the start of an individual token. The only valid value is an exclamation point (!) |
| 2        | an 1      | USER-FLD1<br>Use <i>b</i> in this sub-field                                                                 |
| 3-4      | an 2      | TKN-ID<br>The two-byte ASCII representation of the token ID. The token ID uniquely identifies a token       |
| 5-9      | n 5       | LGTH<br>The length of the token data for the token identified by the TKN-ID field                           |
| 10       | an 1      | USER-FLD2<br>Use <i>b</i> in this sub-field                                                                 |

## 4.2.3 TOKEN B2 EMV REQUEST DATA TOKEN – ASCII FORMAT

The EMV Request Data token contains the thirteen minimum request data elements required for inclusion in request messages, as defined by EMV. For UPI, if this token is included in the original request then it must also be included in the reversal.

For more information about the EMV data elements refer to the Diners/Discover DPAS, MasterCard M/Chip or the Visa Smart Debit Credit (VSDC) documentation sets or the EMVCo specification.

| Position         | Data Type        | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
|------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------|---------|-----------|---|-----------|-----|--|---|------------------|------|---|---|-----|----|---|---|---------|------|---|---|----------|------|---|---|-----------|------|--|---|-----|----|---|---|-----|------|---|---|----------------|------|---|----|----------------|------|---|----|----------|----|---|----|-----------|----|---|----|---------------|------|---|----|---------------|------|---|
| 1-158            |                  | EMV-RQST-TKNX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 1-4              | an 4             | <div>Bit-Map</div> <div>Indicates whether data in each of the remaining fields in the token is present or absent. The token itself is a fixed format structure, so the absence of a data item means that the appropriate field is present but that its contents are undefined.</div> <div>Note that the positions of the bits within the bit map follow the ISO 8583 convention (i.e., the highest order bit represents the first field in the token, following the BIT-MAP field)</div> <table><thead><tr><th>Bit Map Position</th><th>Field Name</th><th>EMV Tag</th><th>Mandatory</th></tr></thead><tbody><tr><td>1</td><td>User-FLD1</td><td>n/a</td><td></td></tr><tr><td>2</td><td>Crypto-Info-Data</td><td>9F27</td><td>Y</td></tr><tr><td>3</td><td>TVR</td><td>95</td><td>Y</td></tr><tr><td>4</td><td>ARQC/TC</td><td>9F26</td><td>Y</td></tr><tr><td>5</td><td>Amt-Auth</td><td>9F02</td><td>Y</td></tr><tr><td>6</td><td>Amt-Other</td><td>9F03</td><td></td></tr><tr><td>7</td><td>AIP</td><td>82</td><td>Y</td></tr><tr><td>8</td><td>ATC</td><td>9F36</td><td>Y</td></tr><tr><td>9</td><td>TERM-CNTRY-CDE</td><td>9F1A</td><td>Y</td></tr><tr><td>10</td><td>TRAN-CRNCY-CDE</td><td>5F2A</td><td>Y</td></tr><tr><td>11</td><td>TRAN-DAT</td><td>9A</td><td>Y</td></tr><tr><td>12</td><td>TRAN-TYPE</td><td>9C</td><td>Y</td></tr><tr><td>13</td><td>UNPREDICT-NUM</td><td>9F37</td><td>Y</td></tr><tr><td>16</td><td>ISS-APPL-DATA</td><td>9F10</td><td>Y</td></tr></tbody></table> | Bit Map Position | Field Name | EMV Tag | Mandatory | 1 | User-FLD1 | n/a |  | 2 | Crypto-Info-Data | 9F27 | Y | 3 | TVR | 95 | Y | 4 | ARQC/TC | 9F26 | Y | 5 | Amt-Auth | 9F02 | Y | 6 | Amt-Other | 9F03 |  | 7 | AIP | 82 | Y | 8 | ATC | 9F36 | Y | 9 | TERM-CNTRY-CDE | 9F1A | Y | 10 | TRAN-CRNCY-CDE | 5F2A | Y | 11 | TRAN-DAT | 9A | Y | 12 | TRAN-TYPE | 9C | Y | 13 | UNPREDICT-NUM | 9F37 | Y | 16 | ISS-APPL-DATA | 9F10 | Y |
| Bit Map Position | Field Name       | EMV Tag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Mandatory        |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 1                | User-FLD1        | n/a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 2                | Crypto-Info-Data | 9F27                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 3                | TVR              | 95                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 4                | ARQC/TC          | 9F26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 5                | Amt-Auth         | 9F02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 6                | Amt-Other        | 9F03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 7                | AIP              | 82                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 8                | ATC              | 9F36                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 9                | TERM-CNTRY-CDE   | 9F1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 10               | TRAN-CRNCY-CDE   | 5F2A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 11               | TRAN-DAT         | 9A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 12               | TRAN-TYPE        | 9C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 13               | UNPREDICT-NUM    | 9F37                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 16               | ISS-APPL-DATA    | 9F10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Y                |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |
| 5-8              | an 4             | <div>USER-FLD1</div> <div>Must contain zeros</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |            |         |           |   |           |     |  |   |                  |      |   |   |     |    |   |   |         |      |   |   |          |      |   |   |           |      |  |   |     |    |   |   |     |      |   |   |                |      |   |    |                |      |   |    |          |    |   |    |           |    |   |    |               |      |   |    |               |      |   |

| Position                 | Data Type                                                                                                                                                                     | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------|---|--------------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9-10                     | an 2                                                                                                                                                                          | <p><b>CRYPTO-INFO-DATA</b></p> <p>The type of cryptogram and the actions to be performed by the terminal. Valid values are shown in the table below.</p> <p>In the EMV specifications, definitions that include bit positions indicate that bit 8 is the leftmost bit</p> <table><tr><th>EMV Defined Bit Position</th><th>Description</th></tr><tr><td>8-7</td><td>Type of cryptogram.<br/><br/>Valid values.<br/>00 = AAC<br/>01 = TC<br/>10 = ARQC<br/>11 = AAR</td></tr><tr><td>6</td><td>Reserved for future use</td></tr><tr><td>5</td><td>Reserved for future use</td></tr><tr><td>4</td><td>Advice required flag. Valid values are as follows:<br/>0 = Advice is not required.<br/>1 = Advice is required.</td></tr><tr><td>3-1</td><td>The reason advice code.<br/><br/>Valid values:<br/>000 = No information given<br/>001 = Service not allowed<br/>010 = PIN try limit exceeded<br/>011 = Issuer authentication failed</td></tr></table>                                                                        | EMV Defined Bit Position | Description | 8-7 | Type of cryptogram.<br><br>Valid values.<br>00 = AAC<br>01 = TC<br>10 = ARQC<br>11 = AAR                                                                     | 6 | Reserved for future use                                                                                                                                       | 5 | Reserved for future use            | 4 | Advice required flag. Valid values are as follows:<br>0 = Advice is not required.<br>1 = Advice is required. | 3-1 | The reason advice code.<br><br>Valid values:<br>000 = No information given<br>001 = Service not allowed<br>010 = PIN try limit exceeded<br>011 = Issuer authentication failed |
| EMV Defined Bit Position | Description                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
| 8-7                      | Type of cryptogram.<br><br>Valid values.<br>00 = AAC<br>01 = TC<br>10 = ARQC<br>11 = AAR                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
| 6                        | Reserved for future use                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
| 5                        | Reserved for future use                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
| 4                        | Advice required flag. Valid values are as follows:<br>0 = Advice is not required.<br>1 = Advice is required.                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
| 3-1                      | The reason advice code.<br><br>Valid values:<br>000 = No information given<br>001 = Service not allowed<br>010 = PIN try limit exceeded<br>011 = Issuer authentication failed |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
| 11-20                    | an 10                                                                                                                                                                         | <p><b>TVR</b></p> <p>The terminal verification results. This field indicates the status of the different functions as seen from the terminal.</p> <p>Valid values are shown in the tables below. The default for all bit settings is a value of 0.</p> <p>In the EMV specification, definitions that include bit positions indicate that bit position 8 is the leftmost bit</p> <p>Bit positions not listed are reserved for future use</p> <p style="text-align: center;">Byte 1</p> <table><tr><th>EMV Defined Bit Position</th><th>Description</th></tr><tr><td>8</td><td>Offline data authentication flag.<br/><br/>Valid values:<br/>0 = Offline data authentication was performed<br/>1 = Offline data authentication was not performed</td></tr><tr><td>7</td><td>Offline static data authentication flag. Valid values<br/><br/>0 = Offline static data authentication passed.<br/>1 = Offline static data authentication failed.</td></tr><tr><td>6</td><td>ICC data flag.<br/><br/>Valid values</td></tr></table> | EMV Defined Bit Position | Description | 8   | Offline data authentication flag.<br><br>Valid values:<br>0 = Offline data authentication was performed<br>1 = Offline data authentication was not performed | 7 | Offline static data authentication flag. Valid values<br><br>0 = Offline static data authentication passed.<br>1 = Offline static data authentication failed. | 6 | ICC data flag.<br><br>Valid values |   |                                                                                                              |     |                                                                                                                                                                               |
| EMV Defined Bit Position | Description                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
| 8                        | Offline data authentication flag.<br><br>Valid values:<br>0 = Offline data authentication was performed<br>1 = Offline data authentication was not performed                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
| 7                        | Offline static data authentication flag. Valid values<br><br>0 = Offline static data authentication passed.<br>1 = Offline static data authentication failed.                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |
| 6                        | ICC data flag.<br><br>Valid values                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |             |     |                                                                                                                                                              |   |                                                                                                                                                               |   |                                    |   |                                                                                                              |     |                                                                                                                                                                               |

| Position | Data Type | Field Name and Description |                          |                                                                                                                                                                                           |
|----------|-----------|----------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |           |                            |                          | 0 = ICC data is present<br>1 = ICC data is missing                                                                                                                                        |
|          |           |                            | 5                        | Card on exception file flag.<br><br>Valid values:<br>0 = Card does not appear on terminal exception file<br>1 = Card appears on terminal exception file                                   |
|          |           |                            | 4                        | Offline dynamic data authentication flag.<br><br>Valid Values:<br>0 = Offline dynamic data authentication passed<br>1 = Offline dynamic data authentication failed                        |
|          |           |                            | 3                        | CDA failure flag<br><br>Valid values:<br>0= CDA did not fail<br>1= CDA faile                                                                                                              |
|          |           |                            | 2                        | SDA selected flag<br><br>Valid values:<br>0= SDA not selected<br>1= SDA selected                                                                                                          |
|          |           |                            | Byte 2                   |                                                                                                                                                                                           |
|          |           |                            | EMV Defined Bit Position | Description                                                                                                                                                                               |
|          |           |                            | 8                        | ICC and Terminal version flag<br><br>Valid values:<br>0 = The ICC and the terminal have the same application versions<br>1 = The ICC and the terminal have different application versions |
|          |           |                            | 7                        | Expired application flag<br><br>Valid Values:<br>0 = The application has not expired<br>1 = The application expired                                                                       |
|          |           |                            | 6                        | Application effective flag<br><br>0 = The application is effective<br>1 = The application is not effective                                                                                |
|          |           |                            | 5                        | Requested service flag<br><br>Valid Values:<br>0 = The requested service is allowed for the card product<br>1 = The requested service is not allowed for the card product                 |
|          |           |                            | 4                        | New Card flag<br><br>Valid Values:<br>0 = The transaction was not initiated with a new card<br>1 = The transaction was initiated with a new card                                          |
|          |           |                            | 2                        | Biometric performed and successful flag<br><br>Valid Values:<br>0 = Not Performed<br>1 = Performed and successful                                                                         |
|          |           |                            | 1                        | Biometric template format not support flag                                                                                                                                                |

| Position | Data Type | Field Name and Description |                                                                                                                                                                                                                                        |
|----------|-----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |           |                            | Valid values:<br>0 = Not supported<br>1 = Supported                                                                                                                                                                                    |
|          |           | Byte 3                     |                                                                                                                                                                                                                                        |
|          |           | EMV Defined Bit Position   | Description                                                                                                                                                                                                                            |
|          |           | 8                          | Cardholder verification was not successful flag<br><br>Valid Values:<br>0 = Cardholder verification was successful<br>1 = Cardholder verification was not successful                                                                   |
|          |           | 7                          | Unrecognised cardholder verification method (CVM) flag.<br><br>Valid Values:<br>0 = The CVM was recognised<br>1 = The CVM was not recognised                                                                                           |
|          |           | 6                          | PIN Try Limit exceeded flag<br><br>Valid Values:<br>0 = The PIN try limit was not exceeded<br>1 = The PIN try limit was exceeded                                                                                                       |
|          |           | 5                          | PIN required/ PIN pad not available or not working condition<br><br>Valid Values:<br>0 = PIN entry is not required or the PIN Pad is present and operable<br>1 = PIN entry is required and the PIN pad is not present or inoperable    |
|          |           | 4                          | PIN required/ PIN Pad present, PIN not entered condition.<br><br>Valid Values:<br>0 = PIN entry is not required or the PIN pad is not present or the PIN was entered<br>1 = PIN entry is required, PIN pad is present, PIN not entered |
|          |           | 3                          | Online PIN captured flag.<br><br>Valid Values:<br>0 = Online PIN not captured<br>1 = Online PIN captured                                                                                                                               |
|          |           | 2                          | Biometric required but Biometric capture device not working flag<br><br>Valid Values:<br>0 = Biometric required and the Biometric capture device is working<br>1 = Biometric required but Biometric capture device is not working      |
|          |           | 1                          | Biometric required, Biometric capture device present, but Biometric Subtype entry was bypassed flag<br><br>Valid Values:<br>0 = Biometric required, Biometric capture device present, but Biometric Subtype entry was not bypassed     |

| Position | Data Type | Field Name and Description |                                                                                                                                                                                        |
|----------|-----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |           |                            | 1 = Biometric required, Biometric capture device present, but Biometric Subtype entry was bypassed                                                                                     |
|          |           | Byte 4                     |                                                                                                                                                                                        |
|          |           | EMV Defined Bit Position   | Description                                                                                                                                                                            |
|          |           | 8                          | Floor limit flag.<br><br>Valid Values:<br>0 = The transaction amount does not exceed the floor limit<br>1 = The transaction amount exceeds the floor limit                             |
|          |           | 7                          | Lower consecutive offline limit flag<br><br>Valid Values:<br>0 = The lower consecutive offline limit was not exceeded<br>1 = The lower consecutive offline limit was exceeded          |
|          |           | 6                          | Upper consecutive offline limit flag<br><br>Valid Values:<br>0 = The upper consecutive offline limit was not exceeded<br>1 = The upper consecutive offline limit was exceeded          |
|          |           | 5                          | Random selection flag<br><br>Valid Values:<br>0 = The transaction was not selected at random for Online processing<br>1 = The transaction was selected at random for Online processing |
|          |           | 4                          | Merchant forced Online flag<br><br>Valid Values:<br>0 = The merchant did not force the transaction online<br>1 = The merchant forced the transaction online                            |
|          |           | 3                          | Biometric Try Limit exceeded flag<br><br>Valid values:<br>0 = The Biometric try limit was not exceeded<br>1 = The Biometric try limit was exceeded                                     |
|          |           | 2                          | A selected Biometric Type not supported flag<br><br>Valid values:<br><br>0 = A selected Biometric Type supported<br>1 = A selected Biometric Type not supported                        |
|          |           | Byte 5                     |                                                                                                                                                                                        |
|          |           | EMV Defined Bit Position   | Description                                                                                                                                                                            |
|          |           | 8                          | Transaction certificate data object list (TDOL) status.<br><br>Valid Values:<br>0 = The default TDOL was not used                                                                      |



| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                       |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |           |                                                                                                                                                                                                                                                                                                                                                                                | 1 = The merchant forced the transaction online Issuer Authentication flag.<br><br>Valid values:<br>0 = Issuer authentication was successful<br>1 = Issuer authentication was not successful                           |
|          |           | 7                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                       |
|          |           | 6                                                                                                                                                                                                                                                                                                                                                                              | Script processing before final GENERATE AC command flag.<br><br>Valid Values:<br>0 = Script processing did not fail before final GENERATE AC command<br>1 = Script processing failed before final GENERATE AC command |
|          |           | 5                                                                                                                                                                                                                                                                                                                                                                              | Script processing after final GENERATE AC flag<br><br>Valid Values:<br>0 = Script processing did not fail after final GENERATE AC command<br>1 = Script processing failed after final GENERATE AC command             |
|          |           | 4                                                                                                                                                                                                                                                                                                                                                                              | Relay resistance threshold exceeded (only for contactless)<br><br>Valid Values:<br>0 = Relay resistance threshold not exceeded<br>1 = Relay resistance threshold exceeded                                             |
|          |           | 3                                                                                                                                                                                                                                                                                                                                                                              | Relay resistance time limits exceeded (only for contactless)<br><br>Valid Values:<br>0 = Relay resistance time limits not exceeded<br>1 = Relay resistance time limits exceeded                                       |
|          |           | 2-1                                                                                                                                                                                                                                                                                                                                                                            | Relay resistance performed (only for contactless)<br>00: Relay resistance not performed (not used by this version of the specification)<br>01: RRP NOT PERFORMED<br>10: RRP PERFORMED<br>11: RFU                      |
| 21-36    | an 16     | ARQC/TC<br>The authorisation request cryptogram. The cryptogram returned by the ICC in response to the GENERATE AC command                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                       |
| 37-48    | an 12     | AMT-AUTH<br>The authorised amount of the transaction (excluding adjustments). Data in this field is right-justified, zero-filled                                                                                                                                                                                                                                               |                                                                                                                                                                                                                       |
| 49-60    | an 12     | AMT-OTHER<br>The authorised cashback amount of the transaction. Data in this field is right-justified, zero-filled                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                       |
| 61-64    | an 4      | AIP<br>The application interchange profile. This field indicates the capabilities of the card to support specific functions in the application.<br>Valid values are shown in the tables below. In the EMV specifications, definitions that include bit positions indicate that bit position 8 is the leftmost bit.<br><br>Bit positions not listed are reserved for future use |                                                                                                                                                                                                                       |

| Position | Data Type | Field Name and Description |                                                                                                                                                                                           |
|----------|-----------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |           | Byte 1                     |                                                                                                                                                                                           |
|          |           | EMV Defined Bit Position   | Description                                                                                                                                                                               |
|          |           | 8                          | Initiate flag<br><br>Valid Values:<br>0 = Do not initiate<br>1 = Initiate                                                                                                                 |
|          |           | 7                          | Offline static data authentication support flag.<br><br>Valid Values:<br>0 = Offline static data authentication is not supported<br>1 = Offline static data authentication is supported   |
|          |           | 6                          | Offline dynamic data authentication support flag.<br>Valid Values<br><br>0 = Offline dynamic data authentication is not supported<br>1 = Offline dynamic data authentication is supported |
|          |           | 5                          | Cardholder verification support flag.<br><br>Valid Values:<br>0 = Cardholder verification is not supported<br>1 = Cardholder verification is supported                                    |
|          |           | 4                          | Terminal risk management support flag.<br><br>Valid Values:<br>0 = Terminal risk management will not be performed<br>1 = Terminal risk management will be performed                       |
|          |           | 3                          | Issuer authentication support flag<br><br>Valid Values:<br>0 = Issuer authentication is not supported<br>1 = Issuer authentication is supported                                           |
|          |           | 1                          | CDA support flag<br><br>Valid Values:<br>0 = CDA not supported<br>1 = CDA supported                                                                                                       |
|          |           | Byte 2                     |                                                                                                                                                                                           |
|          |           | EMV Defined Bit Position   | Description                                                                                                                                                                               |
|          |           | 8                          | EMV mode support flag<br><br>Valid Values:<br>0 = Mag-stripe mode only supported<br>1 = EMV and mag-stripe modes supported                                                                |
|          |           | 7                          | Contactless mobile support flag<br><br>Valid Values:<br>0 = Contactless mobile is not supported<br>1 = Contactless mobile is supported                                                    |
|          |           | 6                          | Reserved for use by the EMV contactless specifications                                                                                                                                    |
|          |           | 1                          | Relay resistance protocol is supported (only for contactless)                                                                                                                             |

[illegible]

## 4.2.4 TOKEN B3 EMV DISCRETIONARY DATA TOKEN – ASCII FORMAT

The EMV Discretionary Request Data token consists of EMV-related data. However, each data element is supported by more than one EMV-compliant interface and, therefore, can be mapped between interfaces by BASE24. For UPI, if this token is included in the original request then it must also be included in the reversal.

| Position                 | Data Type                                                                                                                      | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                               |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------|---------|--------------------------------------------------------------------------------------------------------------------------------|---|--------------|------|--|---|--------------|------|---|---|-----------|-----|--|---|-----------|-----|--|---|---------------|------|---|---|--------------|------|--|---|-----------|------|-----------------------------------------------|---|------------------------------------------------------------------------------------------|----------|----------------------------|
| 1-80                     |                                                                                                                                | EMV-DISCR-TKNX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                               |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 1-4                      | an 4                                                                                                                           | <div><div>BIT-MAP</div><div>Indicates whether data in each of the remaining fields in the token is present or absent. The token itself is a fixed format structure, so the absence of a data item means that the appropriate field is present but that its contents are undefined. Note that the positions of the bits within the bit map follow the ISO 8583 convention (i.e., the highest order bit represents the first field in the token, following the BIT-MAP field). There are 16 bits in the BIT-MAP field, but only 8 fields (excluding the BIT-MAP field) in the token; therefore the lowest order 8 bits in the BIT-MAP field are reserved for future use</div><table><tr><th>Bit Map Position</th><th>Field Name</th><th>EMV Tag</th><th>Mandatory</th></tr><tr><td>1</td><td>TERM-SER-NUM</td><td>9F1E</td><td></td></tr><tr><td>2</td><td>EMV-TERM-CAP</td><td>9F33</td><td>Y</td></tr><tr><td>3</td><td>USER-FLD1</td><td>N/A</td><td></td></tr><tr><td>4</td><td>USER-FLD2</td><td>N/A</td><td></td></tr><tr><td>5</td><td>EMV-TERM-TYPE</td><td>9F35</td><td>Y</td></tr><tr><td>6</td><td>APPL-VER-NUM</td><td>9F09</td><td></td></tr><tr><td>7</td><td>CVM-RSLTS</td><td>9F34</td><td>Y<br/>(contact all and MasterCard contactless)</td></tr><tr><td>8</td><td>This field will contain one of the following data elements:<br/>DF-NAME<br/>APPLICATION ID</td><td>84<br/>4F</td><td>Y<br/>(MasterCard /Maestro)</td></tr></table></div> | Bit Map Position                              | Field Name  | EMV Tag | Mandatory                                                                                                                      | 1 | TERM-SER-NUM | 9F1E |  | 2 | EMV-TERM-CAP | 9F33 | Y | 3 | USER-FLD1 | N/A |  | 4 | USER-FLD2 | N/A |  | 5 | EMV-TERM-TYPE | 9F35 | Y | 6 | APPL-VER-NUM | 9F09 |  | 7 | CVM-RSLTS | 9F34 | Y<br>(contact all and MasterCard contactless) | 8 | This field will contain one of the following data elements:<br>DF-NAME<br>APPLICATION ID | 84<br>4F | Y<br>(MasterCard /Maestro) |
| Bit Map Position         | Field Name                                                                                                                     | EMV Tag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mandatory                                     |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 1                        | TERM-SER-NUM                                                                                                                   | 9F1E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                               |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 2                        | EMV-TERM-CAP                                                                                                                   | 9F33                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Y                                             |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 3                        | USER-FLD1                                                                                                                      | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                               |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 4                        | USER-FLD2                                                                                                                      | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                               |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 5                        | EMV-TERM-TYPE                                                                                                                  | 9F35                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Y                                             |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 6                        | APPL-VER-NUM                                                                                                                   | 9F09                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                               |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 7                        | CVM-RSLTS                                                                                                                      | 9F34                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Y<br>(contact all and MasterCard contactless) |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 8                        | This field will contain one of the following data elements:<br>DF-NAME<br>APPLICATION ID                                       | 84<br>4F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Y<br>(MasterCard /Maestro)                    |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 5-12                     | an 8                                                                                                                           | <div><div>TERM-SERL-NUM</div><div>The interface device (IFD) number, a unique and permanent serial number assigned to the terminal by the manufacturer</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                               |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 13-20                    | an 8                                                                                                                           | <div><div>EMV-TERM-CAP</div><div>The card data input, cardholder verification method (CVM), and security capabilities of the terminal. Valid values are shown in the tables below. The default for all bit settings is a value of 0.<br/>In the EMV specifications, definitions that include bit positions indicate that bit position 8 is the leftmost bit. Bit positions not listed are reserved for future use</div><div>Byte 1 (Card data input capability)</div><table><tr><th>EMV Defined Bit Position</th><th>Description</th></tr><tr><td>8</td><td>Manual Key entry capability.<br/><br/>Valid Values:<br/>0 = The terminal does not support manual key entru to input the card data</td></tr></table></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | EMV Defined Bit Position                      | Description | 8       | Manual Key entry capability.<br><br>Valid Values:<br>0 = The terminal does not support manual key entru to input the card data |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| EMV Defined Bit Position | Description                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                               |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |
| 8                        | Manual Key entry capability.<br><br>Valid Values:<br>0 = The terminal does not support manual key entru to input the card data |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                               |             |         |                                                                                                                                |   |              |      |  |   |              |      |   |   |           |     |  |   |           |     |  |   |               |      |   |   |              |      |  |   |           |      |                                               |   |                                                                                          |          |                            |

| Position | Data Type | Field Name and Description |                                                                                                                                                                                                                                                    |
|----------|-----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |           |                            | 1 = The terminal supports manual key entry to input the card data                                                                                                                                                                                  |
|          |           | 7                          | Magnetic stripe capability.<br><br>Valid Values:<br>0 = The terminal does not support data capture from the magnetic stripe on the card<br>1 = The terminal supports data capture from the magnetic stripe on the card                             |
|          |           | 6                          | IC with contacts capability.<br><br>Valid Values:<br>0 = Terminal does not support data capture from the integrated chip card.<br>1 = The terminal supports data capture from the integrated chip card.                                            |
|          |           | Byte 2 (CVM Capability)    |                                                                                                                                                                                                                                                    |
|          |           | EMV Defined Bit Position   | Description                                                                                                                                                                                                                                        |
|          |           | 8                          | Plaintext PIN for integrated chip card (ICC ) verification capability.<br><br>Valid values:<br>0 = Terminal does not use the plaintext PIN for ICC verification for CVM<br>1 = The terminal uses plaintext PIN for ICC verification for CVM        |
|          |           | 7                          | Enciphered PIN for Online verification capability.<br><br>Valid Values:<br>0 = Terminal does not use the enciphered PIN for Online verification for CVM<br>1 = The terminal uses enciphered PIN for Online verification for CVM                    |
|          |           | 6                          | Signature (paper) capability.<br><br>Valid Values:<br>0 = The terminal does not use signature verification for CVM<br>1 = The terminal uses signature (paper) verification for CVM                                                                 |
|          |           | 5                          | Enciphered PIN for offline verification capability.<br><br>Valid Values:<br>0 = Terminal does not use the enciphered PIN for Offline verification for CVM<br>1 = The terminal uses enciphered PIN for Offline verification for CVM                 |
|          |           | 4                          | No Cardholder Verification Method Required capability.<br><br>Valid Values:<br>0 = Terminal does not use the Cardholder Verification Method Required verification for CVM<br>1 = The terminal uses Cardholder Verification Method Required for CVM |
|          |           | 3                          | Online Biometric<br><br>Valid Values:<br>0 = Terminal does not use Online Biometric<br>1 = The Terminal uses Online Biometrics                                                                                                                     |
|          |           | 2                          | Offline Biometric                                                                                                                                                                                                                                  |

| Position                                    | Data Type                                                                                                                                                                                                    | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------|------------------------------|----------|--------------------------|-------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--|------------------------|--|---------------------------------------------|--|
|                                             |                                                                                                                                                                                                              | <table><tr><td></td><td>Valid Values:<br/>0 = Terminal does not use Offline Biometric<br/>1 = The Terminal uses Offline Biometrics</td></tr></table><br><table><tr><th colspan="2">Byte 3 (Security capability)</th></tr><tr><th>EMV Defined Bit Position</th><th>Description</th></tr><tr><td>8</td><td>Static data authentication capability.<br/><br/>Valid Values:<br/>0 = Static data authentication security is not used by this terminal<br/>1 = Static data authentication security is used by this terminal</td></tr><tr><td>7</td><td>Dynamic data authentication capability<br/><br/>Valid Values:<br/>0 = Dynamic data authentication security is not used by this terminal<br/>1 = Dynamic data authentication security is used by this terminal</td></tr><tr><td>6</td><td>Card capture capability<br/><br/>Valid values:<br/>0 = The terminal does not have the card capture capability<br/>1 = The terminal does have the card capture capability</td></tr><tr><td>5</td><td>RFU</td></tr><tr><td>4</td><td>Combined data authentication capability<br/><br/>Valid Values:<br/>0 = Combined data authentication security is not used by this terminal<br/>1 = Combined data authentication security is used by this terminal</td></tr></table><br><table><tr><th colspan="2">Byte 4</th></tr><tr><th colspan="2">USER-FLD1-EMV-TERM-CAP</th></tr><tr><td colspan="2">This field is used to ensure word alignment</td></tr></table> |  | Valid Values:<br>0 = Terminal does not use Offline Biometric<br>1 = The Terminal uses Offline Biometrics | Byte 3 (Security capability) |          | EMV Defined Bit Position | Description | 8           | Static data authentication capability.<br><br>Valid Values:<br>0 = Static data authentication security is not used by this terminal<br>1 = Static data authentication security is used by this terminal | 7                              | Dynamic data authentication capability<br><br>Valid Values:<br>0 = Dynamic data authentication security is not used by this terminal<br>1 = Dynamic data authentication security is used by this terminal | 6            | Card capture capability<br><br>Valid values:<br>0 = The terminal does not have the card capture capability<br>1 = The terminal does have the card capture capability | 5 | RFU | 4 | Combined data authentication capability<br><br>Valid Values:<br>0 = Combined data authentication security is not used by this terminal<br>1 = Combined data authentication security is used by this terminal | Byte 4 |  | USER-FLD1-EMV-TERM-CAP |  | This field is used to ensure word alignment |  |
|                                             | Valid Values:<br>0 = Terminal does not use Offline Biometric<br>1 = The Terminal uses Offline Biometrics                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| Byte 3 (Security capability)                |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| EMV Defined Bit Position                    | Description                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| 8                                           | Static data authentication capability.<br><br>Valid Values:<br>0 = Static data authentication security is not used by this terminal<br>1 = Static data authentication security is used by this terminal      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| 7                                           | Dynamic data authentication capability<br><br>Valid Values:<br>0 = Dynamic data authentication security is not used by this terminal<br>1 = Dynamic data authentication security is used by this terminal    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| 6                                           | Card capture capability<br><br>Valid values:<br>0 = The terminal does not have the card capture capability<br>1 = The terminal does have the card capture capability                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| 5                                           | RFU                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| 4                                           | Combined data authentication capability<br><br>Valid Values:<br>0 = Combined data authentication security is not used by this terminal<br>1 = Combined data authentication security is used by this terminal |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| Byte 4                                      |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| USER-FLD1-EMV-TERM-CAP                      |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| This field is used to ensure word alignment |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| 21-24                                       | an 4                                                                                                                                                                                                         | USER-FLD1<br>Must contain zeros                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| 25-32                                       | an 8                                                                                                                                                                                                         | USER-FLD2<br>Must contain zeros                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| 33-34                                       | an 2                                                                                                                                                                                                         | EMV-TERM-TYPE<br>The EMV terminal type, indicating the environment of the terminal, its communications capability, and its operational control, as shown in the table below:<br><table><tr><th></th><th>Operational Control</th></tr><tr><th>Environment</th><th>Merchant</th></tr><tr><td colspan="2">Attended terminal</td></tr><tr><td>Online Only</td><td>21</td></tr><tr><td>Offline with online capability</td><td>22</td></tr><tr><td>Offline Only</td><td>23</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  | Operational Control                                                                                      | Environment                  | Merchant | Attended terminal        |             | Online Only | 21                                                                                                                                                                                                      | Offline with online capability | 22                                                                                                                                                                                                        | Offline Only | 23                                                                                                                                                                   |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
|                                             | Operational Control                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| Environment                                 | Merchant                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| Attended terminal                           |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| Online Only                                 | 21                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| Offline with online capability              | 22                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |
| Offline Only                                | 23                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                          |                              |          |                          |             |             |                                                                                                                                                                                                         |                                |                                                                                                                                                                                                           |              |                                                                                                                                                                      |   |     |   |                                                                                                                                                                                                              |        |  |                        |  |                                             |  |

| Position                       | Data Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <table><tr><th colspan="2">Unattended terminal</th></tr><tr><td>Online only</td><td>24</td></tr><tr><td>Offline with Online capability</td><td>25</td></tr><tr><td>Offline Only</td><td>26</td></tr></table> | Unattended terminal    |  | Online only              | 24          | Offline with Online capability | 25                                                                                                                                                                             | Offline Only | 26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Unattended terminal            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Online only                    | 24                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Offline with Online capability | 25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Offline Only                   | 26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 35-38                          | an 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | APPL-VER-NUM<br>The version number assigned by the payment system for the terminal application. If it is not available then '0000' should be populated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 39-44                          | an 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>CVM-RSLTS</p> <p>The results of the last cardholder verification method (CVM) performed. Valid values are shown in the tables below. The default for all bit settings is a value of 0. For Visa and UPI Contactless transactions this field will be zero filled.</p> <p>In the EMV specifications, definitions that include bit positions indicate that bit position 8 is the leftmost bit.</p> <p>Bit positions not listed are reserved for future use</p> <p>Contact transactions:</p> <table><tr><th colspan="2">Byte 1 (CVM performed)</th></tr><tr><th>EMV Defined Bit Position</th><th>Description</th></tr><tr><td>7</td><td>0 = Fail cardholder verification if this cardholder verification method (CVM) is unsuccessful<br/>1 = Apply succeeding card verification rule (CVR) if this CVM is unsuccessful</td></tr><tr><td>6-1</td><td>000000= Fail CVM processing<br/>000001= Plaintext PIN verification performed by ICC<br/>000010= Enciphered PIN verified online<br/>000011= Plaintext PIN verification performed by ICC and signature (paper)<br/>000100= Enciphered PIN verification performed by ICC<br/>000101= Enciphered PIN verification performed by ICC and signature (paper)<br/>000110 = Facial biometric verified offline (by ICC)<br/>000111 = Facial biometric verified online<br/>001000 = Finger biometric verified offline (by ICC)<br/>001001 = Finger biometric verified online<br/>001010 = Palm biometric verified offline (by ICC)<br/>001011 = Palm biometric verified online<br/>001100 = Iris biometric verified offline (by ICC)<br/>001101 = Iris biometric verified online<br/>001110 = Voice biometric verified offline (by ICC)<br/>001111 = Voice biometric verified online<br/>0xxxxx = Values in the range</td></tr></table> |                                                                                                                                                                                                              | Byte 1 (CVM performed) |  | EMV Defined Bit Position | Description | 7                              | 0 = Fail cardholder verification if this cardholder verification method (CVM) is unsuccessful<br>1 = Apply succeeding card verification rule (CVR) if this CVM is unsuccessful | 6-1          | 000000= Fail CVM processing<br>000001= Plaintext PIN verification performed by ICC<br>000010= Enciphered PIN verified online<br>000011= Plaintext PIN verification performed by ICC and signature (paper)<br>000100= Enciphered PIN verification performed by ICC<br>000101= Enciphered PIN verification performed by ICC and signature (paper)<br>000110 = Facial biometric verified offline (by ICC)<br>000111 = Facial biometric verified online<br>001000 = Finger biometric verified offline (by ICC)<br>001001 = Finger biometric verified online<br>001010 = Palm biometric verified offline (by ICC)<br>001011 = Palm biometric verified online<br>001100 = Iris biometric verified offline (by ICC)<br>001101 = Iris biometric verified online<br>001110 = Voice biometric verified offline (by ICC)<br>001111 = Voice biometric verified online<br>0xxxxx = Values in the range |
| Byte 1 (CVM performed)         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| EMV Defined Bit Position       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7                              | 0 = Fail cardholder verification if this cardholder verification method (CVM) is unsuccessful<br>1 = Apply succeeding card verification rule (CVR) if this CVM is unsuccessful                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6-1                            | 000000= Fail CVM processing<br>000001= Plaintext PIN verification performed by ICC<br>000010= Enciphered PIN verified online<br>000011= Plaintext PIN verification performed by ICC and signature (paper)<br>000100= Enciphered PIN verification performed by ICC<br>000101= Enciphered PIN verification performed by ICC and signature (paper)<br>000110 = Facial biometric verified offline (by ICC)<br>000111 = Facial biometric verified online<br>001000 = Finger biometric verified offline (by ICC)<br>001001 = Finger biometric verified online<br>001010 = Palm biometric verified offline (by ICC)<br>001011 = Palm biometric verified online<br>001100 = Iris biometric verified offline (by ICC)<br>001101 = Iris biometric verified online<br>001110 = Voice biometric verified offline (by ICC)<br>001111 = Voice biometric verified online<br>0xxxxx = Values in the range |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |                        |  |                          |             |                                |                                                                                                                                                                                |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Position                                                    | Data Type                                                                                                                                                                                                                                                                                                                                                                                                                           | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
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|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                     | <table><tr><td></td><td>010000-011101 reserved for future use by this specification<br/>011110= Signature (paper)<br/>011111= No CVM required<br/>10xxxx = Values in the range<br/>100000-101111 reserved for use by the individual payment systems<br/>11xxxx = Values in the range<br/>110000-111110 reserved for future use by the issuer<br/>111111= Not available for use( EMV specifications advise to use this value for 'No CVM performed' as valid value)</td></tr></table><br><table><tr><th colspan="2">Byte 2 (CVM Condition)</th></tr><tr><th>Value</th><th>Description</th></tr><tr><td>00</td><td>Always</td></tr><tr><td>01</td><td>If cash or cashback</td></tr><tr><td>02</td><td>If not cash or cashback</td></tr><tr><td>03</td><td>If terminal supports the CVM</td></tr><tr><td>04</td><td>Reserved for future use</td></tr><tr><td>05</td><td>Reserved for future use</td></tr><tr><td>06</td><td>If transaction is in the application currency and is under x value</td></tr><tr><td>07</td><td>If transaction is in the application currency and is over x value</td></tr><tr><td>08</td><td>If transaction is in the application currency and is under y value</td></tr><tr><td>09</td><td>If transaction is in the application currency and is over y value</td></tr><tr><td>0A-7F</td><td>Reserved for future use</td></tr><tr><td>80-FF</td><td>Reserved for future use by individual payment systems</td></tr></table><br><table><tr><th colspan="2">Byte 3 (CVM Result)</th></tr><tr><td colspan="2">Result of the last CVM performed, as known by the terminal.</td></tr><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Unknown (for example, for signature)</td></tr><tr><td>1</td><td>Failed (for example, for offline PIN)</td></tr><tr><td>2</td><td>Successful (for example, for offline PIN)</td></tr></table><br><p>MasterCard Contactless transactions:</p><br><table><tr><th colspan="9">Byte 1</th></tr><tr><th colspan="9">CVM Performed</th></tr><tr><th>b8</th><th>b7</th><th>b6</th><th>b5</th><th>b4</th><th>b3</th><th>b2</th><th>b1</th><th>Meaning</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Mobile CVM Performed</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>No CVM Performed</td></tr></table> |    | 010000-011101 reserved for future use by this specification<br>011110= Signature (paper)<br>011111= No CVM required<br>10xxxx = Values in the range<br>100000-101111 reserved for use by the individual payment systems<br>11xxxx = Values in the range<br>110000-111110 reserved for future use by the issuer<br>111111= Not available for use( EMV specifications advise to use this value for 'No CVM performed' as valid value) | Byte 2 (CVM Condition) |    | Value | Description          | 00 | Always | 01 | If cash or cashback | 02 | If not cash or cashback | 03 | If terminal supports the CVM | 04 | Reserved for future use | 05 | Reserved for future use | 06 | If transaction is in the application currency and is under x value | 07 | If transaction is in the application currency and is over x value | 08 | If transaction is in the application currency and is under y value | 09 | If transaction is in the application currency and is over y value | 0A-7F | Reserved for future use | 80-FF | Reserved for future use by individual payment systems | Byte 3 (CVM Result) |  | Result of the last CVM performed, as known by the terminal. |  | Value | Description | 0 | Unknown (for example, for signature) | 1 | Failed (for example, for offline PIN) | 2 | Successful (for example, for offline PIN) | Byte 1 |  |  |  |  |  |  |  |  | CVM Performed |  |  |  |  |  |  |  |  | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1 | Meaning | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | Mobile CVM Performed | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | No CVM Performed |
|                                                             | 010000-011101 reserved for future use by this specification<br>011110= Signature (paper)<br>011111= No CVM required<br>10xxxx = Values in the range<br>100000-101111 reserved for use by the individual payment systems<br>11xxxx = Values in the range<br>110000-111110 reserved for future use by the issuer<br>111111= Not available for use( EMV specifications advise to use this value for 'No CVM performed' as valid value) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| Byte 2 (CVM Condition)                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| Value                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 00                                                          | Always                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 01                                                          | If cash or cashback                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 02                                                          | If not cash or cashback                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 03                                                          | If terminal supports the CVM                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 04                                                          | Reserved for future use                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 05                                                          | Reserved for future use                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 06                                                          | If transaction is in the application currency and is under x value                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 07                                                          | If transaction is in the application currency and is over x value                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 08                                                          | If transaction is in the application currency and is under y value                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 09                                                          | If transaction is in the application currency and is over y value                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 0A-7F                                                       | Reserved for future use                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 80-FF                                                       | Reserved for future use by individual payment systems                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| Byte 3 (CVM Result)                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| Result of the last CVM performed, as known by the terminal. |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| Value                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 0                                                           | Unknown (for example, for signature)                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 1                                                           | Failed (for example, for offline PIN)                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 2                                                           | Successful (for example, for offline PIN)                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| Byte 1                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| CVM Performed                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |    |       |                      |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| b8                                                          | b7                                                                                                                                                                                                                                                                                                                                                                                                                                  | b6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | b5 | b4                                                                                                                                                                                                                                                                                                                                                                                                                                  | b3                     | b2 | b1    | Meaning              |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 0                                                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0                      | 0  | 1     | Mobile CVM Performed |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |
| 0                                                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1                      | 1  | 1     | No CVM Performed     |    |        |    |                     |    |                         |    |                              |    |                         |    |                         |    |                                                                    |    |                                                                   |    |                                                                    |    |                                                                   |       |                         |       |                                                       |                     |  |                                                             |  |       |             |   |                                      |   |                                       |   |                                           |        |  |  |  |  |  |  |  |  |               |  |  |  |  |  |  |  |  |    |    |    |    |    |    |    |    |         |   |   |   |   |   |   |   |   |                      |   |   |   |   |   |   |   |   |                  |



| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |                       |    |                                       |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|-----------------------|----|---------------------------------------|
|          |           | Byte 2                                                                                                                                                                                                                                                                                                                           |    |    |    |    |    |                       |    |                                       |
|          |           | CVM Condition                                                                                                                                                                                                                                                                                                                    |    |    |    |    |    |                       |    |                                       |
|          |           | b8                                                                                                                                                                                                                                                                                                                               | b7 | b6 | b5 | b4 | b3 | b2                    | b1 | Meaning                               |
|          |           | 0                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 0  | 0  | 0  | 0                     | 0  | Mobile CVM not required               |
|          |           | 0                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 0  | 0  | 0  | 0                     | 1  | Terminal Required CVM                 |
|          |           | Byte 3                                                                                                                                                                                                                                                                                                                           |    |    |    |    |    |                       |    |                                       |
|          |           | CVM Result                                                                                                                                                                                                                                                                                                                       |    |    |    |    |    |                       |    |                                       |
|          |           | b8                                                                                                                                                                                                                                                                                                                               | b7 | b6 | b5 | b4 | b3 | b2                    | b1 | Meaning                               |
|          |           | 0                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 0  | 0  | 0  | 0                     | 0  | Unknown (if Mobile CVM not performed) |
|          |           | 0                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 0  | 0  | 0  | 0                     | 1  | Mobile CVM failed                     |
| 0        | 0         | 0                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 0  | 1  | 0  | Mobile CVM successful |    |                                       |
| 0        | 0         | 0                                                                                                                                                                                                                                                                                                                                | 0  | 0  | 0  | 1  | 1  | Mobile CVM blocked    |    |                                       |
| 45-48    | an 4      | DF-NAME-LGTH<br>The length of the dedicated file name or application identifier in the following field.<br><br>The first 2 digits should always be '00', the last 2 digits will be a decimal representation of the count of hexadecimal bytes in the value provided i.e.<br>Example 1: Length = 0007<br>Example 2: Length = 0016 |    |    |    |    |    |                       |    |                                       |
| 49-80    | an 32     | DF-NAME / Application Id<br>The name of the dedicated file (as described in ISO/IEC 7816-4) or application identifier (as described in ISO/IEC 7816-5). The data is left-justified and padded to the right with zeroes<br>Example 1: Value = 12121212121212<br>Example 2: Value = 12121212121212121212121212121212               |    |    |    |    |    |                       |    |                                       |

## 4.2.5 TOKEN B4 EMV STATUS TOKEN – ASCII FORMAT

The EMV Status token holds data identifying the status of a transaction. The B4 token is mandatory for EMV and technical fallback transactions. For UPI, if this token is included in the original request then it must also be included in the reversal.

For more information about the EMV data elements refer to the Diners/Discover DPAS, MasterCard M/Chip or the Visa Smart Debit Credit (VSDC) documentation sets or the EMVCo specification.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                            |
|----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-20     |           | EMV-STAT-TKNX                                                                                                                                                                                                                                                                                         |
| 1-3      | an 3      | PT-SRV-ENTRY-MDE<br>The point-of-service entry mode. This field indicates the manner in which the card details were entered at the device and the PIN entry capability of the terminal.<br><br>Use '022' for a Chip fallback transaction (mandatory)<br>Map P-22 value for any other chip transaction |
| 4        | an 1      | TERM-ENTRY-CAP<br>The capability of the terminal<br><br>Valid Values:<br>0 = Unknown<br>2 = Magnetic stripe read capability<br>5 = ICC read capability<br>8 = Contactless read capability, but no ICC contact read Capability. Should only be used for Contactless only terminals.                    |
| 5        | an 1      | LAST-EMV-STAT<br>Indicates whether the card used to initiate a magnetic stripe transaction is a chip card. Valid values are as follows:<br><br>0 = Not a chip card<br>1 = A chip card                                                                                                                 |
| 6        | an 1      | DATA-SUSPECT<br>Indicates whether the card authentication method (CAM) data is reliable. This flag is set by the acquiring process. Valid values are,<br><br>0 = CAM data assumed correct<br>1 = CAM data is unreliable                                                                               |
| 7-8      | an 2      | APPL-PAN-SQN-NUM<br>The application PAN sequence number (EMV Tag 5F34). This field identifies and differentiates cards with the same PAN. This field contains <b>b</b> if the card does not include an application PAN sequence number. Map data element P-23 value, if present                       |
| 9-10     | an 2      | UNUSED<br>This field is reserved for future use.<br>Use value '00' or <b>b</b> filled                                                                                                                                                                                                                 |
| 11-14    | an 4      | UNUSED<br>This field is reserved for future use.<br>Use value '0000' or <b>b</b> filled                                                                                                                                                                                                               |

| Position | Data Type                                                | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
|----------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|------|----------------------------------|------|----------------------|------|--------------------------------|------|-------------------------|------|---------------------|------|---------------------------|------|------------------|------|-------------------------------|------|-----------------------------------------------------|------|----------------------------------------------------------|------|----------------------|------|-----------------------------------|-------|-------------|------|--------------------|------|---------------|------|-------------------|------|----------------------------------------------|
| 15-18    | an 4                                                     | <div>RSN-ONL-CDE</div> <div>The message reason code specifies why a transaction is to be authorised online (rather than being completed locally), or why a transaction has been completed locally (rather than being authorised online). Where more than one reason code condition applies, the following order of priority shall exist:</div> <table><tr><th>Value</th><th>Description</th></tr><tr><td>1504</td><td>Terminal not able to process ICC</td></tr><tr><td>1505</td><td>Online forced by ICC</td></tr><tr><td>1506</td><td>Online forced by card acceptor</td></tr><tr><td>1509</td><td>Online forced by issuer</td></tr><tr><td>1511</td><td>Merchant suspicious</td></tr><tr><td>1503</td><td>Terminal random selection</td></tr><tr><td>1510</td><td>Over floor limit</td></tr><tr><td>1508</td><td>Online forced by the terminal</td></tr><tr><td>1500</td><td>ICC application, common data file unable to process</td></tr><tr><td>1501</td><td>ICC application, application data file unable to process</td></tr><tr><td>1502</td><td>ICC random selection</td></tr><tr><td>1507</td><td>Online forced by CAD to be update</td></tr></table> <div>In an advice message that the terminal previously has attempted to send to the acquirer as a request message, this field contains the same value as in the original request message.</div> <div>In an advice message that the terminal previously has not attempted to send to the acquirer as a request message, the valid values are as follows:</div> <table><tr><th>Value</th><th>Description</th></tr><tr><td>1004</td><td>Terminal processed</td></tr><tr><td>1005</td><td>ICC processed</td></tr><tr><td>1006</td><td>Under floor limit</td></tr><tr><td>1007</td><td>Stand-in processing at the acquirer's option</td></tr></table> | Value | Description | 1504 | Terminal not able to process ICC | 1505 | Online forced by ICC | 1506 | Online forced by card acceptor | 1509 | Online forced by issuer | 1511 | Merchant suspicious | 1503 | Terminal random selection | 1510 | Over floor limit | 1508 | Online forced by the terminal | 1500 | ICC application, common data file unable to process | 1501 | ICC application, application data file unable to process | 1502 | ICC random selection | 1507 | Online forced by CAD to be update | Value | Description | 1004 | Terminal processed | 1005 | ICC processed | 1006 | Under floor limit | 1007 | Stand-in processing at the acquirer's option |
| Value    | Description                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1504     | Terminal not able to process ICC                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1505     | Online forced by ICC                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1506     | Online forced by card acceptor                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1509     | Online forced by issuer                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1511     | Merchant suspicious                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1503     | Terminal random selection                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1510     | Over floor limit                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1508     | Online forced by the terminal                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1500     | ICC application, common data file unable to process      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1501     | ICC application, application data file unable to process |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1502     | ICC random selection                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1507     | Online forced by CAD to be update                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| Value    | Description                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1004     | Terminal processed                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1005     | ICC processed                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1006     | Under floor limit                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 1007     | Stand-in processing at the acquirer's option             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 19       | an 1                                                     | <div>ARQC-VRFY</div> <div>The result of the authorisation request cryptogram verification.</div> <div>0 = Authorisation request cryptogram not verified</div> <div>9 = Authorisation request cryptogram not verified; transaction processed as magnetic stripe instead of chip</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |
| 20       | an 1                                                     | <div>ISO-RC-IND (Use value '0')</div> <div>The ISO 8583 (1987) Response Code Indicator. This field indicates whether the ISO response code sent to the interchange should be used in generating the Authorisation Response Cryptogram (ARPC), or if the ISO response code received from the interchange should be returned to the terminal as the Authorisation Response Code. Valid value:</div> <div>0 = Do not use interchange response code</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |      |                                  |      |                      |      |                                |      |                         |      |                     |      |                           |      |                  |      |                               |      |                                                     |      |                                                          |      |                      |      |                                   |       |             |      |                    |      |               |      |                   |      |                                              |

## 4.2.6 TOKEN B5 EMV RESPONSE DATA TOKEN – ASCII FORMAT

The EMV Response Data token contains the response cryptogram, data required to generate the response cryptogram, and flags used to identify the scripts to be returned to the acquirer.

Any unused fields or parts of fields must contain ASCII zeroes rather than spaces.

For more information about the EMV data elements refer to the Diner/Discover DPAS, MasterCard M/Chip or the Visa Smart Debit Credit (VSDC) documentation sets or the EMVCo specification.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-38     |           | EMV-RESP-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1-4      | an 4      | ISS-AUTH-DATA-LGTH<br>This field contains the length of the issuer authentication data. The ASCII version of the token must contain the decimal (not hexadecimal) representation of the length value                                                                                                                                                                                                                                                             |
| 5-20     | an 16     | ARPC<br>The Authorisation Response Cryptogram computed by the card application for online issuer authentication                                                                                                                                                                                                                                                                                                                                                  |
| 21-36    | an 16     | ADDL-DATA<br>Additional issuer authentication data used in the algorithm to compute the authorisation response cryptogram.<br><br>The first two bytes contain the issuer response code used when generating the ARPC and the remaining are ASCII zero filled                                                                                                                                                                                                     |
| 37       | an 1      | SEND-CRD-BLK (normally value 'N' used in the response)<br>A code indicating whether a card block script is to be generated by the Authorisation process and sent to the ICC<br><br>C = Send a PIN change script<br>N = No, do not send a card block script<br>U = Send a PIN unblock script<br>Y = Yes, send a card block script<br>This field is ignored. If a script is in token B6 it should be sent back by the merchant regardless what is set in this bit. |
| 38       | an 1      | SEND-PUT-DATA (normally value 'N' used in the response)<br>A code indicating whether a put data script is to be generated by the Authorisation process and sent to the ICC<br><br>Y = Yes, send a put data script<br>N = No, do not send a put data script<br>This field is ignored. If a script is in token B6 it should be sent back by the merchant regardless what is set in this bit.                                                                       |

#### 4.2.7 TOKEN B6 EMV SCRIPT DATA TOKEN – ASCII FORMAT

The EMV Script Data token holds EMV script data. This token is present only if the transaction response contains script data.

Note: The EMV Script Data token is variable length. The values given are the maximum lengths. Any unused fields or parts of field will not be zero filled as this token is defined to be of variable length.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                             |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-260    |           | EMV-SCRIPT-TKNX                                                                                                                                                                                                                        |
| 1-4      | an 4      | ISS-SCRIPT-DATA-LGTH<br>Decimal representation of the count of hexadecimal values in the Iss-Script-Data                                                                                                                               |
| 5-260    | an 256    | ISS-SCRIPT-DATA<br>The Issuer Script Templates (EMV Tag 71 or 72) sent to the terminal for processing by the card application. Each template may contain a script ID and one or more script commands. This field is of variable length |

## 4.2.8 TOKEN C0 BASE24-POS RELEASE 5.1 TOKEN

The fields in the BASE24-pos Release 5.1 token are described below:

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-26     |           | PS51-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1-4      | an 4      | CVD-FLD<br>Manually entered values such as the Diners/Discover Card Verification Value (CVV2), MasterCard Card Validation Codes (CVC2), Visa Card Verification Value 2 (CVV2) and UPI Card Validation Number (CVN2). Only the leftmost three bytes are used in CVC2, CVV2 and CVN2 verification processing. A  indicates that no information is available |
| 5        | an 1      | RESUB-STAT<br>A code indicating the status for resubmission. Valid values are as follows:<br> = Normal transaction (no information is available)<br>A = Resubmission approved<br>D = Resubmission hard declined<br>R = Resubmission<br>S = Store and Forward submission<br>T = Resubmission for tries or time out condition                               |
| 6-8      | an 3      | RESUB-CNTR<br>The number of times the transaction has been submitted for processing.  filled = No information available                                                                                                                                                                                                                                  |
| 9-18     | an 10     | TERM-POSTAL-CDE<br>The 10-byte postal ZIP code of the location of the terminal. A  indicates that no information is available                                                                                                                                                                                                                         |

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19       | an 1      | <p>E-COM-FLG<br/>The electronic commerce flag. Valid values are as follows:</p> <p>4 = In-App Authentication (Diners/Discover only)<br/> 5 = Secure electronic commerce transaction with cardholder authentication or Mastercard DSRP<br/> 6 = Encrypted electronic commerce transaction where the merchant is capable of authenticating the cardholder, but was unable to complete the authentication, (e.g., because the issuer or cardholder does not participate in the appropriate authentication program)<br/> 7 = Encrypted electronic commerce transaction<br/> A = Non-Authentication (UPI only)<br/> <del>h</del> = Not applicable (Customer present transactions only)</p> <p>MOTO-FLG <span style="float: right;">REDEFINES E-COM-FLG</span></p> <p>The mail/telephone order flag. Valid values are as follows:<br/> 0 = Not a mail or telephone order transaction<br/> 1 = Single transaction<br/> 4 = Unknown classification<br/> <del>h</del> = Not applicable (Customer present transactions only)</p> |
| 20       | an 1      | Reserved for future use- <del>h</del> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 21       | an 1      | Reserved for future use- <del>h</del> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 22       | an 1      | <p>CVD-FLD-PRESENT<br/>A code indicating whether the Card Verification Value is on the card. Valid values are as follows:</p> <p>0 = CVV2 value is deliberately bypassed or not provided by the merchant<br/> 1 = CVV2 value is present<br/> 2 = CVV2 value is on the card, but is illegible<br/> 3 = The expiration date is not required to be sent from the interchange, so the CVD2 field cannot be validated (for example, a BASE24-billpay transaction)<br/> 9 = Cardholder states that the card has no CVV2 imprint<br/> <del>h</del> = No information available</p>                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 23       | an 1      | <p>SAF-OR-FORCE-POST<br/>A code indicating whether the transaction is a store-and forward or force-post transaction. Valid values are as follows:</p> <p>F = Force-post transaction<br/> S = Store-and-forward transaction<br/> <del>h</del> = Neither a force-post nor a store-and-forward transaction</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24       | an 1      | <p>AUTHN-COLL-IND</p> <p>The authentication collection indicator. For 3DS transactions including In-App with 3DS use the below appropriate values as follows:</p> <p>0 = UCAF data collection is not supported by the merchant or an Identity Check merchant has chosen not to undertake Identity Check on this transaction or Identity Check failed authentication and merchant desires to proceed with transaction</p> <p>1 = Universal Cardholder Authentication field (UCAF) data collection is supported by the merchant and UCAF data must be present</p> <p>2 = Universal Cardholder Authentication field (UCAF) data collection is supported by the merchant, and UCAF data must be present containing a fully authenticated AAV</p> <p>4 = Mastercard Identity Check Insights; UCAF data if available then should be present.</p> <p>5 = RFU</p> <p>*6 = Merchant Risk Based Decisioning. This includes European PSD2 Acquirer Exemption transactions.</p> <p>*7 = Recurring payments or MITs (i.e. partial shipment). Liability will depend on the original UCAF values provided and matching with the initial transaction.</p> <p> = Not applicable (Customer present transactions only)</p> <p><b>*If Transaction is Mastercard In-App/Tokenised with/without DSRP and 3DS is not undertaken then the following values should be used</b></p> <p>6 = Tokenised transaction with Digital Secure Remote Payment (DSRP) cryptogram without UCAF data.</p> <p>7 = Tokenised subsequent MIT, no 3DS undertaken and without DSRP present.</p> |
| 25       | an 1      | <p>FRD-PRN-FLG</p> <p>The fraud prone outlet flag. This flag indicates the fraud rating of the retailer (field 48 in the inter-member message). Valid value:</p> <p>0 = Not fraud prone</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 26       | an 1      | <p>CAVV-AAV-RSLT-CDE</p> <p>This field should be  filled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



## 4.2.9 TOKEN C4 POINT OF SERVICE DATA TOKEN

This token is mandatory except for 0420/0800 messages. The fields in the Point of service data token are described below:

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-12     |           | PT-SRV-DATA-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1        | n 1       | <b>TERM-ATTEND-IND</b><br>A code indicating whether the terminal is attended by the card acceptor. Valid values are as follows:<br>0 = The terminal is attended.<br>1 = The terminal is unattended (for example, a cardholder activated terminal or a personal computer).<br>2 = No terminal used (for example, voice or ARU authorisation, subsequent recurring payments).                                                                                                                                                                                                          |
| 2        | n 1       | <b>TERM-OPER-IND</b><br>Reserved for future use. This field is zero filled.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3        | n 1       | <b>TERM-LOC-IND</b><br>A code indicating the location of the terminal. Valid values are as follows:<br><br>0= The terminal is on the premises of the card acceptor facility.<br>1 = The terminal is off the premises of the card acceptor facility (merchant terminal in a remote location).<br>2 = The terminal is off the premises of the card acceptor facility (home personal computer).<br>3 = No terminal was used (voice or ARU authorisation, subsequent recurring payments).<br>4 = The terminal is on the premises of the card acceptor facility (home personal computer). |
| 4        | n 1       | <b>CRDHLD-PRESENT-IND</b><br>A code indicating whether the cardholder is present at the POS terminal. Valid values are as follows:<br><br>0 = The cardholder is present<br>1 = The cardholder is not present—unspecified reason<br>2 = The cardholder is not present—mail or fax order<br>3 = The cardholder is not present—telephone or ARU order<br>4 = The cardholder is not present—standing order or subsequent recurring transaction<br>5 = The cardholder is not present—electronic order (home personal computer or Internet)                                                |
| 5        | n 1       | <b>CRD-PRESENT-IND</b><br>A code indicating whether the card is present at the POS terminal. Valid values are as follows:<br><br>0 = The card is present<br>1 = The card is not present                                                                                                                                                                                                                                                                                                                                                                                              |

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6        | n 1       | <p>CRD-CAPTR-IND</p> <p>A code indicating whether the terminal has card capture capabilities. Valid values are as follows:</p> <p>0 = The terminal does not have card capture capabilities.<br/>1 = The terminal has card capture capabilities.</p>                                                                                                                                                                                                                |
| 7        | n 1       | <p>TXN-STAT-IND</p> <p>A code indicating a purpose or status of the request. Valid values are as follows:</p> <p>0 = Normal Request<br/>4 = Preauthorised Request<br/>9 = Account Status Enquiry Service</p>                                                                                                                                                                                                                                                       |
| 8        | n 1       | <p>TXN-SEC-IND</p> <p>A code indicating the card acceptor's security level. Valid values are,</p> <p>0 = No security concern<br/>1 = Suspected fraud (merchant suspicious – code 10)<br/>2 = Identification verified</p>                                                                                                                                                                                                                                           |
| 9        | n 1       | <p>TXN-RTN-IND</p> <p>Field used to flag final authorisation flagging. Valid values are as follows:</p> <p>9= Final Authorisation<br/>0= Other</p>                                                                                                                                                                                                                                                                                                                 |
| 10       | n 1       | <p>CRDHLD-ACVT-TERM-IND</p> <p>A code indicating whether the cardholder activated the terminal with a card, and if so, the level of security. Valid values are as follows:</p> <p>0 = The transaction is not a cardholder activated terminal transaction<br/>1 = Cardholder Activated Terminal (CAT) with PIN or CDCVM required—level 1 security<br/>2 = Cardholder Activated Terminal (CAT) with No CVM required—level 2 security<br/>6 = Electronic commerce</p> |

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11       | n 1       | <p>TERM-INPUT-CAP-IND</p> <p>A code indicating the terminal capabilities for transferring the data on the card to the terminal. Valid values are as follows:</p> <p>0 = Unknown or unspecified<br/> 1 = No terminal used (voice or audio response unit authorisation, subsequent recurring payments)<br/> 2 = Magnetic stripe reader<br/> 3 = Contactless chip (EMV)<br/> 4 = Contactless magnetic stripe<br/> 5 = Magnetic stripe reader and EMV-compatible ICC reader<br/> 6 = Key entry only<br/> 7 = Magnetic stripe reader and key entry<br/> 8 = Magnetic stripe reader, key entry, and EMV-compatible ICC reader<br/> 9 = EMV-compatible ICC reader</p> <p><b>* N.B. EMV Contactless capability</b><br/> <b>VISA</b> – 3 should be used when the POS device is ONLY contactless capable. If the device is also contact capable then the relevant value, e.g. 8 or 5, should be used</p> <p><b>Diners, Discover, MasterCard and UPI</b> – 3 should be used for every device that is contactless capable (including those that have contact capability)</p> |
| 12       | n 1       | <p>CRDHLDR-ID-METHOD</p> <p>A code indicating how the cardholder was verified at the point-of-service. Valid values are as follows:</p> <p>0 = Unknown (default)<br/> 1 = Signature<br/> 2 = PIN or CDCVM<br/> 3 = None (Cardholder present)<br/> 4 = None (Cardholder not present)<br/> 5 = Authentication value<br/> 7 = Biometrics<br/> 9 = Other</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## 4.2.10 TOKEN C6 TRANS STAIN XID TOKEN

The Trans Stain XID token carries the authentication information for Diners/Discover, Visa and UPI secure internet transactions. The fields in the Trans Stain XID token are described below. This is a fixed length token, all fields should be initialised to spaces. All positions, including spaces must be included in the Token length.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-80     |           | TRANS-STAIN-XID-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1-40     | ans 40    | <p>A unique transaction identifier assigned:</p> <ul style="list-style-type: none"> <li>• For a Visa transaction with a tokenised PAN this will contain the token cryptogram. Value must be sent in as Hex.</li> <li>• For a UPI transaction using 3D Secure 2.0 this will contain the DS Transaction ID ans36 left justified and space fill the remainder.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 41-80    | ans 40    | <p>TRANS-STAIN</p> <p>A value used to authenticate the cardholder:</p> <ul style="list-style-type: none"> <li>• A Visa transaction using the 3D Secure (Three-Domain Secure Electronic Commerce Verification Service) method. Value must be sent in as Hex.</li> <li>• For a Diners/Discover or Visa transaction using the 3D Secure method, this field contains the Cardholder Authentication Verification Value (CAVV). Value must be sent in as Hex (an 40).</li> <li>• For Diners In App the token CAVV should be populated in this field. Value must be sent in as Hex (an 40).</li> <li>• For a UPI transaction using 3D Secure 2.0 this will contain the Authentication Value (AV) of UP3DS program ans28 (Base64 encoded) left justified and space fill the remainder.</li> </ul> |

## 4.2.11 TOKEN CE AUTHENTICATION DATA TOKEN

The Authentication Data token standardises the transport cardholder authentication data for MasterCard and Maestro e-commerce transactions. The fields in the Authentication Data token are described below. This is a fixed length token, all fields should be initialised to spaces. All positions, including spaces must be included in the Token length.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-202    |           | AUTHN-DATA-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1-2      | an 2      | AUTHN-IND-FLG<br>The authentication indicator flag. Valid value:<br><br>01 = UCAF                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3-34     | ans 32    | AUTHN-IND-DATA - AAV<br><br>For a MasterCard transaction using the Secure Payment Application Universal Cardholder Authentication Field (UCAF) method, this field contains the Accountholder Authentication Value (AAV) which is Base64 encoded<br>If data is not available, then must be  filled                                                                                                                                                                              |
| 35       | an 1      | AUTHN-IND-DATA – PRG – PROTOCOL<br><br>For a Mastercard transaction using 3D secure this field will contain the Program Protocol.<br><br>Valid values are:<br><br>2 = EMV 3D Secure (3DS) 2.2<br>3 = EMV 3D Secure (3DS) 2.3<br>4 = EMV 3D Secure (3DS) 2.4<br>5 = EMV 3D Secure (3DS) 2.5<br>6 = EMV 3D Secure (3DS) 2.6<br>7 = EMV 3D Secure (3DS) 2.7<br>8 = EMV 3D Secure (3DS) 2.8<br>9 = EMV 3D Secure (3DS) 2.9<br><br>If data is not available, then must be  filled |
| 36-71    | ans 36    | AUTHN-IND-DATA – SER-TRAN-ID<br><br>For a Mastercard transaction using 3D secure version 2.x this field contains the Directory Server Transaction ID. Must be sent in as the same format as received. Example below:<br>f38e6948-5388-41a6-bca4-b49723c19437<br><br>If data is not available, then must be  filled                                                                                                                                                           |
| 72-99    | ans 28    | AUTHN-IND-DATA - DSRP<br><br>For a MasterCard In App transaction this will contain the token cryptogram which is Base64 encoded<br>If data is not available, then must be  filled                                                                                                                                                                                                                                                                                            |
| 100-202  | an 103    | FILLER – must be  filled                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## 4.2.12 TOKEN CH POS DATA1 TOKEN

The fields in the binary format POS Data1 token are described below. This is a fixed length token, all fields should be initialised to spaces. All positions, including spaces must be included in the Token length. The fields in the POS Data1 token are described below. Position 2 carries the result of the Diners/Discover CVV2, Visa CVV2, MasterCard CVC2 or UPI CVN2 validation.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-40     |           | POS-DATA1-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1        | an 1      | RESP-SRC-RSN-CDE<br>Returned in the response  fill in request<br>A code indicating the response source or reason code. This field is set by an interchange. Valid values are as follows:<br>3 = Issuer is in suppress inquiries mode<br>4 = Response provided by scheme (Diners/Discover only)<br>5 = Response provided by issuer<br>7 = Reversal advice provided by interchange to identify a potential duplicate transaction<br>8 = Reversal advice provided by interchange to identify a probable duplicate authorisation<br>A = Third party agent<br>B = Transaction met Visa Transaction Advisor Service criteria<br> = fill in Response |

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2        | an 1      | <p>CRD-VRFY-FLG2</p> <p>Returned in the response  fill in request</p> <p>Indicates whether the card involved in the card-read transaction has already been verified using the CVV2/CVD2. Processes that generate this token but do not use this field should initialize the value to . Valid values are as follows:</p> <p>0 = Card verification was not performed because the transaction was denied before card verification processing started</p> <p>C = Card verification was performed and the card verification digits (CVD) were invalid. The situation was noted and the transaction processing continued</p> <p>D = Card verification was performed and the CVD was invalid. The transaction was denied and the ERR-FLG field was set to C</p> <p>N or  = Card verification was not attempted or a security device error occurred</p> <p>Y = Card verification was performed and the CVD was valid</p> <p>O = Card verification was not performed, CVD was not on the card. Not all cards have a CVD value encoded. The card expiration date must be equal to or greater than an expiration date defined on the CPF to insure that the CVD field has been encoded. If the card expiration date is equal to or greater than the CPF date, the CVD checks are performed</p> <p>P = Card verification was not performed. Either the merchant ignored the CVD on purpose or the user falsely indicated no CVD was on the card</p> <p>R = Card verification was performed and the CVD was invalid. The situation was noted and the transaction should be referred</p> <p>S = CVV2 should be on the card but the merchant indicates that it is not.</p> <p>U = Issuer has not certified or has not provided the encryption keys to the interchange</p> |
| 3-14     | an 12     | <p>ONLINE-LMT</p> <p>Returned in the response  fill in request</p> <p>The value against which the transaction amount is compared to determine whether under limit or over limit authorisation is performed, using information configured in the Routing Table File (RTBL). The value is retrieved from the POS Terminal Data File (PTD) or the Authorisation Selection Table File (AST)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 15-18    | an 4      | <p>RETL-CLASS-CDE</p> <p> fill in request</p> <p>Classification code of the retailer from the POS Retailer Definition File (PRDF)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19       | an 1      | <b>EMV-CAPABLE-OUTLET</b><br>ⓑ fill in request<br><br>Indicates the EMV capability of the outlet from the POS Retailer Definition File (PRDF) or the Interchange. Valid values are as follows:<br><br>N = No, the outlet is not EMV capable<br>Y = Yes, the outlet is EMV capable                                                                                                                                                                   |
| 20       | an 1      | <b>PMNT-IND</b><br>Payment indicator, Valid values are:<br><br>I = Instalment payment (Visa only)<br>R = Initial Recurring payment<br>C = Credential on File payment<br>L = Incremental payment (optional)<br>S = Re-submission payment<br>D = Delayed Charges payment<br>A = Re-authorisation payment (Visa only)<br>X = No Show<br>P = Partial/Split Shipment (Currently only used for Mastercard/Maestro partial shipment)<br>ⓑ = Not applicable |
| 21-22    | an 2      | Reserved for future use- ⓑ filled                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 23-24    | an 2      | Reserved for future use- ⓑ filled                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 25-27    | an 3      | Reserved for future use- ⓑ filled                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 28       | an 1      | <b>TERM-OUTPUT-CAP-IND</b><br>Returned in the response ⓑ fill in request<br><br>This field indicates the ability of the terminal to print or display messages. Valid values are:<br><br>ⓑ = Not applicable<br>0 = Unknown<br>1 = None<br>2 = Print<br>3 = Display<br>4 = Print and display                                                                                                                                                          |



| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29       | an 1      | <p>CRDHLDR-AUTHN-CAP-IND</p> <p>Returned in the response <del>b</del> fill in request</p> <p>This field indicates the primary means of verifying the cardholder at the terminal. Valid values are:</p> <p>0 = No electronic authentication<br/> 1 = PIN<br/> 2 = Electronic signature analysis<br/> 3 = Biometrics<br/> 4 = Biographics<br/> 5 = Electronic authentication inoperative<br/> 6 = Other<br/> 7 = Reserved for future use<br/> 8 = Reserved for future use<br/> 9 = Authentication value</p> |
| 30-40    | an 11     | Reserved for future use- <del>b</del> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## 4.2.13 TOKEN CT TRANSACTION SPECIFIC DATA TOKEN

The Transaction Specific Data token carries the Mastercard Gaming Payment of Winnings indicator as well as the Visa and Mastercard Account Funding business application indicator. The fields in the Transaction Specific Data token are described below.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-4      |           | TXN-SPCF-DATA-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1-2      | an 2      | DATASET-ID<br><br>Contains the value used to identify the data that follows. Valid values are:<br>57 = Related transaction data                                                                                                                                                                                                                                                                                                                                                                                         |
| 3-4      | an 2      | BUS-APPL-ID<br><br>This subelement contains the Business Application Identifier used for Visa Account Funding and Original Credit transactions. This subelement is also used to map to/from the Mastercard Payment Transaction Type Identifier for Banknet and MasterCard payment transactions.<br>Valid values are:<br><br>OG = Online gambling payout<br>AA = Account to Account<br>FT = Fund Transfer<br>PD = Payroll and Pension Disbursement<br>FD = Fund Disbursement (Visa only)<br>TU = Top-Up for Prepaid-load |

## 4.2.14 TOKEN FC INTERCHANGE PROGRAM TOKEN

The Interchange Program Token carries the additional data required for Indian Recurring/MIT transactions. The fields in the Interchange program Token are described below. This is a variable length token, with varying lengths per scheme (Mastercard/Visa). All positions, including spaces must be included in the Token length.

| Position                                           | Data Type | Field Name and Description                                                                                                                                                                                                                                          |
|----------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-56                                               |           | ICHG-PGM-TKN                                                                                                                                                                                                                                                        |
| 1-2                                                | an 2      | FRMT-CDE<br>Valid values are:<br>07 - MasterCard recurring payment arrangement data, currently used in e-commerce transactions in India.<br>09 - Visa recurring payment arrangement data                                                                            |
| <b>MC-E-COMM-RECUR-PMNT-DATA (MasterCard Data)</b> |           |                                                                                                                                                                                                                                                                     |
| 3                                                  | an 1      | RECUR-PMNT-MNDT-TYP<br>Valid Values are:<br>1 - Recurring Payment with fixed amount<br>2 - Recurring Payment with maximum amount                                                                                                                                    |
| 4                                                  | an 1      | RECUR-PMNT-TXN-TYP<br>Valid Values are:<br>1 - Registration/first transaction<br>2 - Subsequent Transaction                                                                                                                                                         |
| 5                                                  | an 1      | VALID-FLG<br>A value that indicates whether or not the recurring payment transaction has been pre-validated by the merchant before the subsequent recurring authorisation request is sent to the issuer.<br>Valid Values are:<br>0 - Not Validated<br>1 - Validated |
| 6-15                                               | an 10     | MAX-RECUR-PMNT-AMT<br>The maximum recurring amount agreed with the cardholder in the currency of the source location of the transaction. If no maximum amount has been agreed (fixed amount) then this amount should match the transaction amount in P-4.           |
| 16-25                                              | an 10     | RGSTRD-TXN-ID<br>A unique reference number that is generated by the issuer before the recurring payment registration and needs to be passed at the time of the registration transaction and in every subsequent transaction.                                        |
| 26-27                                              | an 2      | NUM-RECUR-PMNTS<br>Valid values are:<br>02 to 99: Count<br>ND - Not defined<br>UC - Until cancelled                                                                                                                                                                 |
| 28                                                 | an 1      | RECUR-PMNT-FREQ<br>Valid Values are:<br>0 - Annually<br>1 - Daily<br>2 - Weekly<br>3 - Fortnightly<br>4 - Monthly<br>5 - Quarterly                                                                                                                                  |

| Position                                | Data Type | Field Name and Description                                                                                                                                                                                                                                          |
|-----------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         |           | 6 - Twice-Yearly<br>8 - Ad Hoc                                                                                                                                                                                                                                      |
| <b>VISA-RECUR-PMNT-ARNG (VISA Data)</b> |           |                                                                                                                                                                                                                                                                     |
| 3                                       | AN1       | PMNT-TYP<br>Valid values are:<br>1 - Registration/first transaction<br>2 - Subsequent transaction<br>3 - Modification<br>4 - Cancellation<br><br>A cancellation (value = '4') can only be sent in an 0100 account verification or authorisation request.            |
| 4                                       | AN1       | PMNT-AMT-IND-PER-TXN<br>Valid values are:<br>1 - Recurring Payment with fixed amount<br>2 - Recurring Payment with maximum amount                                                                                                                                   |
| 5-6                                     | AN2       | NUM-PMNT<br>Valid Values are:<br>02 – 99: Count<br>ND - Not defined<br>UC - Until cancelled                                                                                                                                                                         |
| 7-8                                     | AN2       | FREQ-PMNT<br>Valid values are:<br>01 - Daily<br>02 - Twice Weekly<br>03 - Weekly<br>04 - Ten Days<br>05 - Fortnightly<br>06 - Monthly<br>07 - Every Two Months<br>08 - Trimester<br>09 - Quarterly<br>10 - Twice Yearly<br>11 - Annually<br>12 - Unscheduled        |
| 9-43                                    | AN35      | RGSTR-REF-NUM<br>A unique reference number that is generated by the issuer before the recurring payment registration and needs to be passed at the time of the registration transaction and in every subsequent transaction.                                        |
| 44-55                                   | AN12      | MAX-PMNT-AMT<br>The maximum recurring amount agreed with the cardholder in the currency of the source location of the transaction. If no maximum amount has been agreed (fixed amount) then this amount should match the transaction amount in P-4.                 |
| 56                                      | AN1       | VALID-IND<br>A value that indicates whether or not the recurring payment transaction has been pre-validated by the merchant before the subsequent recurring authorisation request is sent to the issuer.<br>Valid Values are:<br>0 – Not Validated<br>1 – Validated |

## 4.2.15 TOKEN F1 ADDITIONAL DATA TOKEN

The Additional Data Token carries the device type indicator. The fields in the Additional Data token are described below. This is a fixed length token, all fields should be initialised to spaces. All positions, including spaces must be included in the Token length.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-30     |           | ADDL-DATA-TKN                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1-15     | an 15     | Reserved for future use-  filled                                                                                                                                                                                                                                                                                                          |
| 16       | an 1      | Reserved for future use-  filled                                                                                                                                                                                                                                                                                                          |
| 17       | an 1      | DEVICE-TYPE<br>Indicates the type of device used. Valid values are:<br><br>0 = Dedicated mPOS Terminal with PCI compliant dongle (with or without key pad), used when the reader is external to the COTS device<br>1 = Off the Shelf Mobile Device, used when the reader is embedded in the COTS device<br> = Not a mobile payment device |
| 18-30    | an 13     | Reserved for future use-  filled                                                                                                                                                                                                                                                                                                          |

## 4.2.16 TOKEN F3 TRANSIT DATA TOKEN

The Transit Data Token carries Transit Transaction Type Indicator and Transportation Mode Indicator which must be submitted by merchants in MasterCard transit request and advice messages.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-4      |           | TRANSIT-DATA-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1-2      | an 2      | TRANSIT-TRAN-TYP<br>Indicates the transit transaction type. Valid values are:<br><br>01 = Prefunded<br>02 = Real-time Authorised<br>03 = Post-Authorised Aggregated<br>04 = Authorised Aggregated Split Clearing<br>05 = Other<br>06 = Post-authorized aggregated Maestro<br>07 = Debt Recovery<br>08-99 Reserved for Future Use                                                                                                                                                                                                                                                                                                         |
| 3-4      | an 2      | TRANSPORT-MDE<br>Indicates the mode of transportation. Valid values are:<br><br>00 = Unknown<br>01 = Urban Bus<br>02 = Interurban Bus<br>03 = Light Train Mass Transit (Underground Metro, LTR)<br>04 = Train<br>05 = Commuter Train<br>06 = Water Borne Vehicle<br>07 = Toll<br>08 = Parking<br>09 = Taxi<br>10 = High Speed Train<br>11 = Rural Bus<br>12 = Express Commuter Train<br>13 = Para Transit<br>14 = Self Drive Vehicle<br>15 = Coach<br>16 = Locomotive<br>17 = Powered Motor Vehicle<br>18 = Trailer<br>19 = Regional Train<br>20 = Inter City<br>21 = Funicular Train<br>22 = Cable Car<br>23-99 Reserved for Future Use |

## 4.2.17 TOKEN SE ADDITIONAL AMOUNTS TOKEN

The Additional Amounts Token carries Related Amount Currency Code and Related amount for Visa OCT transactions. The fields in Additional Amounts Token are described below. This is a fixed length token.

| Position | Data Type | Field Name and Description                                                                                                         |
|----------|-----------|------------------------------------------------------------------------------------------------------------------------------------|
| 1-22     |           | CSM-PAYMENT-DATA-TKN                                                                                                               |
| 1-4      | AN 4      | Reserved for future use<br><br>Must be  filled.   |
| 5-7      | N 3       | CRNCY-CDE<br>The numeric ISO currency which defines the currency used in positions 9-20 of this token                              |
| 8        | AN 1      | Reserved for future use<br><br>Must be  filled.   |
| 9-20     | N 12      | AMT<br>The amount value in minor currency units, right justified with leading zeros                                                |
| 21-22    | AN 2      | Reserved for future use<br><br>Must be  filled. |

## 4.2.18 TOKEN SO PAYMENT ACCOUNT REFERENCE TOKEN

The PAR is a non-financial reference assigned to each unique PAN and used to link a Payment Account represented by that PAN to affiliated Payment Tokens.

| Position | Data Type | Field Name and Description                                                                                          |
|----------|-----------|---------------------------------------------------------------------------------------------------------------------|
| 1-40     |           | PAR-TKN                                                                                                             |
| 1-29     | an 29     | PAR - The Payment Account Reference number.                                                                         |
| 30-40    | an 11     | Reserved for Future Use-  filled |

## 4.2.19 TOKEN S8 PAN MAPPING TOKEN

The PAN Mapping Token carries the actual cardholder Primary Account Number (PAN) where it is different from the one submitted in the auth request/reversal message under the PAN mapping service. It is used when a transaction uses a different PAN to the embossed number visible to the cardholder. It is returned to merchants in response messages only when the card has an alternate PAN.

| Position | Data Type | Field Name and Description                                                                                                                    |
|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1-30     |           | PAN-MAPPING-TKN                                                                                                                               |
| 1        | an 1      | ACCT-NUM-IND<br>Indicates the type of PAN mapping account. Valid values are:<br><br>E = Embossed account number<br>M = Primary account number |
| 2-20     | an 19     | ACCT-NUM<br>Indicates the card's embossed PAN (The submitted PAN held on the chip will be returned in P-2)                                    |
| 21-24    | an 4      | EXP-DAT<br>Expiry Date of the PAN Mapping Account for this card (YYMM)                                                                        |
| 25-30    | an 6      | Reserved for Future Use- <del>b</del> filled                                                                                                  |



## 4.2.20 TOKEN 01 ADDRESS VERIFICATION TOKEN

The fields in the Address Verification token are described below.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-30     |           | ADDR-VER-TKN                                                                                                                                                                                                                                                                                                                                                               |
| 1-20     | an 20     | ADDR<br>The billing address for the cardholder. Both the full street address or only numeric values are acceptable. If it is only numeric values it is left-justified and <b>b</b> filled.                                                                                                                                                                                 |
| 21-29    | an 9      | ZIP-CDE<br>The ZIP code or postal code for the billing address for the cardholder. The ZIP code must be five or nine digits long. If the ZIP code is five digits long, it is left-justified and <b>b</b> filled<br>The postcode can be the full postcode or just the numeric values. If the postcode is just the numeric values, it is left-justified and <b>b</b> filled. |
| 30       | an 1      | Reserved for Future Use- <b>b</b> filled                                                                                                                                                                                                                                                                                                                                   |

## 4.2.21 TOKEN 04 BASE24-POS RELEASE 5.0 TOKEN

The BASE24-pos Release 5.0 Token carries the MasterCard Merchant Advice Codes for recurring payment and CNP transactions.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                          |
|----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-20     |           | PS50-TKN                                                                                                                                                                                                                                                                                                                                                                            |
| 1        | an 1      | A code used to provide additional information regarding the disposition of the transaction. Valid values are as follows:<br>1 = Updated/additional information needed<br>2 = Cannot approve at this time; try again later<br>3 = Do not try again<br>T= Token error (returned for all schemes if P-63 structure is invalid)<br>U = Payment blocked by card scheme; do not try again |
| 2-12     | an 11     | Reserved for Future Use- <b>b</b> filled                                                                                                                                                                                                                                                                                                                                            |
| 13       | an 1      | Duplication of CV2 result code taken from CH Token                                                                                                                                                                                                                                                                                                                                  |
| 14-20    | an 7      | Reserved for Future Use- <b>b</b> filled                                                                                                                                                                                                                                                                                                                                            |

## 4.2.22 TOKEN 20 INTERCHANGE COMPLIANCE TOKEN

The fields in the Interchange Compliance token are described below.

This token is used to carry interchange compliance data for Diners, Discover, MasterCard, Maestro and Visa.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-22     |           | ICHG-COMPLIANCE TKN                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1        | an 1      | LIFE-CYCLE-IND – No longer in use<br>The indicator acquirers will be required to forward in their clearing transactions<br>This field should be  filled                                                                                                                                                                                 |
| 2-16     | an 15     | TRACE-ID / TRAN-ID / NRID<br>The code assigned by the interchange to a transaction that has met the required compliance edits. For MasterCard/Maestro a combination of the Network ID, Reference number, and settlement date is filled into this field, depending on the interchange<br><br> indicates that no information is available |
| 17-20    | an 4      | VALID-CDE (VISA only) – No longer in use<br>This field should be  filled                                                                                                                                                                                                                                                                |
| 21       | an 1      | MONITORING-STAT – No longer in use<br>A code returned from the interchange indicating whether MasterCard changed the Point of Service Entry Mode from 90 to 02<br><br>This field should be  filled                                                                                                                                    |
| 22       | an 1      | ERR-IND – No longer in use<br>A code returned from the interchange indicating an error condition that may have occurred. Valid values are as follows:<br><br>This field should be  filled                                                                                                                                             |

## 4.2.23 TOKEN QE PAYMENT TOKEN

The fields in the Payment token are described below. This token is used to carry device data and contactless CVM for Diners, Discover, MasterCard, Maestro and Visa.

| Position | Data Type | Field Name and Description                                                                                                                                                                 |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-142    |           | PYT-TKN                                                                                                                                                                                    |
| 1-4      | an 4      | FORM-FACTOR-TAG<br>This field contains the form factor tag 9F6E                                                                                                                            |
| 5-6      | an 2      | FORM-FACTOR-LGTH<br>This field contains the length of the form factor field, this is a decimal representation of the bytes within the form factor<br><br>Example 1: 04                     |
| 7-70     | an 64     | FORM-FACTOR<br>A form factor value is assigned to the device by the issuer, this value must be passed in the authorisation request when present on the device.<br>Example 1: 01010101      |
| 71-74    | an 4      | DECRETIONARY-DATA-TAG<br>This field contains the discretionary data tag 9F7C                                                                                                               |
| 75-76    | an 2      | DECRETIONARY-DATA-LGTH<br>This field contains the length of the discretionary data field, this is a decimal representation of the bytes within the discretionary data<br><br>Example 1: 05 |
| 77-140   | an 64     | DECRETIONARY-DATA<br>The discretionary data is assigned to the device, this value must be passed in the authorisation request when present on the device.<br><br>Example 1: 0101010101     |
| 141      | an 1      | CONTACTLESS SIGNATURE<br>Indicates whether contactless signature CVM was performed at the POS. Valid values are as follows:<br><br>N = No<br>Y = Yes                                       |
| 142      | an 1      | CONTACTLESS CDCVM<br>Indicates whether consumer device cardholder verification method was performed at POS.<br><br>N = No<br>Y = Yes                                                       |

## 4.2.24 TOKEN QG VISA &amp; MASTERCARD ADDITIONAL AUTHORISATION DATA

The fields in the Visa and Mastercard additional authorisation data token are described below. This token is used to carry additional data for Visa and Mastercard.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-30     |           | VMAAD-TKN                                                                                                                                                                                                                                                                                                                                                                  |
| 1-8      | n 8       | <p>DATE OF BIRTH</p> <p>This is a numeric field containing the primary recipients date of birth in YYYYMMDD format</p> <p>Where the data is not available please <i>b</i> fill</p>                                                                                                                                                                                         |
| 9-18     | an 10     | <p>PARTIAL PAN or ACCOUNT NUMBER</p> <p>This is an alphanumeric field containing the primary recipients partial PAN or account number. If partial PAN only the first 6 and last 4 digits should be entered, if account number only up to 10 characters should be provided</p> <p>Where the data is not available or is not maximum length please <i>b</i> fill</p>         |
| 19-24    | an 6      | <p>PARTIAL POST CODE</p> <p>This is an alphanumeric field containing the primary recipients post code information. If this is a UK post code only the first part of the post code should be provided. For Non-UK post code up to 6 characters of the post code should be provided</p> <p>Where the data is not available or is not maximum length please <i>b</i> fill</p> |
| 25-30    | a 6       | <p>SURNAME</p> <p>This is an alpha field containing the primary recipient's surname. Only up to 6 characters should be provided.</p> <p>Where the data is not available or is not maximum length please <i>b</i> fill</p>                                                                                                                                                  |

## 4.2.25 TOKEN QI CSM DATA TOKEN

The CSM Data Token indicates support for partial approvals and balance responses in request messages. This token is also used to carry payment indicators for In-App, debt repayment, cryptocurrency, high risk securities, credential on file, PAR, deferred authorisation and SCA exemption indicators. This is a fixed length token, all fields should be initialised to spaces. All positions, including spaces must be included in the Token length.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                    |
|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-32     |           | CSM-DATA-TKN                                                                                                                                                                                                                                                                                                                                  |
| 1        | an 1      | PARTIAL-APPRV-FLG<br>Indicates that the terminal supports Partial Approvals and Balance Responses.<br><br>Valid values are:<br><br>1 = Partial Approvals and Balance Responses are supported<br>0 = Partial Approvals and Balance Responses are NOT supported                                                                                 |
| 2-5      | an 4      | Reserved for future use- <b>b</b> fill<br>Must be <b>b</b> filled                                                                                                                                                                                                                                                                             |
| 6        | an 1      | DSRP-TYPE<br>Indicates whether the transaction is In App or not<br><br>3 = In-App/Tokenised transactions<br><br>If data is not available, then must be <b>b</b> filled                                                                                                                                                                        |
| 7        | an 1      | SPCL-PAY-IND<br>Indicator to support Specialised Payment Transaction Indicator including Debt Repayment, Cryptocurrency and High Risk Securities transactions.<br><br>Valid values are:<br><br>C = Cryptocurrency<br>D = Debt Repayment Transaction<br>S = High Risk Securities<br><br>If data is not available, then must be <b>b</b> filled |
| 8        | an 1      | COF-IND<br>Credential on File Indicator<br><br>Valid values are:<br><br>F = First<br>S = Stored<br><br>If data is not available, then must be <b>b</b> filled                                                                                                                                                                                 |

| Position | Data Type                                              | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |                       |      |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
|----------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------|------|------------|--------|-----|----|-----------|---|---|---|---|----|---------------------------------|---|---|---|---|----|------------------------------------------------|---|-----|-----|---|----|--------------------------------------------------|-----|-----|-----|-----|----|--------------------------------------------------------|-----|---|-----|---|----|-------------------------------------|-----|---|-----|---|----|----------------------------------|-----|---|---|---|----|---------------------------------|---|---|-----|-----|
| 9        | an 1                                                   | PAR-IND<br>Payment account reference indicator is set if PAR data is supported<br><br>Valid values are:<br><br>Y = PAR data supported<br><br>If data is not available, then must be <b>b</b> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                       |      |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 10       | an 1                                                   | Reserved for future use<br>Must be <b>b</b> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                       |      |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 11-12    | an 2                                                   | SCA Exemption Indicator <table><tr><th>Value</th><th>Exemption Description</th><th>Visa</th><th>Mastercard</th><th>Diners</th><th>UPI</th></tr><tr><td>10</td><td>Low Value</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>20</td><td>Transaction Risk Analysis (TRA)</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>40</td><td>Trusted Merchant Beneficiary <sup>Note 1</sup></td><td>✓</td><td>N/A</td><td>N/A</td><td>✓</td></tr><tr><td>80</td><td>Secure Corporate Payment (SCP) <sup>Note 1</sup></td><td>N/A</td><td>N/A</td><td>N/A</td><td>N/A</td></tr><tr><td>01</td><td>Merchant Initiated Transaction (MIT) <sup>Note 3</sup></td><td>N/A</td><td>✓</td><td>N/A</td><td>✓</td></tr><tr><td>02</td><td>Recurring Payment <sup>Note 2</sup></td><td>N/A</td><td>✓</td><td>N/A</td><td>✓</td></tr><tr><td>04</td><td>SCA Delegation <sup>Note 1</sup></td><td>N/A</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>08</td><td>Authentication Outage Exception</td><td>✓</td><td>✓</td><td>N/A</td><td>N/A</td></tr></table><br>If data is not available, then must be <b>b</b> filled | Value      | Exemption Description | Visa | Mastercard | Diners | UPI | 10 | Low Value | ✓ | ✓ | ✓ | ✓ | 20 | Transaction Risk Analysis (TRA) | ✓ | ✓ | ✓ | ✓ | 40 | Trusted Merchant Beneficiary <sup>Note 1</sup> | ✓ | N/A | N/A | ✓ | 80 | Secure Corporate Payment (SCP) <sup>Note 1</sup> | N/A | N/A | N/A | N/A | 01 | Merchant Initiated Transaction (MIT) <sup>Note 3</sup> | N/A | ✓ | N/A | ✓ | 02 | Recurring Payment <sup>Note 2</sup> | N/A | ✓ | N/A | ✓ | 04 | SCA Delegation <sup>Note 1</sup> | N/A | ✓ | ✓ | ✓ | 08 | Authentication Outage Exception | ✓ | ✓ | N/A | N/A |
| Value    | Exemption Description                                  | Visa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mastercard | Diners                | UPI  |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 10       | Low Value                                              | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ✓          | ✓                     | ✓    |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 20       | Transaction Risk Analysis (TRA)                        | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ✓          | ✓                     | ✓    |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 40       | Trusted Merchant Beneficiary <sup>Note 1</sup>         | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | N/A        | N/A                   | ✓    |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 80       | Secure Corporate Payment (SCP) <sup>Note 1</sup>       | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | N/A        | N/A                   | N/A  |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 01       | Merchant Initiated Transaction (MIT) <sup>Note 3</sup> | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ✓          | N/A                   | ✓    |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 02       | Recurring Payment <sup>Note 2</sup>                    | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ✓          | N/A                   | ✓    |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 04       | SCA Delegation <sup>Note 1</sup>                       | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ✓          | ✓                     | ✓    |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 08       | Authentication Outage Exception                        | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ✓          | N/A                   | N/A  |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 13-16    | an 4                                                   | Reserved for future use<br>Must be <b>b</b> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                       |      |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 17       | an 1                                                   | Deferred Authorisations<br><br>D = Deferred Authorisation<br><br>If data is not available, then must be <b>b</b> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |                       |      |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 18-21    | an 4                                                   | Reserved for future use<br>Must be <b>b</b> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                       |      |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |
| 22       | an 1                                                   | Single Tap and PIN<br>"0" – Terminal not capable of single tap and PIN (Request)<br>"1" – Terminal capable of single tap and PIN (Request)<br>"3" – Transaction with PIN (Request)<br><br>If data is not available, then must be <b>b</b> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |                       |      |            |        |     |    |           |   |   |   |   |    |                                 |   |   |   |   |    |                                                |   |     |     |   |    |                                                  |     |     |     |     |    |                                                        |     |   |     |   |    |                                     |     |   |     |   |    |                                  |     |   |   |   |    |                                 |   |   |     |     |

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                   |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23       | an 1      | TRAN-AMT-VAR-FIX-IND<br>"F" – Fixed Amount (Request)<br>"V" – Variable Amount (Request)<br><i>b</i> = Not applicable<br><br>Values of 'F' and 'V' must be used in Mastercard Initial and subsequent recurring transactions, for all non-recurring transactions value must be set to 'space'. |
| 24       | an 1      | CSM-MISC-RESP-DATA-TKN-FLAG<br>"Y" – Miscellaneous Response Data Token (QX) Supported (Request)<br><br>If data is not available, then must be <i>b</i> filled                                                                                                                                |
| 25       | an 1      | PMNT-CRD-IND<br>"C" – Credit Transaction<br><br>Value of 'C' can only be used for Visa OCTs (0200 message types).<br>If data is not available, then must be <i>b</i> filled                                                                                                                  |
| 26-32    | an 7      | Reserved for future use<br>Must be <i>b</i> filled                                                                                                                                                                                                                                           |

Note 1: Not supported by Barclaycard Payments.

Note 2: For Mastercard/Maestro/UPI this is set automatically by Barclays host on receipt of a correctly identified subsequent recurring authorisation request, merchants must not send in any SCA exemption indicators on subsequent recurring message types i.e. C4 Token pos 4 = '4'.

Note 3: For Mastercard/Maestro this is set automatically by Barclays host when the authorisation request contains Trace ID, merchants must not send in any SCA exemption indicators on Merchant Initiated Transactions (MITs). For UPI as Scheme Reference Data is not supported, merchants must populate the Merchant Initiated Transaction exemption indicator to signify the transaction is a MIT.

## 4.2.26 TOKEN QJ POS BALANCE TOKEN

The POS Balance Token provide the cardholder balance in the response. This is a MasterCard functionality.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                       |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-28     |           | POS-BAL-TKN                                                                                                                                                                                                                                                                      |
| 1-13     | an 13     | TXN-AMT-1<br>A balance specified in the transaction currency.<br>If the balance amount is identified as a Debit amount then this field will be pre-fixed with "-". If the amount is a credit amount there will be no "+" prefix. The field will just contain the credit balance. |
| 14-16    | n 3       | TXN-CRNCY-CDE<br>The ISO numeric currency code of the transaction currency.                                                                                                                                                                                                      |
| 17-28    | an 12     | Reserved for future use- <del>b</del> filled                                                                                                                                                                                                                                     |

## 4.2.27 TOKEN QL DYNAMIC CURRENCY CONVERSION

This token is used when a transaction amount is converted to the cardholder's billing currency using dynamic currency conversion (DCC). This is a Visa only functionality.

| Position | Data Type | Field Name and Description                                                                                                                                                                             |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-16     |           | CSM-DCC-TKN                                                                                                                                                                                            |
| 1-11     | n 11      | DCC-CRD-ACCPT-CRNCY-AMT<br>Transaction amount prior to DCC, in the card acceptor currency minor denomination                                                                                           |
| 12-14    | n 3       | DCC-CRD-ACCPT-CRNCY-CDE<br>The ISO numeric currency code of the transaction currency prior to DCC.                                                                                                     |
| 15       | n 1       | DCC-IND<br>Code to indicate whether or not DCC was accepted by the cardholder<br><br>Valid Values:<br><br>0 = DCC offered and accepted by cardholder<br>1 = DCC offered and not accepted by cardholder |
| 16       | an 1      | Reserved for future use must be <del>b</del> filled                                                                                                                                                    |



## 4.2.28 TOKEN QV PAYMENT FACILITATOR TOKEN

The Payment Facilitator token carries the Sub-Merchant ID. This token should only be included in Payment Facilitator transactions.

| Position | Data Type | Field Name and Description                                                                         |
|----------|-----------|----------------------------------------------------------------------------------------------------|
| 1-26     |           | CSM-PFID-TKN                                                                                       |
| 1-15     | an 15     | SUBMERCH-ID<br>This will contain the sub merchant ID, left justified and space fill the remainder. |
| 16-26    | an 11     | Reserved for future use<br>Must be <del>b</del> filled.                                            |

## 4.2.29 TOKEN QW RECIPIENT DATA TOKEN

The Recipient Data token carries the sender data for Mastercard Gaming Payment of Winnings and Visa and Mastercard Account Funding Transactions. The values given are the maximum lengths. Any unused fields or parts of field will not be zero or space filled as this token is defined to be of variable length.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-500    |           | CSM-RECIPIENT-DATA-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1-500    | ans..500  | <p>RECIPIENT DATA<br/>This will contain the recipient data. This field is of variable length and should follow TLV.</p> <p>Tags received should be of 3 bytes Alpha-Numeric.<br/>Lengths received should be of 3 bytes Numeric.<br/>The TLV format may optionally be repeated as many times as will fit in the overall token length.</p> <p>The following Tags are valid:</p> <p>001 - Recipient First Name<br/>002 - Recipient Middle Name<br/>003 - Recipient Surname<br/>004 - Recipient Street Address<br/>005 - Recipient City<br/>006 - Recipient State/Province<br/>008 - Recipient Country<br/>010 - Recipient Account Number<br/>011 - Recipient Account Number Type</p> <p>Please refer to the AFT, Gaming and Visa Direct guides for lengths and usages of the above tags.</p> <p>Example:<br/>P-63 {043}[&amp; 0000200043! QW00021 001003JOE003006BLOGGS]</p> |

## 4.2.30 TOKEN QX MISCELLANEOUS RESPONSE DATA TOKEN

| Position | Data Type | Field Name and Description                                                                                                                                                                                                           |
|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-40     |           | CSM-MISC-RESP-DATA-TKN                                                                                                                                                                                                               |
| 1-2      | an 2      | ADDITIONAL PAYMENT DECLINE CODE<br>A code used to provide additional information regarding the disposition of the transaction. Valid values are as follows:<br>22 = Merchant does not qualify for product code<br>b = Not applicable |
| 3-7      | an 5      | CSM-MISC-TKN-ANI-RSLT-DATA-ACCT-NAME-RQST-RSLT                                                                                                                                                                                       |
| 3        | an 1      | Account Name Request Result<br><br>A – Name Match Performed<br>B – Name Match Not Performed<br>C – Name Match Not Supported<br>b = Not applicable                                                                                    |
| 4        | an 1      | Account Name Match Decision<br><br>M - Match<br>P – Partial Match<br>N – No Match<br>X – Not Performed or not supported<br>b = Not applicable                                                                                        |
| 5        | an 1      | Last Name Account Match Decision<br><br>M - Match<br>P – Partial Match<br>N – No Match<br>X – Not Performed or not supported<br>b = Not applicable                                                                                   |
| 6        | an 1      | Middle Name Account Match Decision<br><br>M - Match<br>P – Partial Match<br>N – No Match<br>X – Not Performed or not supported<br>b = Not applicable                                                                                 |
| 7        | an 1      | First Name Account Match Decision<br><br>M - Match<br>P – Partial Match<br>N – No Match<br>X – Not Performed or not supported<br>b = Not applicable                                                                                  |
| 8-40     | an 33     | Reserved for future use<br>Must be b filled.                                                                                                                                                                                         |

## 4.2.31 TOKEN QY SENDER DATA TOKEN

The Sender Data token carries the sender data for Mastercard Gaming Payment of Winnings and Visa Account Funding Transactions. The values given are the maximum lengths. Any unused fields or parts of field will not be zero or space filled as this token is defined to be of variable length.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-500    |           | CSM-SENDER-DATA-TKN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1-500    | ans..500  | <p><b>SENDER DATA</b><br/>This will contain the sender data. This field is of variable length and should follow TLV.</p> <p>Tags received should be of 3 bytes Alpha-Numeric.<br/>Lengths received should be of 3 bytes Numeric.<br/>The TLV format may optionally be repeated as many times as will fit in the overall token length.</p> <p>The following Tags are valid:</p> <p>001 - Sender First Name<br/>002 - Sender Middle Name<br/>003 - Sender Surname<br/>004 - Sender Street Address<br/>005 - Sender City<br/>006 - State/Province<br/>008 - Sender Country<br/>010 - Sender Account Number<br/>011 - Sender Account Number Type<br/>019 - Sender Purpose Of Payment Code</p> <p>Please refer to the AFT, Gaming and Visa Direct guides for lengths and usages of the above tags.</p> <p>Example:<br/>P-63 {043}[&amp; 0000200043! QY00021 001003JOE003006BLOGGS]</p> |

#### 4.2.32 TOKEN R3 PAYMENT DATA TOKEN

The Payment Data Token carries the Mastercard only Merchant Payment Gateway ID (MPG ID). This is a fixed length token and the fields are described below.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                |
|----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-11     |           | CSM-PAYMENT-DATA-TKN                                                                                                                                                                                                      |
| 1-11     | N 11      | Merchant Payment Gateway ID.<br><br>The value provided during registration, or default value used, should be fixed length 11, right justified with leading zeroes. This field should not be populated in 0420/1 messages. |

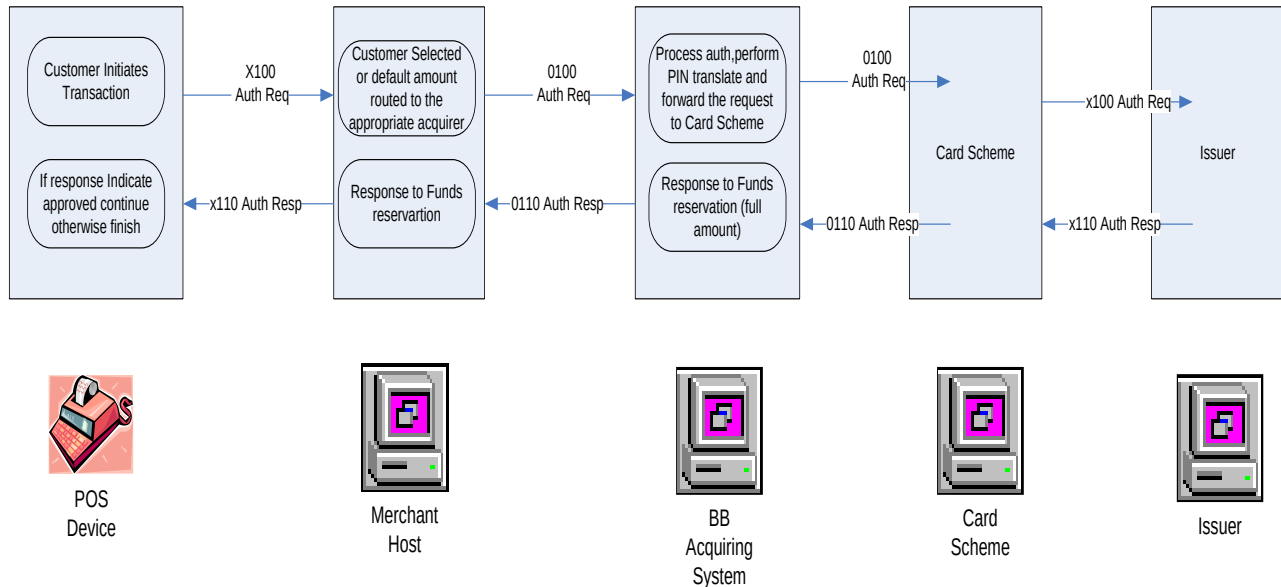
#### 4.2.33 TOKEN R4 INSTALMENT TOKEN

The Instalment Token carries the plan registration system identifier for Visa Instalment Service transactions. The fields in the Instalment Token are described below. This is a fixed length token.

| Position | Data Type | Field Name and Description                                                                                                                                                                                                                                                                   |
|----------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-35     |           | CSM-INSTALMENT-R4-TKN                                                                                                                                                                                                                                                                        |
| 1-35     | AN 35     | PL-REG-SYS-ID<br>This field contains the Visa instalment service (VIS) Plan Registration System Identifier. The Visa instalment service (VIS) Plan Registration System Identifier is an alpha numeric value with a length of 10. This should be left justified with the remainder as spaces. |

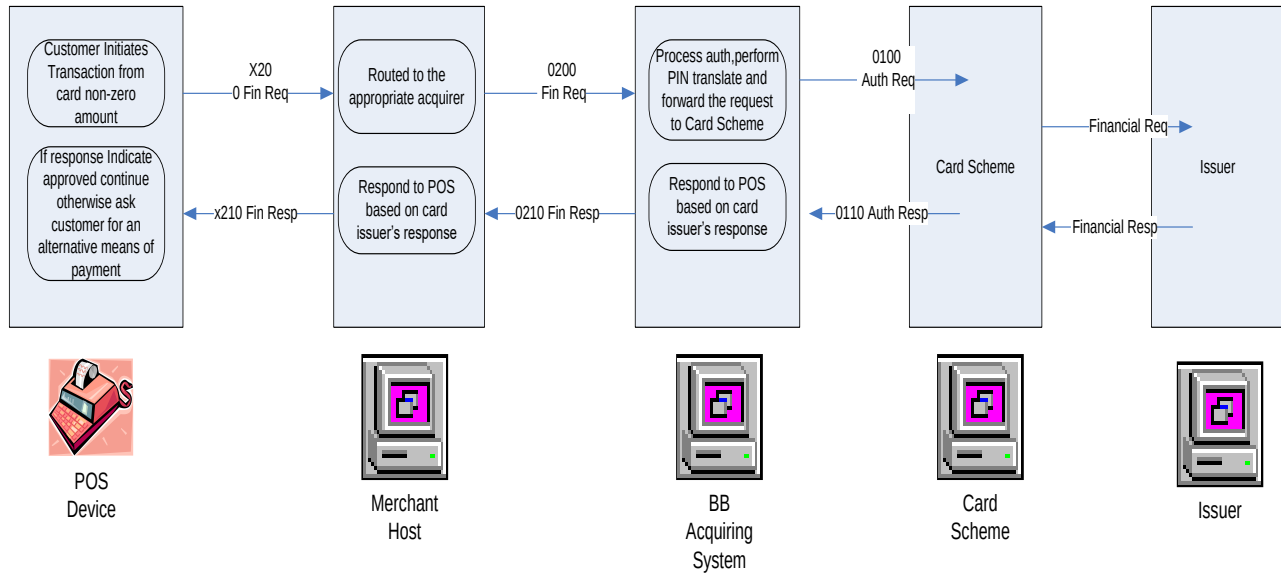
## 5 TRANSACTION FLOWS

### 5.1 AUTHORISATION TRANSACTION FLOW



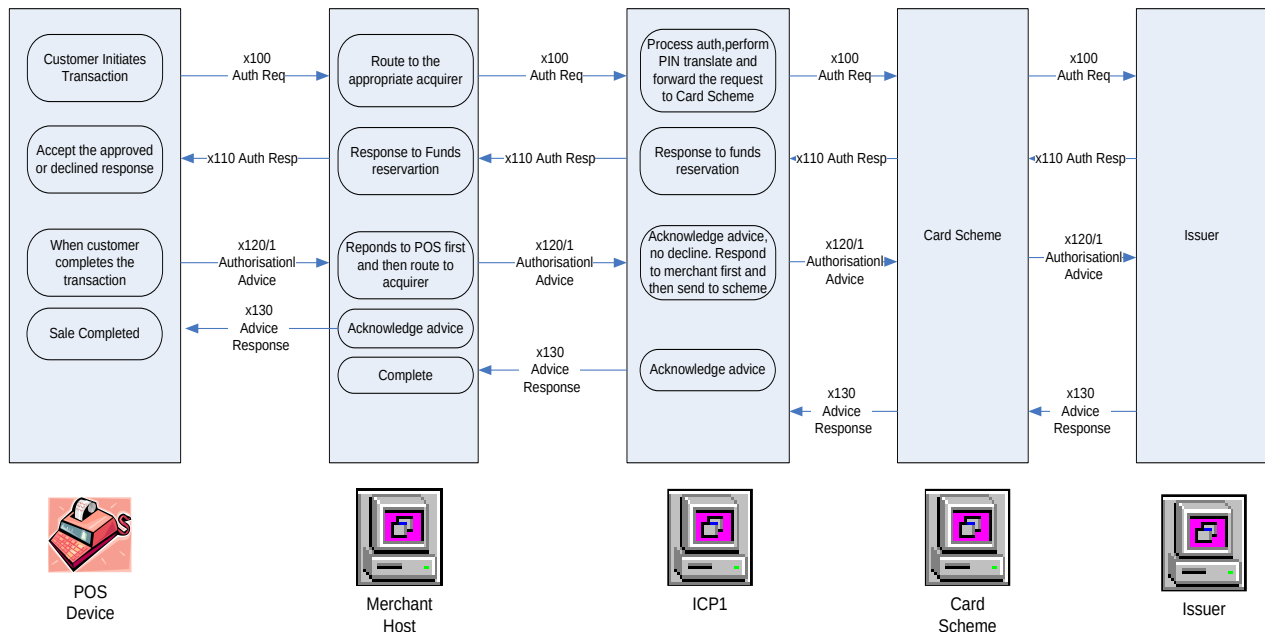
- 1) The merchant submits a authorisation request (0100) to the Barclay's acquiring system
- 2) Validate the message as per ISO 8583 message specification
- 3) Reformat the valid authorisation requests as per the appropriate Card Scheme rules and forward the transaction (0100) to the corresponding Card Scheme network (Diners/Discover/MasterCard/Visa)
- 4) The authorisation response is received from the relevant Card Schemes
- 5) Forward back the approved or declined authorisation response (0110) to the merchant host

## 5.2 FINANCIAL TRANSACTION FLOW



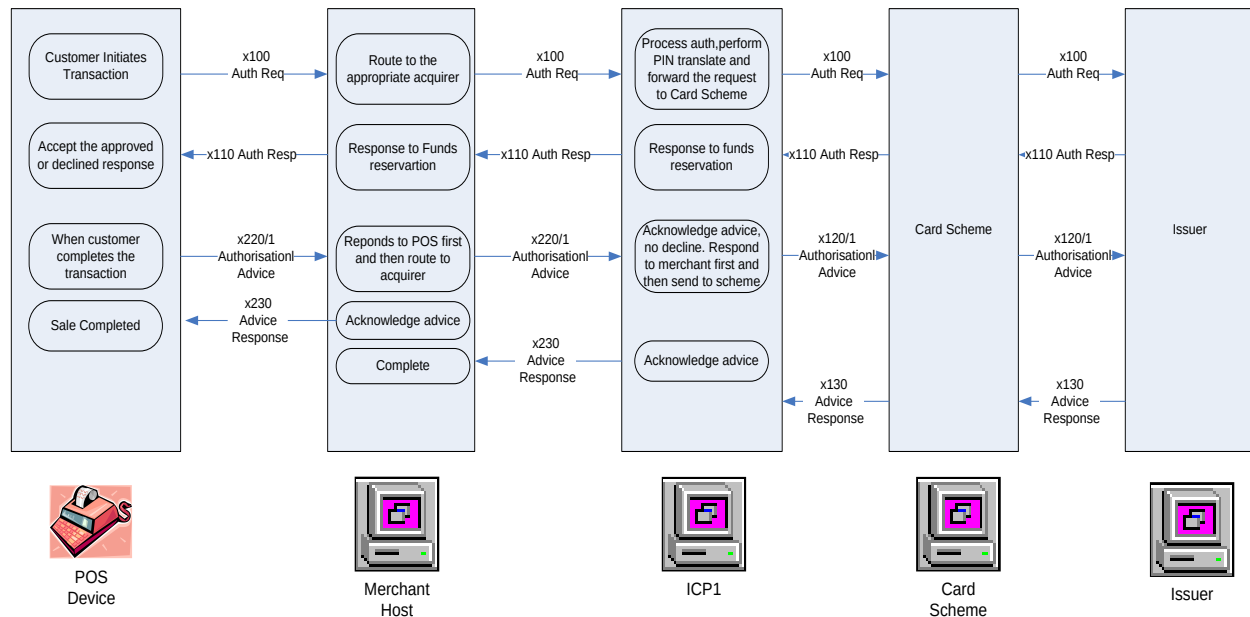
- 1) The merchant submits a Financial request (0200) to the Barclay's acquiring system
- 2) Validate the message as per ISO 8583 message specification
- 3) Reformat the valid financial requests as per the appropriate Card Scheme rules and forward the transaction (0100) to the corresponding Card Scheme network (Diners/Discover/MasterCard/Visa)
- 4) The financial response is received from the relevant Card Schemes
- 5) Forward back the approved or declined Financial response (0210) to the merchant host

### 5.3 AUTHORISATION ADVICE TRANSACTION FLOW



- 1) The merchant submits a authorisation advice request/advice repeat (0120/0121) to the Barclays acquiring system
- 2) Validate the message as per ISO 8583 message specification and accept the advice message.
- 3) Send an advice acknowledgement (0130) response to the merchant host
- 4) AFD advices are forwarded to schemes when required by scheme rules. At present all other authorisation advice messages are not forwarded to Card Scheme networks
- 5) Barclays acquiring system receives an advice acknowledgement (0130) response from MasterCard for each submitted advice message.

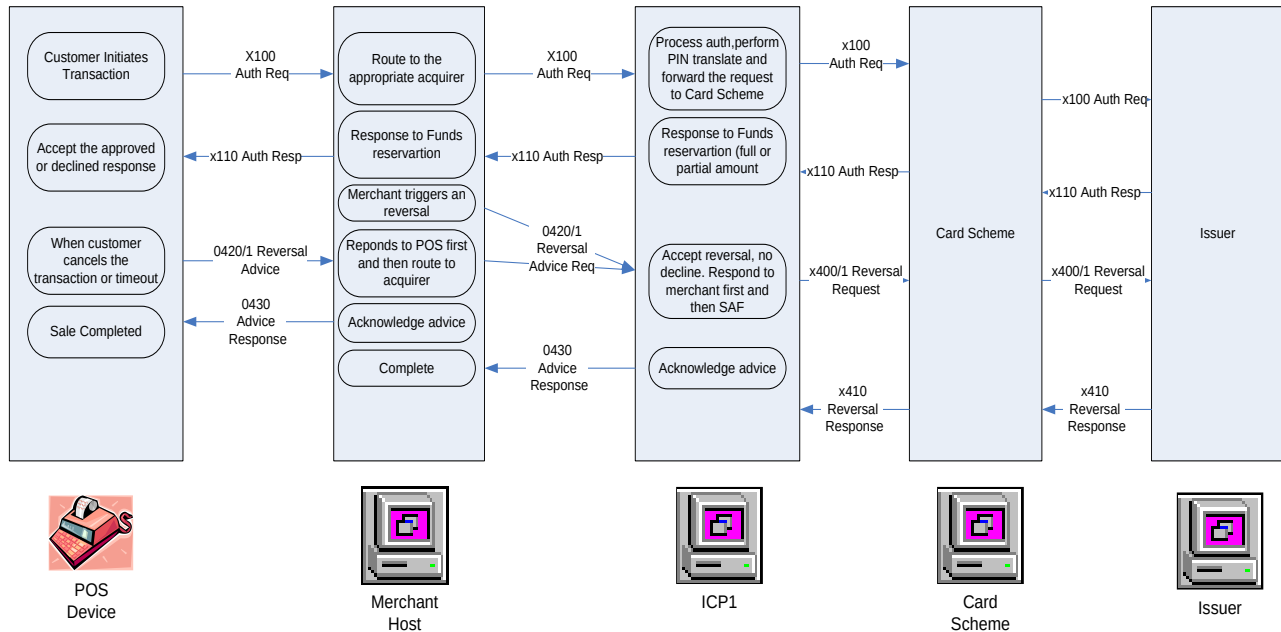
## 5.4 FINANCIAL ADVICE TRANSACTION FLOW



- 1) The merchant submits a Financial advice request/advice repeat (0220/0221) to the Barclay's acquiring system
- 2) Validate the message as per ISO 8583 message specification and accept the advice message.
- 3) Send an advice acknowledgement (0230) response to the merchant host
- 4) AFD advices are forwarded to schemes when required by scheme rules. At present all other authorisation advice messages are not forwarded to Card Scheme networks
- 5) Barclays acquiring system receives an advice acknowledgement (0130) response from MasterCard for each submitted advice message.

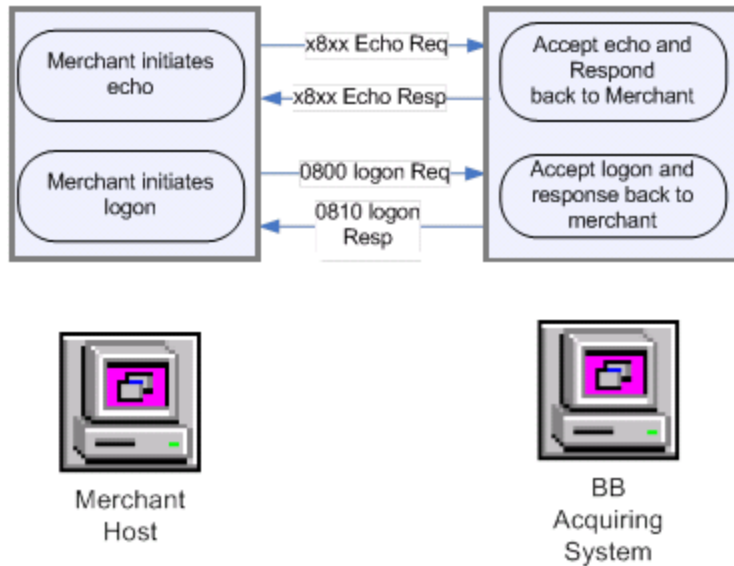


## 5.5 REVERSAL TRANSACTION FLOW



- 1) The merchant submits a Reversal advice/ repeat request (0420/0421) to the Barclay's acquiring system
- 2) Validate the message as per ISO 8583 message specification
- 3) Reverse the original authorisation message
- 4) Forward back the approved or declined reversal response (0430) to the merchant host
- 5) And then reformat the valid reversal requests as per the appropriate Card Scheme rules and forward the transaction (0400/0401) to the corresponding Card Scheme network (Diners/Discover/MasterCard/Visa)
- 6) The reversal response is received from the relevant Card Schemes

## 5.6 NETWORK MANAGEMENT MESSAGE FLOW



- 1) The merchant submits logon/ echo Network Management request (0800/0810) to the Barclay's acquiring system
- 2) Validate the message as per ISO 8583 message specification and accept the request.
- 3) Respond with a 0810 Network Management response message to the merchant host.

## 6 APPENDIX A: PIN ENCRYPTION METHODS

### 6.1 3DES ENCRYPTION

Triple Data Encryption Standard, The term "Triple DES" is used to denote DES cryptography performed with double length keys. This is sometimes known as EDE mode (Encrypt-Decrypt-Encrypt)

The Triple DES PIN block encryption method supported using the TDEA Electronic Code Book (TECB) as described in ANSI X9.52 standard.

The supported PIN block format is ISO 9564 format '0' is also known as ANSI X9.8 PIN block format.

PIN Translation:

PIN Translation refers to the "decryption and re-encryption" of PIN data from one security zone to another. Usually this is necessitated when there are more than one security zones between the PIN originating and verification location.

Decrypt the PIN using the secret PIN encryption key that exists between the merchant and Barclay's. Re-encrypt the PIN under another secret key between Barclay's and the Card Scheme before forwarding to the Card Schemes.

## 7 APPENDIX B: EXAMPLE FOR AUTH REQUEST AND RESPONSE

### 0100 Auth request Message Log:

```

13036 618 08/12/18 14:05:11.06          188870 7 - Pro
SOURCE S1B- -X25 (STA 1)
DEST P1B- (PRO 12)

HEADER [ISO026000050] TYPE [0100] PMAP [B238440128E19002] SMAP [
00000000000000008] P-3 [000000] P-4 [0000000000100] P-7 [1218150530]
P-11 [000388] P-12 [150442] P-13 [1218] P-18 [5542] P-22 [021]
P-32 {06}[492900] P-35 {37}[454638*****0009=12091060912345678901] P-37
[181504000388] P-41 [34900001 ] P-42 [7819056 ] P-43 [Mozart Ltd.
LangstrassBerlin GE] P-48 {027}[7819056
A A ] P-49 [978] P-52 [3AF5C1FA2254882C] P-63 {034}[6 0000200034! C400012 100001400 172] S-125
{012}[00HOSTB24 00]

```

### 0110 Auth response Message Log:

```

13040 369 08/12/18 14:05:11.67          188874 6 - Pro
SOURCE P1B- (PRO 12)
DEST S1B- -X25 (%177777)

HEADER [ISO026000057] TYPE [0110] PMAP [723800010EC18000] P-2 {1
6}[454638*****0009] P-3 [000000] P-4 [0000000000100] P-7 [121814051 1] P-11
[000388] P-12 [150442] P-13 [1218] P-32 {06}[492900] P-37
[181504000388] P-38 [487329] P-39 [00] P-41 [34900001 ] P-42 [7819056
] P-48 {027}[7819056 A A ] P-49 [978]

```

## 8 MODIFICATION HISTORY

| Version | Date       | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Author |
|---------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| v 0.1   | 20/01/2009 | Initial Creation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | VJ     |
| v 0.2   | 10/02/2009 | Added P-2 Primary Account Number entry to section 3.5.2 0110 Authorisation response<br><br>Added definition to B4 token positions 9-10 and 11-14 value can be <del>b</del> filled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | VJ     |
| v 1.0   | 10/03/2009 | Amended front sheet to say HISO not ISO and added the version no (87)<br><br>Amended data element field definition of 0420/1 message class P-38 from conditional to mandatory in section 3.5 message structure and content<br><br>Amended data element field definition of 0420/1 message class P-22 from mandatory to conditional in section 3.5.9 0420 reversal advice & 0421 reversal advice repeat request<br><br>Amended data element field definition of 0420/1 message class P-38 from conditional to mandatory in section 3.5.9 0420 reversal advice & 0421 reversal advice repeat request<br><br>Added P-43 Card Acceptor name/location to section 3.5.10 0430 reversal advice response<br><br>Added 'This data element is conditional in 0420/1' to section 3.6.12 point of service entry mode<br><br>Added 'Optional' in front of format for P-37 – DDHHMM + STAN (6 digits) to section 3.6.17 Retrieval reference number (p-37)<br><br>Added 'The Authorisation Identification Response data element is mandatory for 0110, 0120/1, 0210, 0220/1 and 0420/1 messages.' to section 3.7.11 authorisation identification response (P-38) | NO     |
| v 1.1   | 10/09/2009 | Added the following 4 token formats<br>C0 BASE24-pos Release 5.1 Token,<br>01 Address Verification Token<br>CE Authentication Data Token<br>C6 Trans Stain XID Token                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | NO     |

|         |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |    |
|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| v 1.2   | 19/01/2010 | Added MOTO definition positions 1-2 to the processing code (P-3)<br>Removed the 01 Address Verification Token<br>Added the CH POS Data1 Token                                                                                                                                                                                                                                                                                                                                                                | NO |
| v 1.3   | 06/07/2010 | Added Visa Account Verification definition positions 1-2 to the processing code (P-3)<br>Added Visa Account Verification statement to transaction amount (P-4)<br>Removed Duplicated values 0 to 4 in CO token position 19<br>Added (not supported in Europe) comment to values 8 and 9 in CO token position 19<br>Added Static AAV value of 3 to CO token position 24<br>Added values 8,9,A,B,C,D,W,X,Y,Z to CO token position 26<br>Corrected reference to 'CH' token within header text of section 4.2.10 | NO |
| v 1.4   | 17/08/2010 | Corrected length definition of P-12 in all 3.5 tables from n12 to n6<br>Added the following token format<br>F1 E-commerce Additional Data Token                                                                                                                                                                                                                                                                                                                                                              | NO |
| v 1.5   | 27/06/2011 | Removed reference to (Visa Only) of the definition of positions 1-2 to the processing code (P-3)<br>Added MasterCard and Maestro Account Status Enquiry Service statement to transaction amount (P-4)<br>Added Visa '96' value to point of service entry mode (P-22)<br>Added value '8' to 3 <sup>rd</sup> digit of point of service entry mode (P-22)<br>Amended P-25 to a conditional field and added text to section 3.6.14                                                                               | NO |
| V 1.6   | 20/02/2012 | Added N.B. for contactless capable devices in the TERM-INPUT-CAP-IND definition of the C4 token<br>Added F3 & S8 token descriptions                                                                                                                                                                                                                                                                                                                                                                          | SS |
| V 1.6.1 | 12/04/2012 | Updated Advice and Reversal diagrams and tidied up some grammar                                                                                                                                                                                                                                                                                                                                                                                                                                              | SS |

|         |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
|---------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| V 1.6.2 | 20/04/2012 | <p>Corrected 3.1 message types identifier table</p> <p>Corrected 3.5 P-17 0100, 0120/1, 0200, 0220/1 and 0420/1 data condition code from O to M</p> <p>Corrected 3.5 P-63 0420/1 data condition code from M to C</p> <p>Corrected 3.5.1, 3.5.3, 3.5.5, 3.5.7, 3.5.9 P-17 data condition code from O to M</p> <p>Corrected 3.6.10 from optional to mandatory for the listed message types</p> <p>Added comment to 3.6.14 codes 02 and 27 mandatory for CAT terminals.</p> <p>Amended 3.6.29 pos 1-4 to reference message not transaction type</p> <p>Amended spelling of reversed to reserved for response codes 45-50 in 3.7.12</p> <p>Amended numbering structure of 3.7.18</p> <p>Updated B2 token to include TC support as well as ARQC.</p> <p>Updated B2 token to include bit 4 definition of advice required flag.</p> <p>Updated B3 Token EMV Terminal Capabilities to include No CVM and CDA</p> <p>Updated B4 Token pos 4 Terminal Entry Capabilities to include value 8 for contactless only capability</p> <p>Updated C4 Token pos 7 Transaction Status Indicator to include value 9 for Account Status Enquiry Service</p> <p>Amended ref to C0 token in text of 4.2.10 CH token</p> <p>Amended token CH 4.2.10 position numbering</p> | NO |
| V 1.7   | 12/06/2012 | <p>Added 01 Token and 20 Token descriptions and to table 4.1 List of Supported Tokens</p> <p>Added P-44 Additional Response Data in Data Field Description and to Data Element tables</p> <p>Amended section and page numbering after 3.7.16</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MP |

|        |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
|--------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| V 1.8  | 31/01/2013 | <p>Updated section 1.3 reference token version number</p> <p>Replaced LLL in section 1.4 with ..NN and NNN</p> <p>Clarified section 3.2.1</p> <p>Added value 000 to position 5-7 of message header within section 3.2.2</p> <p>Deleted reference to EBCDIC in section 3.4</p> <p>Corrected format of P-56 in all 3.5 tables. P-56 was an34 now an37.</p> <p>Amended condition of P-12 and P-13 from M to ME in all 3.5 response tables</p> <p>Removed entry of P-17 throughout the specification.</p> <p>Added text in 3.6.12 on clarification of cards without a PSN then omit field</p> <p>Added text to section 3.6.13 pre-auth value 06</p> <p>Added clarification to section 3.6.16</p> <p>Added wording to include retrieval reference number to section 3.6.28 position 5-16</p> <p>Added 'application id' to position 49-80 of B3 token</p> <p>Added (mandatory) to positions 1-3 value '022' of B4 token</p> <p>Removed 'chip authentication program token format from token CE</p> <p>Added new '07' value meaning Debt Recovery to positions 1-2 of the F3 token</p> <p>Added Proximity Payment Token QE to list of supported tokens in section 4.1 and created token definition to new section 4.2.18</p> | NO |
| V 1.81 | 05/04/2013 | <p>Added value 34 to the reversal code table within section 3.6.18.</p> <p>Added definition of last field position 22 to token 20 Interchange Compliance Token section 4.2.17.</p> <p>Added additional wording to token 20.</p> <p>Amended format of Proximity Payment Token QE token in section 4.2.18.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | NO |
| V 1.82 | 18/10/2013 | <p>MPOS indicator – Device Type in F1 token now called 'additional data token'</p> <p>Update tables on pages 15 to 26 - P25 mandatory (addition in response message) and removal of P17 and other changes</p> <p>Update to TVR,CVMR and other fields in B3 token</p> <p>Changes to Additional response data tables</p> <p>Addition of 'space' as a valid field where applicable</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AM |



|       |            |                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
|-------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| V.2   | 05/02/2014 | QG VMAAD token<br>Pre-authorisation and Final authorisation in C4 token<br>F4 Digital Wallet Token- Staged Digital Wallets,<br>MasterPass and V.Me<br>New response codes L2 & T8<br>CH token update<br>M indicator in S8 token<br>Update to C0 token to reflect MasterCard April 2014<br>changes<br>3.5 Message structure and content – primary<br>account number conditional indicator changed for<br>responses | AM |
| V 2.1 | 20/06/2014 | Addition of Partial Approvals and Balance responses<br>with QI and QJ tokens and update to P4 in response.<br>Other amendments to B2, C4 and F1 tokens.<br>Updates to section 3. Introduction fo new V.Me<br>token                                                                                                                                                                                               | AM |
| V2.2  | 07/08/2014 | Update to the QD token structure                                                                                                                                                                                                                                                                                                                                                                                 | AM |
| V2.3  | 23/10/2014 | Updates to token section where the list of tokens is<br>lengths and requests vs response presence added,<br>new values in C0 token position 24. Updates to the<br>CH token as now used in the request for recurring<br>and instalments.                                                                                                                                                                          | AM |
| V2.4  | 04/11/2014 | Changes to QD token layout                                                                                                                                                                                                                                                                                                                                                                                       | AM |
| V2.5  | 15/07/2015 | Edited to remove unsupported elements and various<br>minor amendments.                                                                                                                                                                                                                                                                                                                                           | AM |
| V2.6  | 01/03/2016 | Update to;<br>QE token – to generalise as not contactless only<br>PIC X changed to 'an'<br>PIC 9 changed to 'n'<br>Removed 'Level' column from tokens<br>Removal of V.Me                                                                                                                                                                                                                                         | AM |
| V2.7  | 16/05/2016 | Addition of value of 4 to postion 3 of C4 token<br>QI DSRP Type, DF name mandate, review of<br>response codes, mandatory elements B2/B3 tokens                                                                                                                                                                                                                                                                   | AM |
| V2.8  | 13/06/2017 | Updates for ICP Enhancements<br>Account Funding<br>Payment of Winnings<br>Recurring<br>JCB<br>Added CT and O4 Tokens<br>Updated TVR and CVMR for Biometrics<br>Updated AIP for RRP<br>V1 Response code added                                                                                                                                                                                                     | SC |
| V2.9  | 03/07/2017 | Updated Tokens CTand CE to be fixed length<br>Removed text in section 3.6.13 for pre-auth value 06                                                                                                                                                                                                                                                                                                               | SC |

|        |            |                                                                                                                                                                                                                                                                            |    |
|--------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| V3.0   | 09/10/2017 | Updated Token 04 to include the duplication of the CV2 result Code in Position 13.<br>Updated Token 01 to include additional clarification.<br>Updated CH Token for 'Changes to Support Stand-In processing'<br>Updated C0 Token for Mastercard DSRP                       | SC |
| V3.1   | 06/02/2018 | Updated C4 Token Position 12 to include value 5 = Authentication value.<br>Updated C0 Token for clarification.<br>Update to QI Token to include Debt Repayment Indicator.<br>Updated CH and QI Token for Credential on File.<br>Updated CH token to include new MIT flags. | SC |
| V3.2   | 09/11/2018 | Update to C6 token to include CAVV for Visa In-App 3D secure transactions.<br>Update to the CE token to include PRG-PROTOCOL and SER-TRAN-ID for Mastercard 3D secure transactions.                                                                                        | SC |
| V3.3   | 24/01/2019 | Added Token SO for PAR and PAR indicator to QI token.<br>Added default values for P-32 and S-125 and Draft capture flag for Bank owned terminals.<br>Added P-15 Date, Settlement for 0210/0230/0430 response messages.<br>Updated CE and CT tokens to be variable length.  | SC |
| V3.4   | 29/03/2019 | Updated to include UPI ECOMM TOKEN C6<br>Updated CH token to include UPI CVN2 and S-125 field Draft capture flag discription.<br>Updated B3 token CVM-RSLTS clarification.                                                                                                 | SC |
| V3.4.1 | 18/04/2019 | Corrected P-15 to be 'DDMM'                                                                                                                                                                                                                                                | SC |

|        |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|--------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| V3.5   | 30/05/2019 | <p>Update to glossary of terms added NRID</p> <p>Updated Originator Code values in Message Header</p> <p>Amended B2 token TVR Byte 3 bit 3 meaning from Online CVM captured flag to Online PIN captured flag</p> <p>Updated C0 token to include new value '4' In-App Authentication (Diners/Discover only)</p> <p>Updated C6 token to include reference to Diners/Discover for the CAVV value</p> <p>Updated C6 token to state XID not used for Diners/Discover</p> <p>Updated CH token to include new value '4' – Response provided by scheme (Diners/Discover only)</p> <p>Updated 20 token in include reference to NRID</p> <p>B3 token CVMR deleted UPI CDCVM and amended existing sentence to state UPI aswell as Visa</p> <p>B4 token clarified population required for EMV and technical fallback transactions only. Removed a sentence in the header of the B4 token.</p> <p>B4 token positions 9-10 changed to UNUSED</p> <p>B4 token RSN-ONL-CDE values put into priority order, added statement where more than one condition applies</p> | NO       |
| V3.6   | 01/07/2019 | <p>Renamed QG Token to extend its use from Visa only to Visa and Mastercard</p> <p>Updated field P-15 to be conditional and added clarification as to when it is returned.</p> <p>Update to glossary of terms to include SCA.</p> <p>Update to QI Token to variable length, amend DEBT-REPAY-IND to SPCL-PAY-IND and added cryptocurrency, added SCA Exemption Indicators, added Deferred Authorisation.</p> <p>Amendment to Response Code 65 and addition of Response Code V2 in P-39 field.</p> <p>Added note to Token C6 to remove the requirement of XID for Visa 3DSv2 transactions</p> <p>Added 7 as a option for C0 Token Pos 24 to support Mastercard SLI217</p>                                                                                                                                                                                                                                                                                                                                                                             | DC/SC/JL |
| V3.7   | 30/08/2019 | <p>Updated Field P-38 to clarify when it should be present in message types 0120/0220/0420.</p> <p>Updated descriptions in C0 token position 24 AUTHN-COLL-IND</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SC       |
| V3.7.1 | 10/09/2019 | Added default value for Application Ver Number in Token B3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | SS       |

|        |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                    |
|--------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| V3.7.2 | 04/10/2019                   | Note added regarding the removal of support for response code 01/02 - Referral<br>Removal of Referral reference in B2 Token Pos 9-10<br>Updated CE and C6 tokens to clarify format of authentication cryptograms.<br>Updated QI token SCA SCP exemption to N/A for Visa                                                                                                                                                                                                                                                            | DT & SC                            |
| V3.7.3 | 09/12/2019                   | Updated QI Token position 7 SPCL-PAY-IND to include a value of "S" for High Risk Securities<br><br>Updated Token CE to reflect the separation of the 3DS AAV and In-App Token Cryptogram                                                                                                                                                                                                                                                                                                                                           | JL<br><br>SS                       |
| V3.7.4 | 29/01/2020                   | Updated Data types column for VMAAD & MDAAD in QG token<br>Removed P-56 ZKA security related information from Table 3.5.9 0420/1 request messages.                                                                                                                                                                                                                                                                                                                                                                                 | SC                                 |
| V3.7.5 | 20/03/2020<br><br>09/06/2020 | Clarification of the values in CO Token Position 24<br><br>Update P-12 and P-13 to be optional fields for 0800/0810 messages. Removal of Logoff messages.                                                                                                                                                                                                                                                                                                                                                                          | JL<br><br>SC                       |
| V3.7.6 | 17/07/2020                   | Corrected C6 token table                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | SC                                 |
| V3.7.7 | 02/10/2020                   | Updated QI token to add position 22 – Single Tap and PIN support.<br>Updated P-39 Response codes to include 66 - PIN REQUIRED - 1TAP<br>Amended P-38 to be fixed length in section 3.5                                                                                                                                                                                                                                                                                                                                             | SC                                 |
| V3.7.8 | 19/11/2020                   | Updated descriptions for values '1' and '2' in C4 token position 10.                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SC                                 |
| V3.7.9 | 16/02/2021                   | Updated section 3.7.14 Response codes (P-39) to include a note for certain response codes that if received the card acceptor must not resubmit the transaction for authorisation.<br>Updated notes 1 & 2 in section 3.7.14 Response codes (P-39.)<br>Updated text description for Token C4 Pos 12 CRDHLDR-ID-METHOD value 2 to include CDCVM.<br>Updated B2/B3/B4 tokens for UPI to be included in reversals<br>Clarification on Diners In App CAVV in C6 token<br>Tokens QI/CT & CE tokens all changed to be fixed length tokens. | SC<br><br>ST<br>SC<br><br>DC<br>SC |
| V3.8.0 | 18/02/2021                   | Updated F1/C6/CE/CH/QI token descriptions.<br>Updated QI (pos 11-12)SCA indicator field to be alphanumeric.                                                                                                                                                                                                                                                                                                                                                                                                                        | SC                                 |

|        |            |                                                                                                                                                                                                                                                                                                                                                                                                                           |                |
|--------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| V3.8.1 | 30/04/2021 | Updated QI (pos 11-12) SCA indicator field to include Authentication Outage Exception.<br>Updated QI token pos 8 COF indicator description for 'S' to be Stored as previous description was incorrect.<br>Added QL token for DCC (Visa only)                                                                                                                                                                              | SC<br>DT       |
| V3.8.2 | 15/06/2021 | Minor update to mask PAN in transaction examples.                                                                                                                                                                                                                                                                                                                                                                         | JL             |
| V3.8.3 | 07/07/2021 | Updated to clarify CH token pos 20 = 'R' Initial Recurring transaction and C4 pos 4 = 4 Subsequent Recurring transaction.                                                                                                                                                                                                                                                                                                 | SC             |
| V3.8.4 | 23/08/2021 | Updated C0 token to include ECI 'A' (POS 19) for UPI and ECI '4' (POS 24) for Mastercard. Updated QI token (POS 11-12) to include Diners/UPI SCA exemption indicators.                                                                                                                                                                                                                                                    | SC             |
| V3.8.5 | 25/11/2021 | Updated C0 token position 24 value = '4' (Mastercard ECI) to allow UCAF data to be present. Added new response codes 25/26/27 and updated the descriptions in Token 04 for MAC's.<br>Updated F4 token to remove Masterpass.                                                                                                                                                                                               | SC             |
| V3.8.6 | 12/01/2022 | Added QV token for Payment Facilitator. Updated P-18 MCC Code to be conditional for 0100 and 0200 messages, as this field is mandatory for Payment Facilitator transactions.<br>Allow space fill for C0 Token Pos 19 and 24 for customer present transactions.<br>Added QY token (Sender Data) and QW token (Recipient Data) for Gaming POW                                                                               | DC             |
| V3.8.7 | 03/05/2022 | Updated QY token (Sender Data) and QW token (Recipient Data) for Gaming POW to be ANS<br>Added default/unknown value for Mastercard Address Verification Status listing.                                                                                                                                                                                                                                                  | DC<br>ST       |
| V3.8.8 | 29/06/2022 | Updated Sections 3.5, 3.5.2, 3.5.6 & 3.7.4 for field P-4 in 0110 and 0210 messages from 'ME' to be 'M'.<br>Due to partially approved transactions receiving a different amount in the response to the request amount.<br>Added Response Code 83 to section 3.7.14<br>Clarified 0810 message that P-12 and P-13 are conditional echoes for IFSF petrol interface<br>Added CE to section 1.5 data element field definitions | SC<br>NO<br>SS |

|        |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                         |
|--------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| V3.8.9 | 21/03/2023 | <p>Updated C6 token TRANS-STAIN-XID-TKN to remove references related to 3DS v1</p> <p>Updated CE token AUTHn-IND-DATA-PRG-PROTOCOL to remove reference to 3DS v1</p> <p>Updated QI token POS 6 indicator to now be for Inapp and Tokenised transactions</p> <p>Added values for Mastercard CIT/MIT changes in CH token pos 20 (partial/split shipment indicator) and QI token pos 23 for Fixed/variable indicators for Mastercard recurring.</p> <p>Removed 'between 196 and' from section 3.3 Handling of Erroneous Messages</p> <p>Update tokens CT, QW and QY to include BAI and sender/recipient data for Visa AFT transactions</p> <p>Added new mPOS indicators to P-22 pos 3 and token F1.</p> <p>Added new QX token, used for additional payment decline codes.</p> <p>Removed P-56 and token F4 from the spec as no longer supported by BP.</p> <p>Added new field P-54 additional amounts and values in P-3 (09 = cashback) to support PWCB and CBWP transactions.</p> | <p>ST</p> <p>SC</p> <p>NO</p> <p>SC</p> |
| V3.9.0 | 03/04/2023 | Removed value '07' Telephone device request from P-25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | NO                                      |
|        | 19/04/2023 | Updated to include Visa response codes 46 and 6P. Updated the description for response code 82.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SC                                      |
| V3.9.1 | 30/05/2023 | <p>Removed P-44 Response values for Visa table: 'B, C, G, I, M, P, D'</p> <p>Updated note for partial shipment in CH token</p> <p>Updated P-54 length to be n15 in section 3.5 and section 3.6.25 to include a structure table for P-54.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>TW</p> <p>SC</p>                     |
| V3.9.2 | 20/07/2023 | Updated CT and QW tokens to allow support for Mastercard AFTs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SC                                      |
| V3.9.3 | 18/09/2023 | <p>Added new FC token for Visa/Mastercard India recurring requirement.</p> <p>Added statement to P-18 Merchant Type</p> <p>Removed invalid note 3 for RC 62 in P-39</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>SC</p> <p>NO</p> <p>SC</p>           |
| V3.9.4 | 14/11/2023 | Updated 4.2.11 Added new program protocol values to support new 3D Secure versions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | TW                                      |
| V3.9.5 | 01/02/2024 | Updated field P-2 presence from O (optional) to CE (conditional echo) for response messages 0110, 0130, 0210 & 0230 and to ME (mandatory echo) for 0430.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ST                                      |

|        |            |                                                                                                                                                                                                                                                                                            |    |
|--------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| V3.9.6 | 25/04/2024 | Added positions 3-40 on Token QX Misc Response Data.<br>Updated 4.2.30 Added new Tag 019 Sender Purpose Of Payment Code to Token QY Sender Data.                                                                                                                                           | TW |
| V3.9.7 | 10/06/2024 | Clarified C0 token position 19 value '5' requires to be set for Mastercard DSRP.<br>Clarified C0 token position 24 values '6' and '7'.                                                                                                                                                     | NO |
| V3.9.8 | 30/07/2024 | Added 4.2.31 new Token R3 Payment Data Token.<br>Added new token to 4.1 list of Tokens.<br><br>Added MPG ID to 1.2 Glossary of terms                                                                                                                                                       | TW |
| V3.9.9 | 15/11/2024 | Added token R3 for Visa instalment service (VIS) transactions. Added token SE and updated tokens QI POS 25 = C to support Visa Direct OCTs.<br>Additional tags added to QW recipient data.<br>Updated note for P-52 to not include with OCTs.<br>Updated Token 04 to allow T = Token error | SC |